LLM Inference Optimization

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Disclaimer

Machine learning systems is a broad and rapidly evolving field. The course material has been developed using a broad spectrum of resources, including research papers, lecture slides, blogposts, research talks, tutorial videos, and other materials shared by the research community. External pictures and are heavily reused.

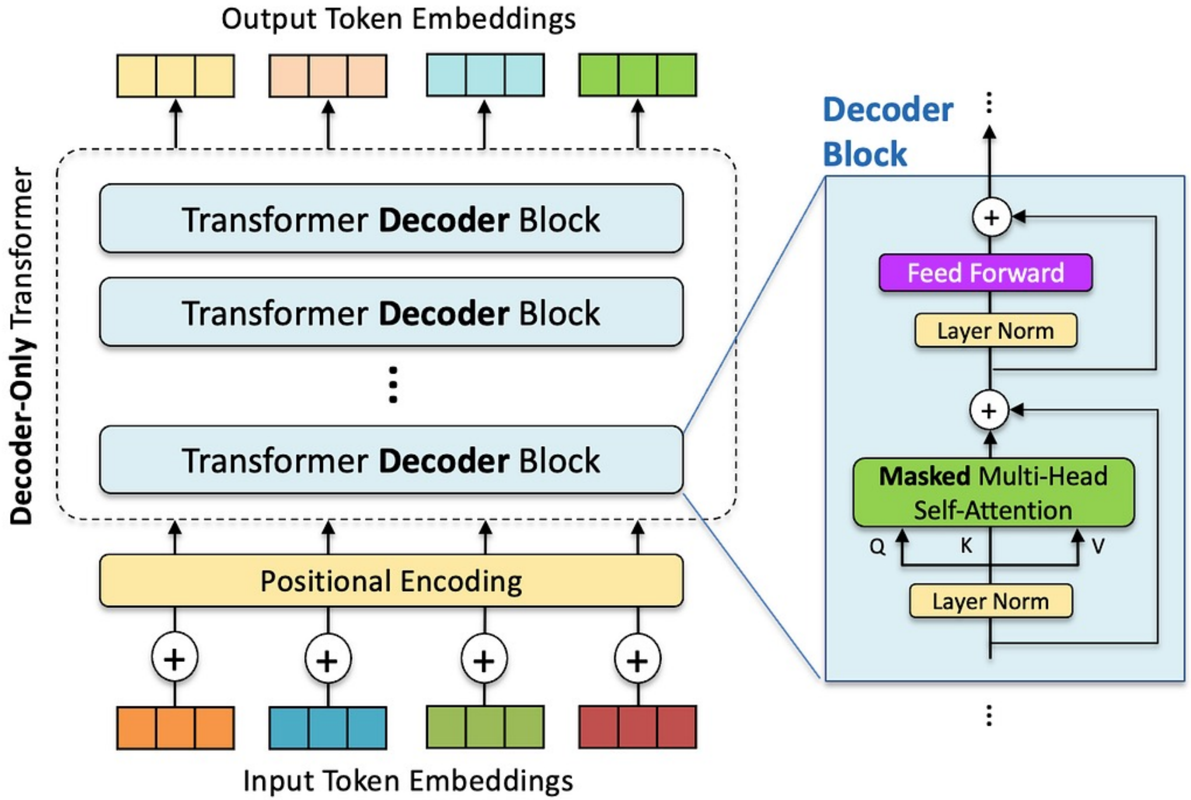
Inference Optimization: Outline

- Overview
- Attention Optimization
- Continuous Batching
- KV Cache Optimization
- Speculative Decoding
- Distributed Serving

Overview

- Decoder-only Transformer

GPT

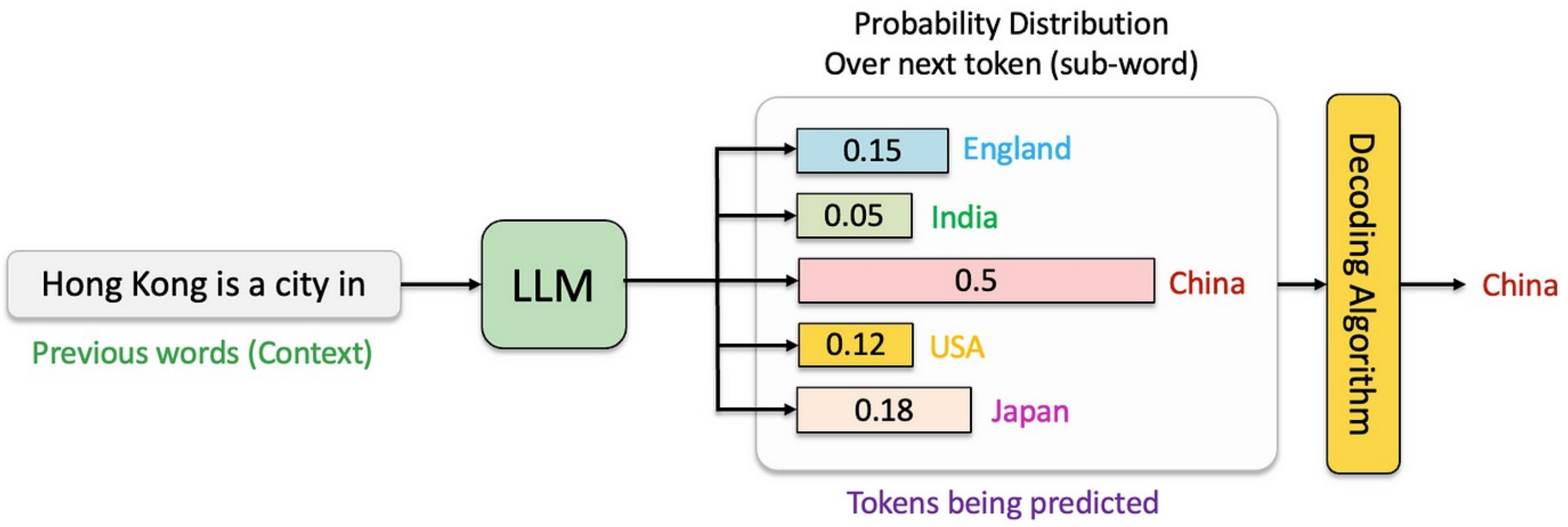


GPT (Generative Pre-trained Transformer) is the first decoder-only Transformer model

Source: The Rise of LLMs: From GPT to Modern Innovations

Overview

- Decoder-only Transformer
  - Generating a probabilistic distribution over possible next token, and a decoding algorithm is employed to select the actual output token



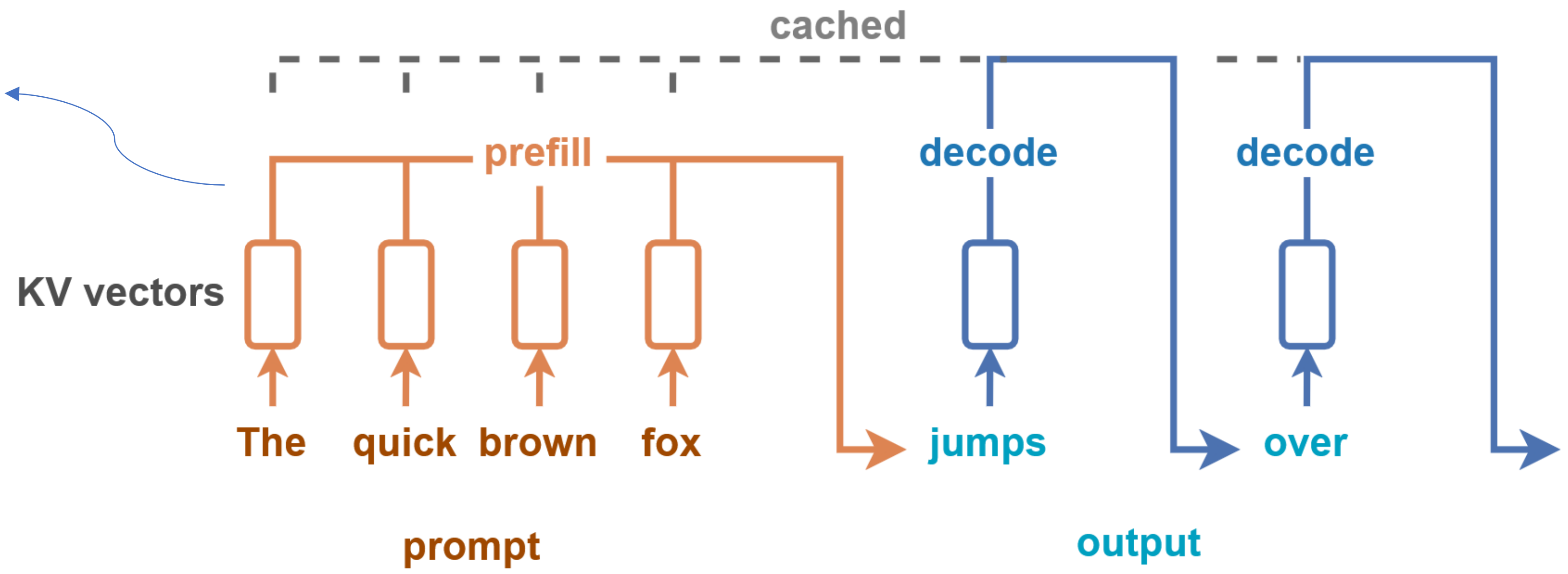
Next-token prediction, i.e. generating output tokens one by one

Source: The Rise of LLMs: From GPT to Modern Innovations

Overview

- Autoregressive Decoding
  - **“Prefill”** refers to the **initial parallel computation phase** where the model processes the entire input prompt for subsequent token-by-token generation

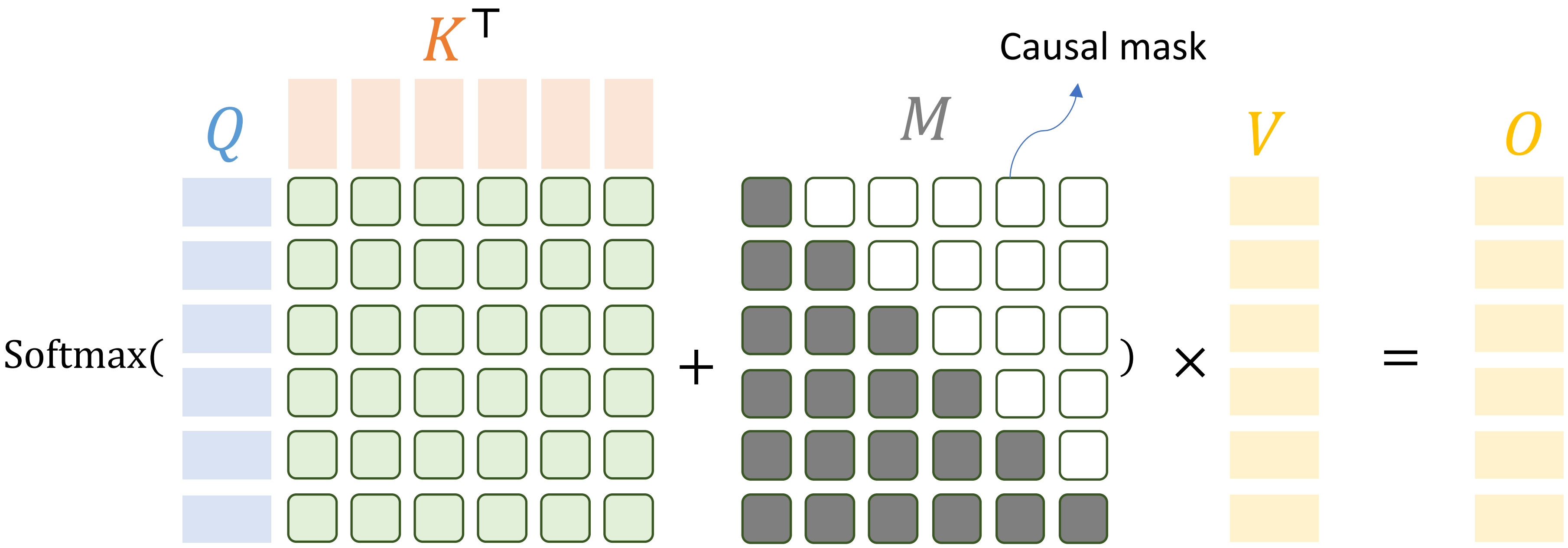
Parallelizable



Source: [Prefill and Decode for Concurrent Requests - Optimizing LLM Performance]()

Overview

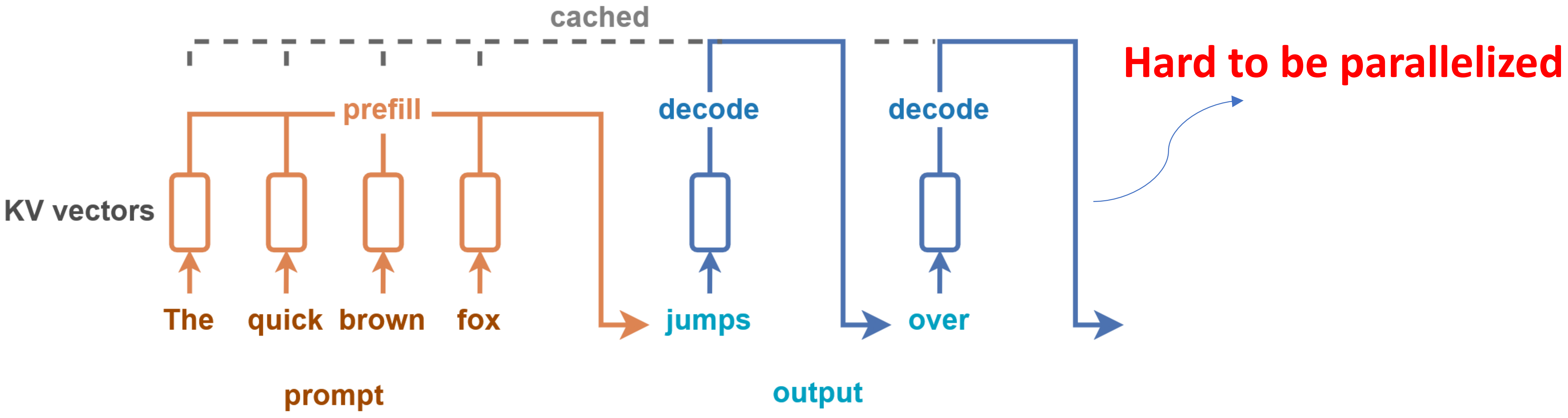
- Prefill: highly parallelizable
  - A small batch size can ”saturate” GPU computation
    - Parallelization over batch size, header size, sequence length and thread-block tiling



Source: Jia-Bin Huang from University of Maryland College Park

Overview

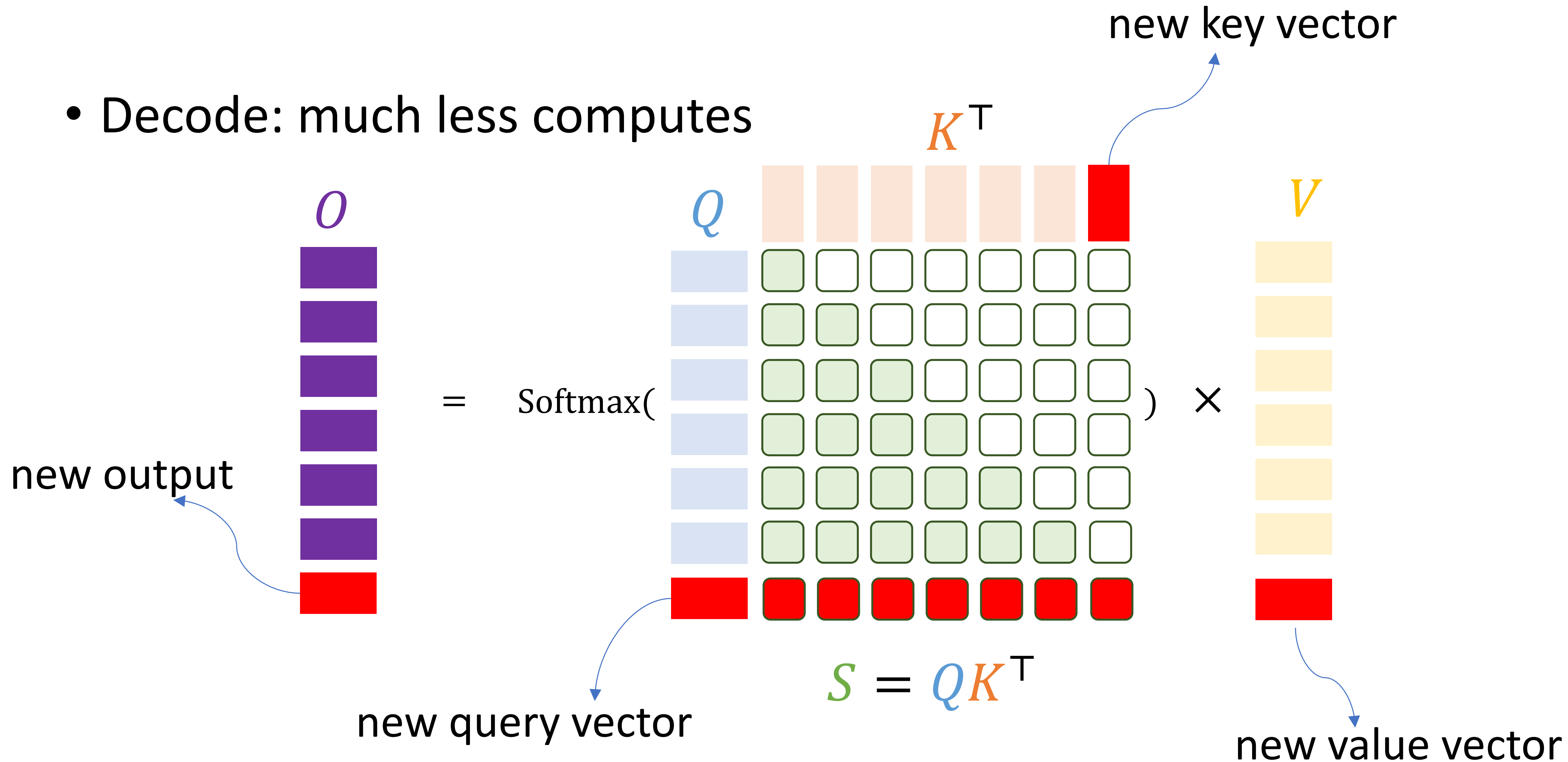
- Autoregressive Decoding
  - **"Decode"** refers to the **iterative process of generating output tokens** one at a time, where each new token is predicted based on the input prompt and all previously generated tokens



Source: [Prefill and Decode for Concurrent Requests - Optimizing LLM Performance]

Overview

- Decode: much less computes

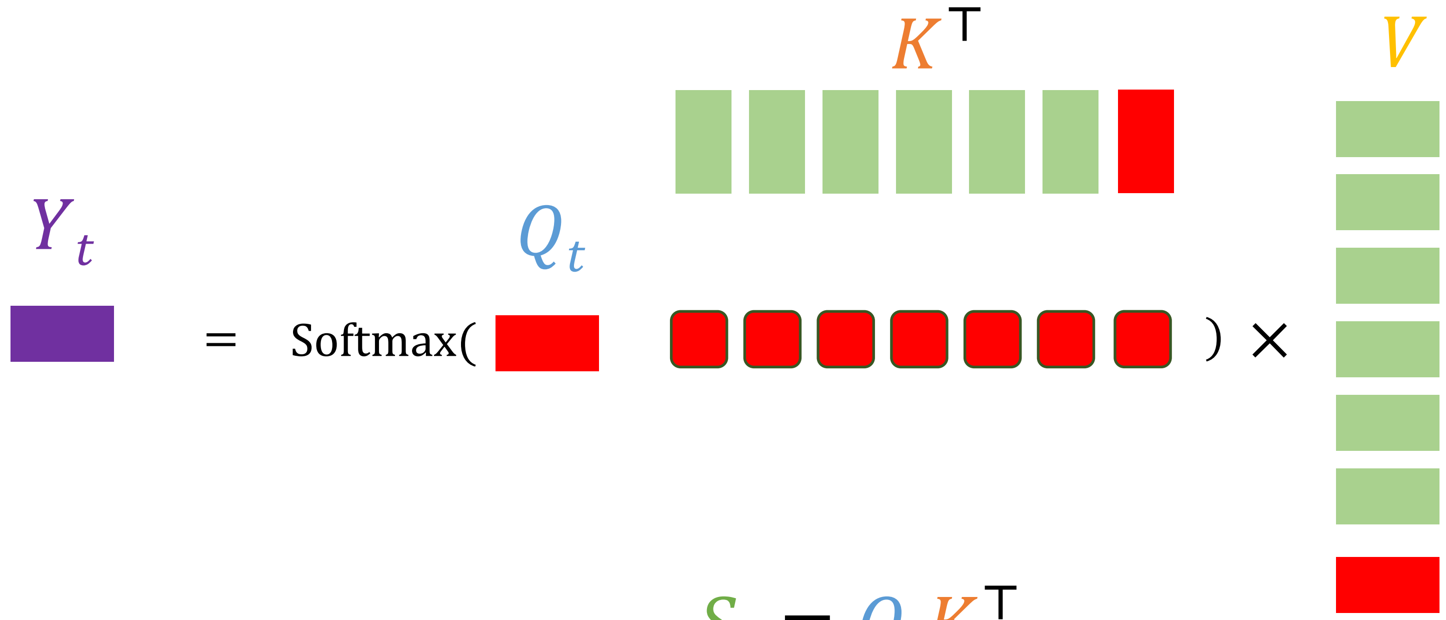


Decode without caching

Overview

Newly computed
Could be cached

- Decode: much less computes
  - Why caching Key and Value, other than Query

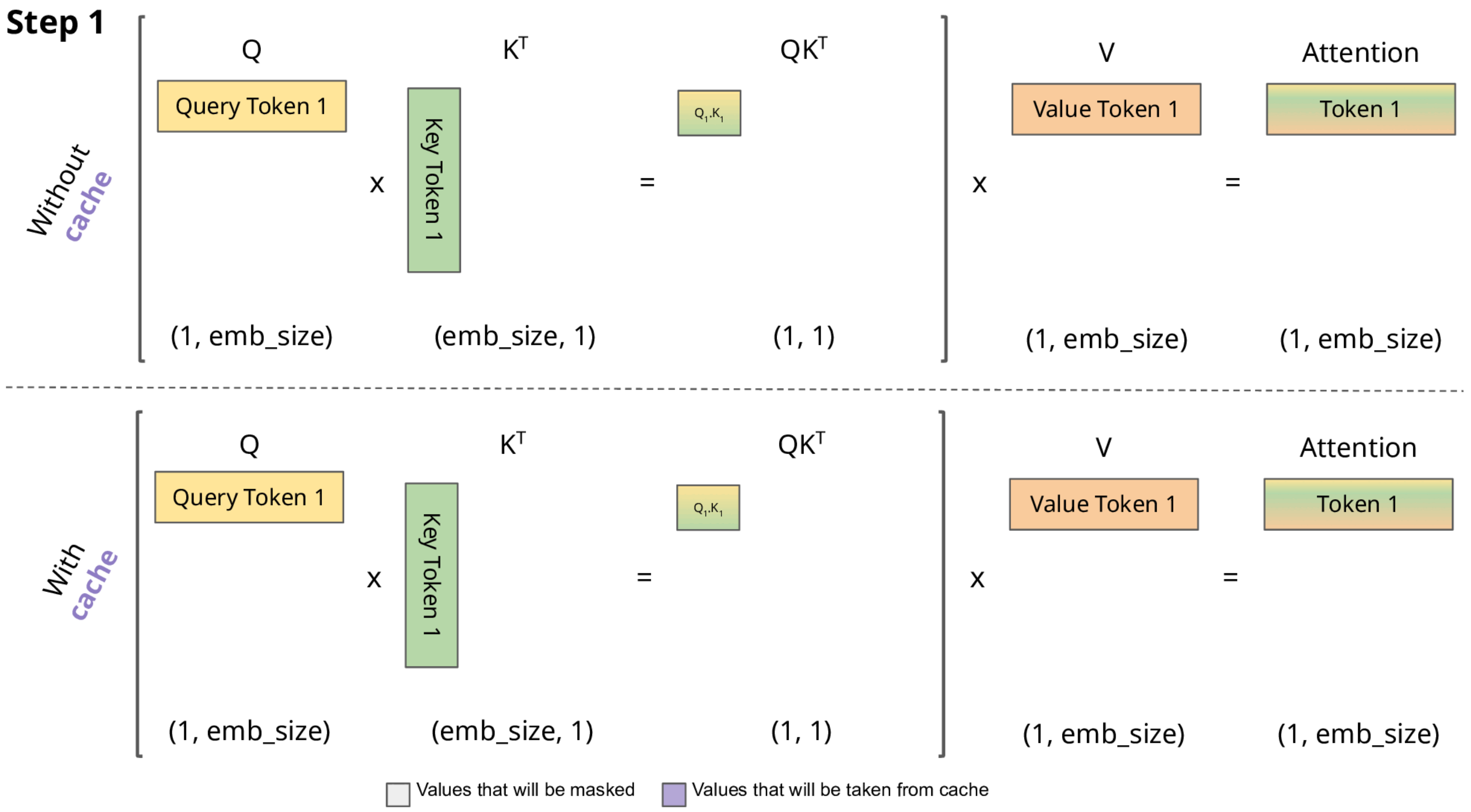


$$S_t = Q_t K^\top$$

Decode Compute output token one by one

Overview

- An animation



Source: KV cache, paged attention / NLP&RL seminars RU S06 | 25s | girafe-ai

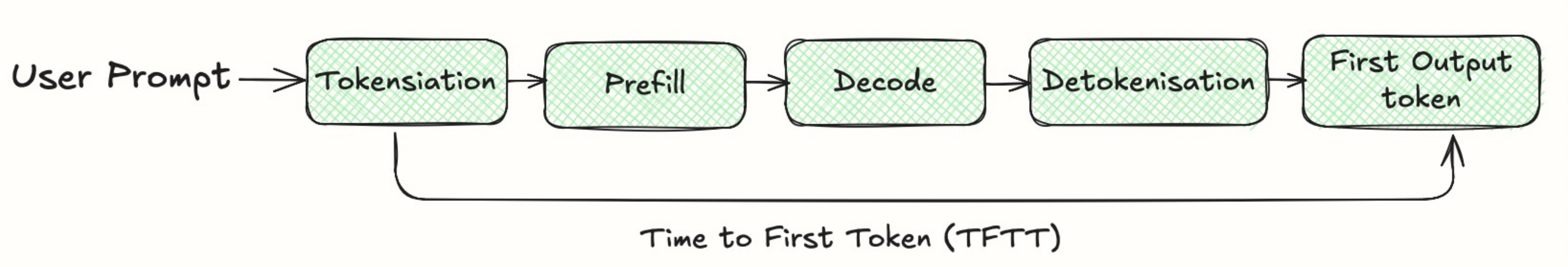
Overview

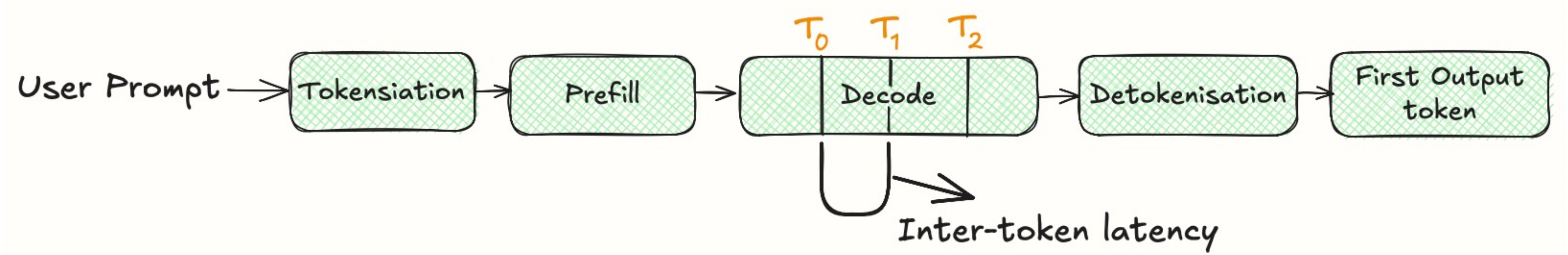
- Why Caching KV?
  - Reducing redundant computations for KEYs and VALUEs
  - Increasing memory consumption

2	$\times$	2	$\times$	8096	$\times$	80	$\times$	64	$\times$	64	
K/V		**Float16**		**Sequence length**		**# of layers**		**# of heads**		**dimension**	**~10.6 GB!**

Overview

- Autoregressive Decoding
  - Token generation core metrics: **TTFT** and **TPOT** (time per output token)



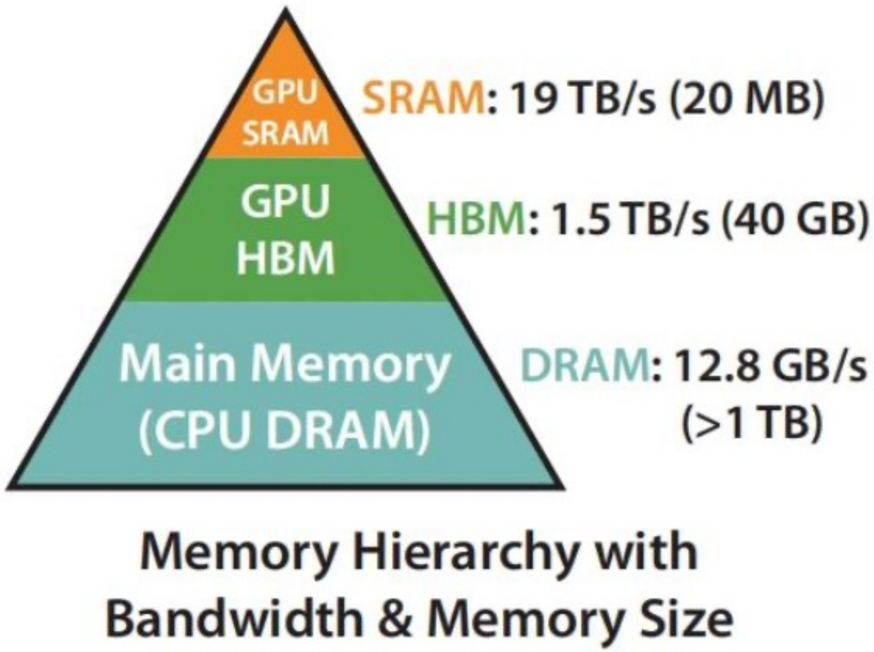


Source: [Prefill and Decode for Concurrent Requests - Optimizing LLM Performance]

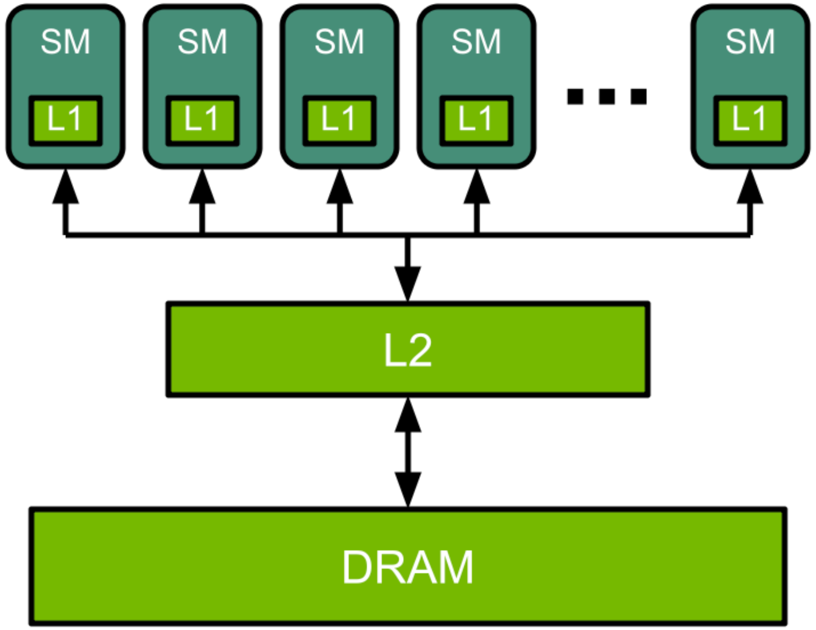
Overview

Insufficient to store data at SRAM: load from HBM to SRAM and write back to HBM

- Hierarchical GPU Memory
  - I/O throughput versus memory size



A macroscope view of I/O tradeoff (V100)



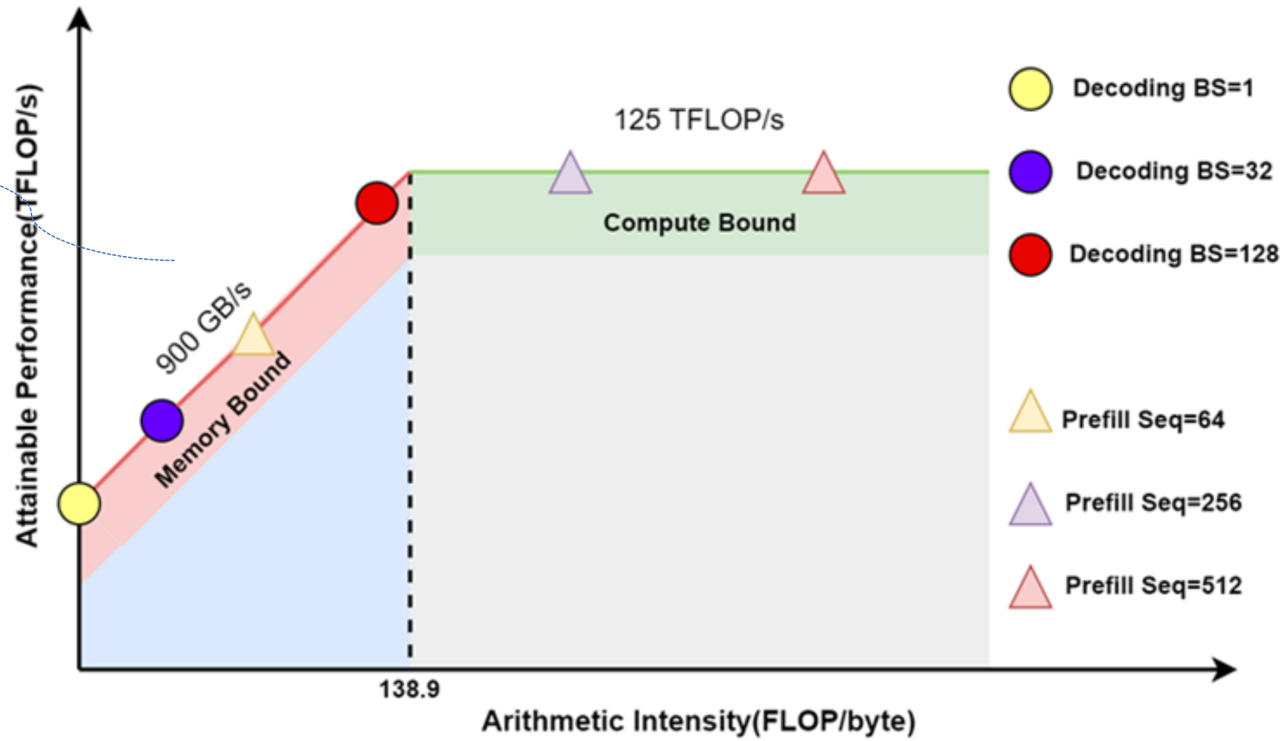
L1 instruction cache: 192KB per SM * 108 SM ~20MB
Data flow: DRAM/L2 Cache to/from L1 Cache, much slower than computing

Source: https://docs.nvidia.com/deeplearning/performance/dl-performance-gpu-background/index.html

Overview

- Prefill and Decode
  - Prefill: Compute intensive with GEMM, Decode: memory I/O intensive with GEMV

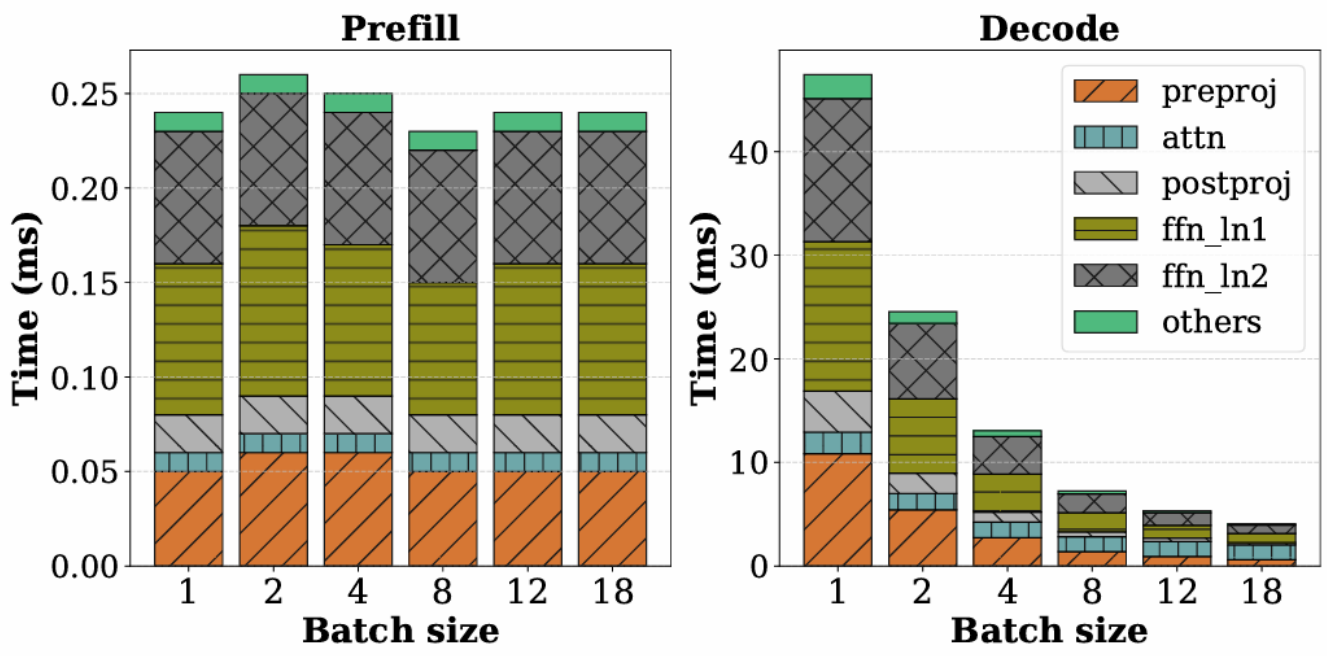
Wasted FLOPS



Roofline model of NVIDIA V100 GPU

Overview

- Prefill and Decode
  - Prefill: Compute intensive, Decode: memory I/O intensive
    - Prefill saturates GPU compute even at batch size of 1
    - Decode under-utilizes GPU compute and costs as much as 200 times prefill for bs=1



Per-token prefill and decode time with different batch sizes (sequence length = **1024**) for LLaMa-13B on A6000 GPU

Source: SARATHI: Efficient LLM Inference by Piggybacking Decodes with Chunked Prefills

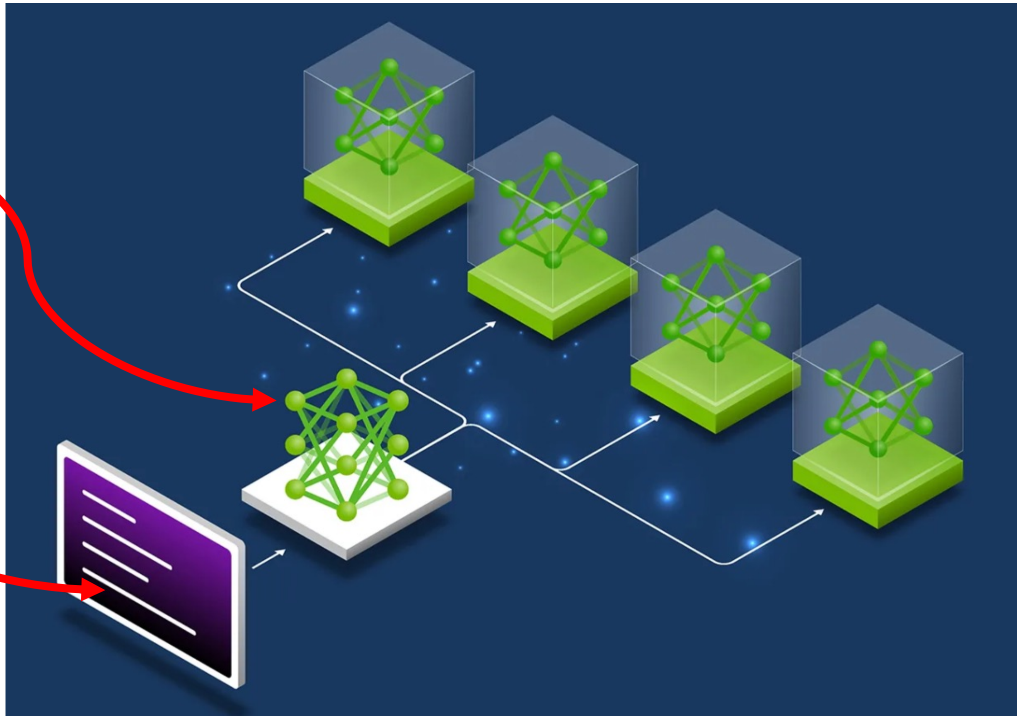
Overview

- Very Large-scale LLM Inference

deepseek

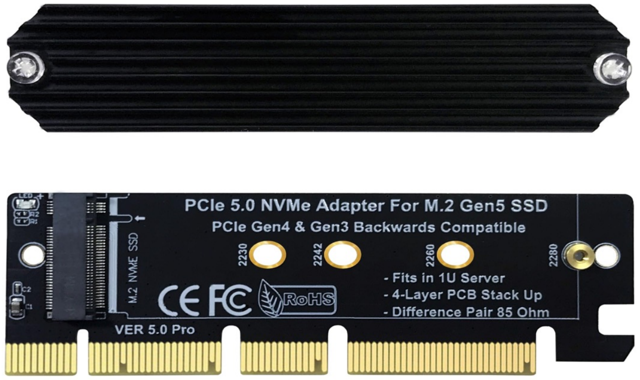
Full model size with 256 experts

Long context and high request loads

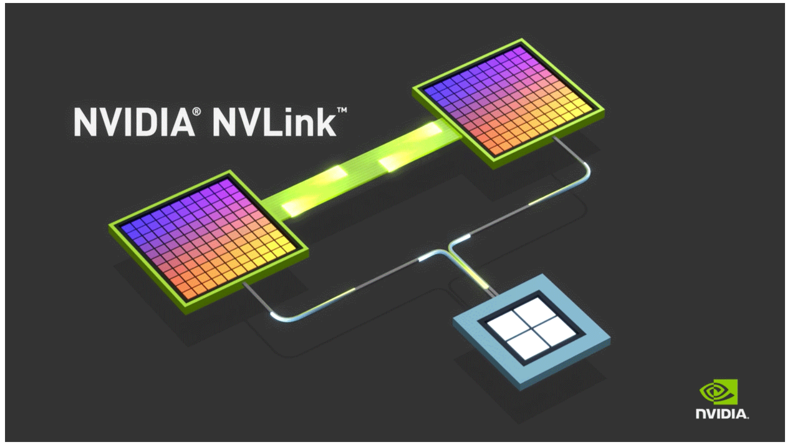


Overview

- Versatile communication medium
  - Transmission delay emerges



PCIe Link
e.g. CPU – GPU with
up to 128GB/s bandwidth



NVLink
e.g. GPU – GPU
in total 1.8TB/s bandwdith

RoCE
e.g. Machine – Machine
up to 800 Gbps

Overview

- Challenges of LLM Inference
  - New token generation paradigm
    - **Prefill** and **Decode**
  - Hierarchical memory
    - **Fast** I/O small size versus **slow** I/O large size
  - Heterogeneous bandwidth
    - **High** intra-machine bandwidth versus **low** inter-machine bandwidth
- To emphasize
  - many optimization methods, e.g. **attention optimization** and **prefill optimization** can be employed for model **training**

Inference Optimization: Outline

- Overview
- Attention Optimization
    - **Sparse Attention**
    - Linear Attention
    - Flash Attention
    - Continuous Batching
- Continuous Batching
- KV Cache Optimization
- Speculative Decoding
- Distributed Serving

Attention Optimization

Key matrix: $K \in \mathbb{R}^{N \times d}$

$$\begin{bmatrix} \alpha_{1,1} & \alpha_{1,2} & \alpha_{1,3} & \alpha_{1,4} & \alpha_{1,5} \\ \alpha_{2,1} & \alpha_{2,2} & \alpha_{2,3} & \alpha_{2,4} & \alpha_{2,5} \\ \alpha_{3,1} & \alpha_{3,2} & \alpha_{3,3} & \alpha_{3,4} & \alpha_{3,5} \\ \alpha_{4,1} & \alpha_{4,2} & \alpha_{4,3} & \alpha_{4,4} & \alpha_{4,5} \\ \alpha_{5,1} & \alpha_{5,2} & \alpha_{5,3} & \alpha_{5,4} & \alpha_{5,5} \end{bmatrix} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{bmatrix} \begin{bmatrix} k_1^\top & k_2^\top & k_3^\top & k_4^\top & k_5^\top \end{bmatrix}$$

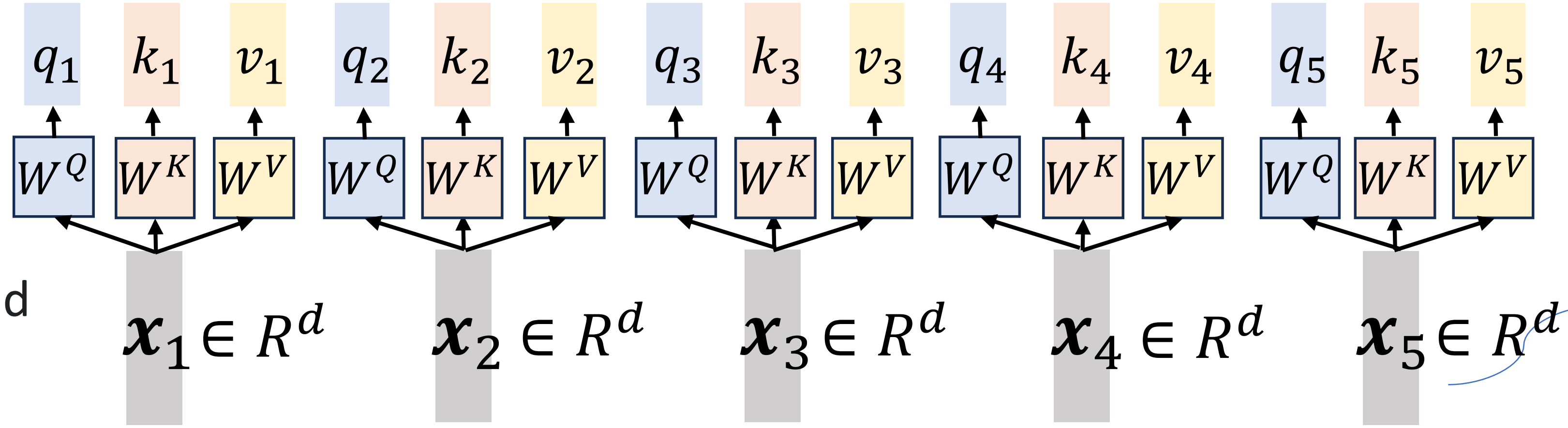
Attention score matrix: $\mathbb{R}^{N \times N}$

Query matrix: $Q \in \mathbb{R}^{N \times d}$

$W^Q \in R^{d \times d}$

$W^K \in R^{d \times d}$

$W^V \in R^{d \times d}$



Parameters

Embedded Tokens

Token

Source: Jia-Bin Huang from University of Maryland College Park

Attention Optimization

- Real-world operations
  - Transferring both matrices from global memory to shared memory for computing, and write the result back to global memory progressively
- Complexity
  - Computation: $O(N^2 d)$
    - Considering multiplications and additions → computing time $T_C = \frac{2N^2 d}{C}$
  - Communication: $O(N^2)$
    - Considering bidirectional I/O read/write → communication time $T_{I/O} = \frac{2*(N^2+2Nd)}{B}$
  - Global storage: $O(N^2)$

(partially) overlapped

GTP3 with 10K tokens in FP16:
200MB per head per layer

Attention Optimization

- Real-world operations
  - Transferring both matrices from global memory to shared memory for

Computation, I/O and storage should all be optimized!

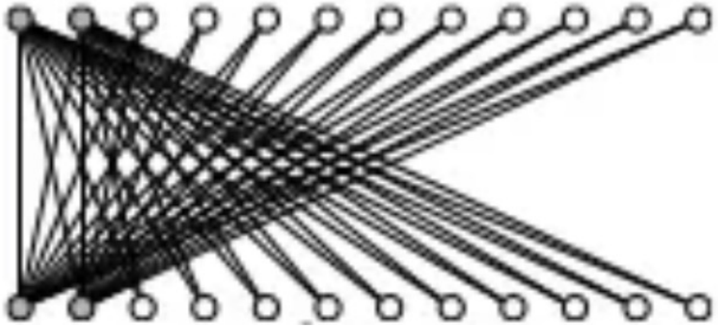
  - Communication: $O(N^2)$
    - Considering bidirectional I/O read/write → communication time $T_{I/O} = \frac{2*(N^2+2Nd)}{B}$
  - Global storage: $O(N^2)$

Sparse Attention

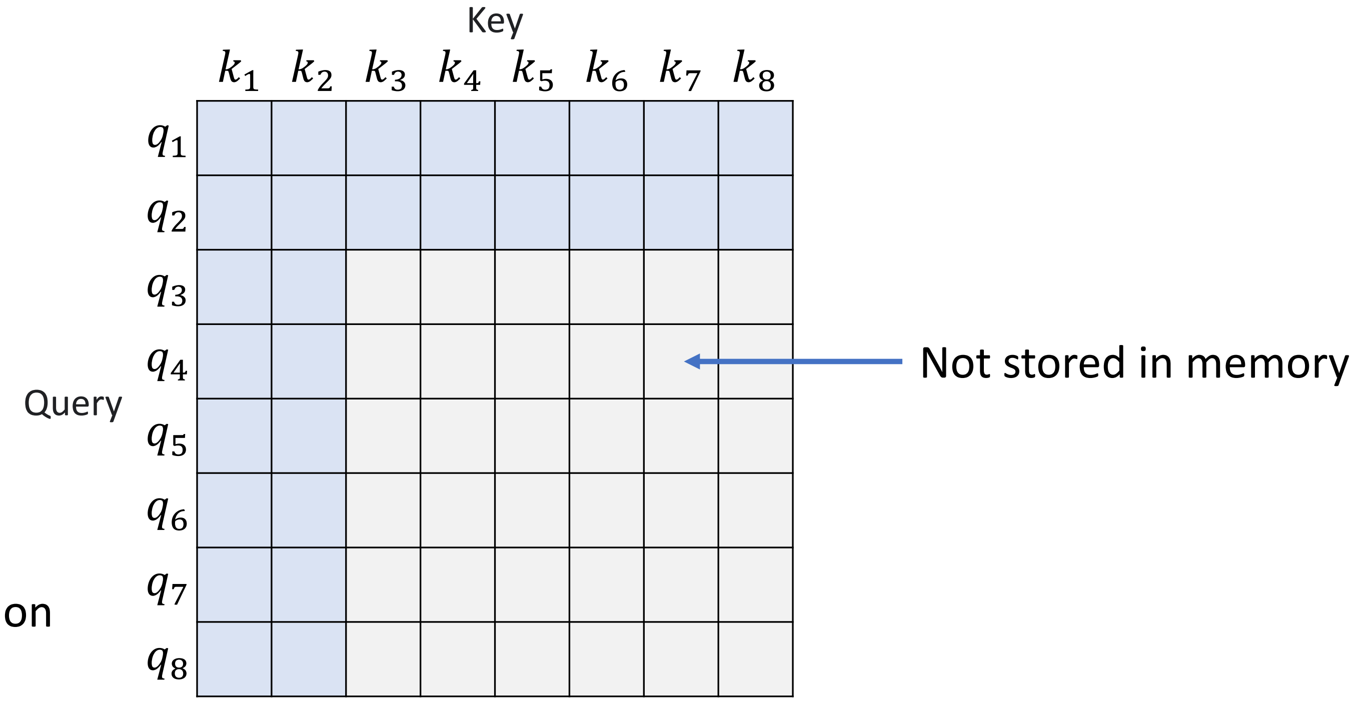
- Sparse attention
  - reduce computational and memory cost by only computing attention for a **subset of token pairs** instead of all pairs
  - evidence from machine learning community (e.g. Rewon Child, 2019)
- Sparse attention patterns
  - Position-based sparse attention: **rule-based or heuristic methods** to decide the positions where the interactions of these pair-wise tokens are important
  - Content-based sparse attention: tokens selectively attending only to other tokens that are **relevant based on their representations**

Position-based Sparse Attention

- Global attention
  - global nodes act as an information hub, allowing them to attend to every other node in the sequence

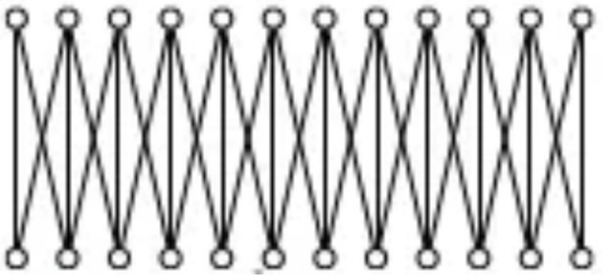


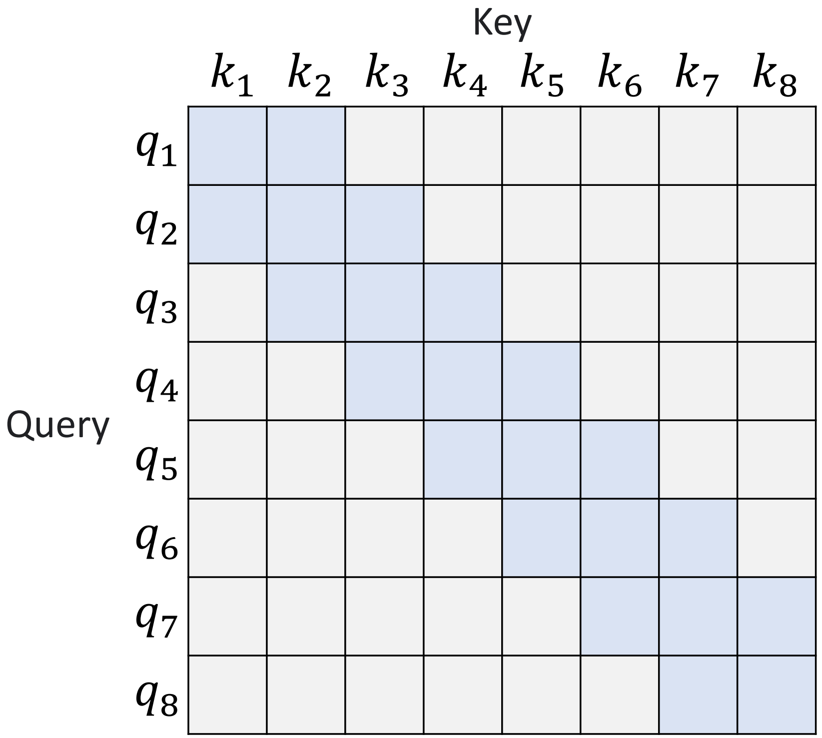
A bipartite graph illustration of QK^T computation



Position-based Sparse Attention

- Band attention
  - often termed “local attention” or “sliding window attention”, in which a node’s attentions are confined to neighboring nodes in a local window

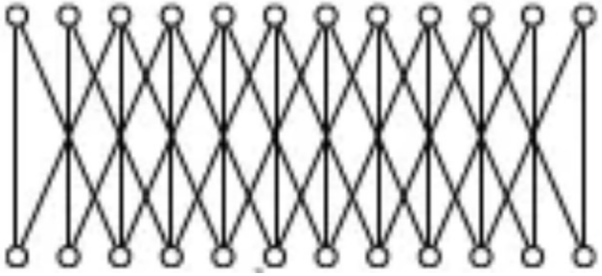


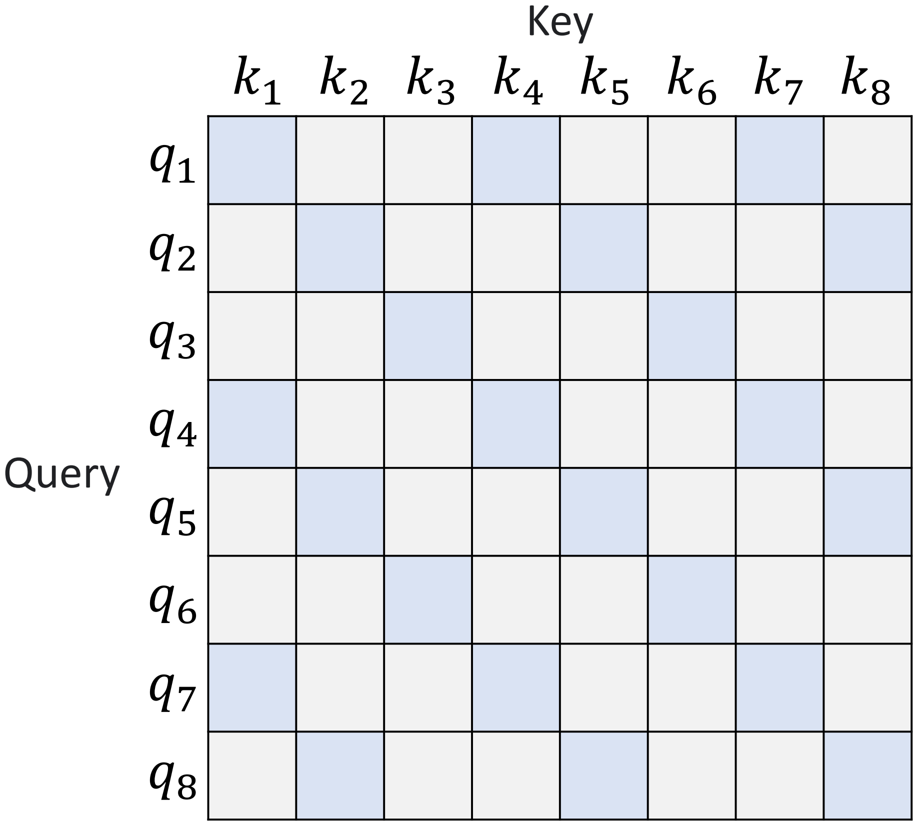


- Reducing complexity from $O(N^2)$ to $O(kN)$
- Limited receptive field (cannot capture long-range dependencies)
- Sacrificing context modeling ability: slow information propagation

Position-based Sparse Attention

- Dilated attention
  - using a dilated window with gaps of dilation equal to or greater than 1

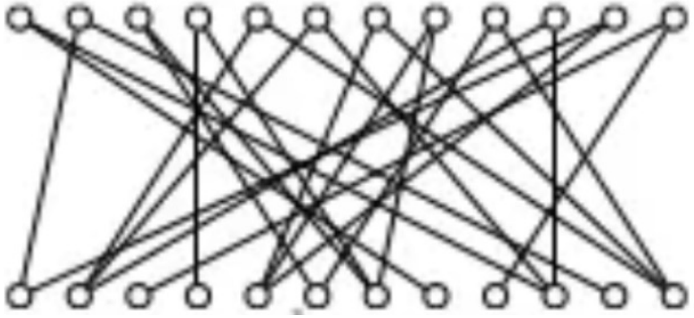


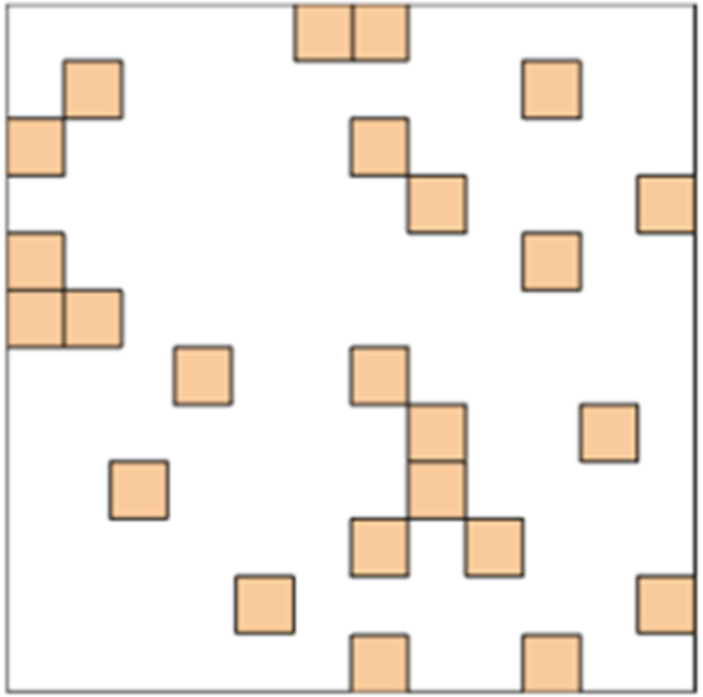


- Reducing complexity from $O(N^2)$ to $O(kN)$
- Missing fine-grained, short-range dependencies (failure to capture important neighbors in those gaps)

Position-based Sparse Attention

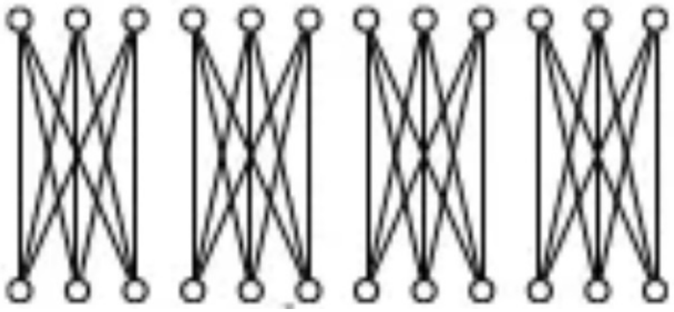
- Random attention
  - each query randomly samples a few keys for better capturing non-local interactions

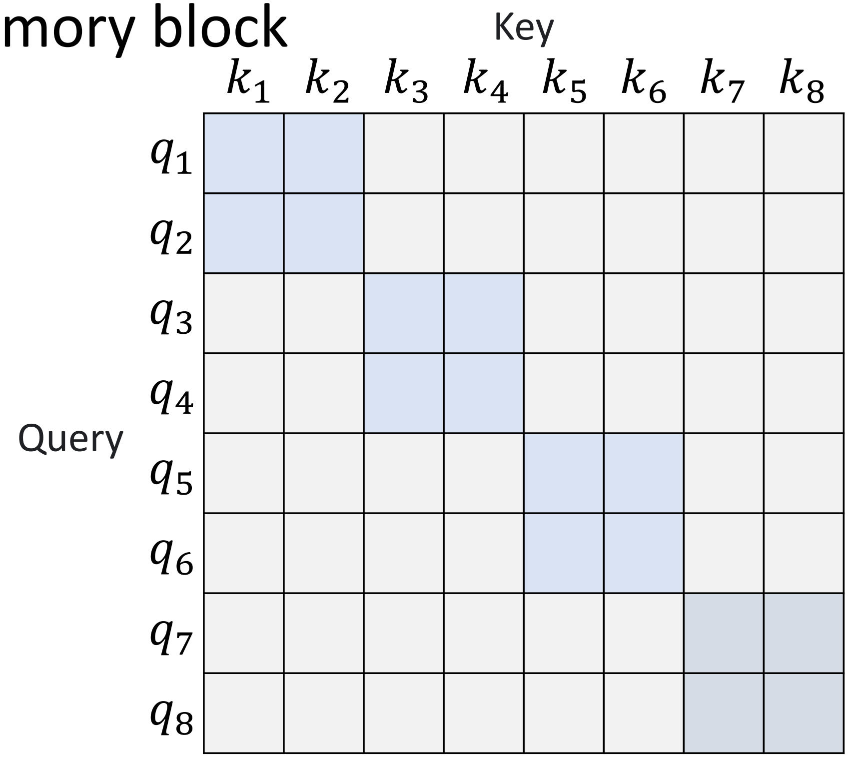




Position-based Sparse Attention

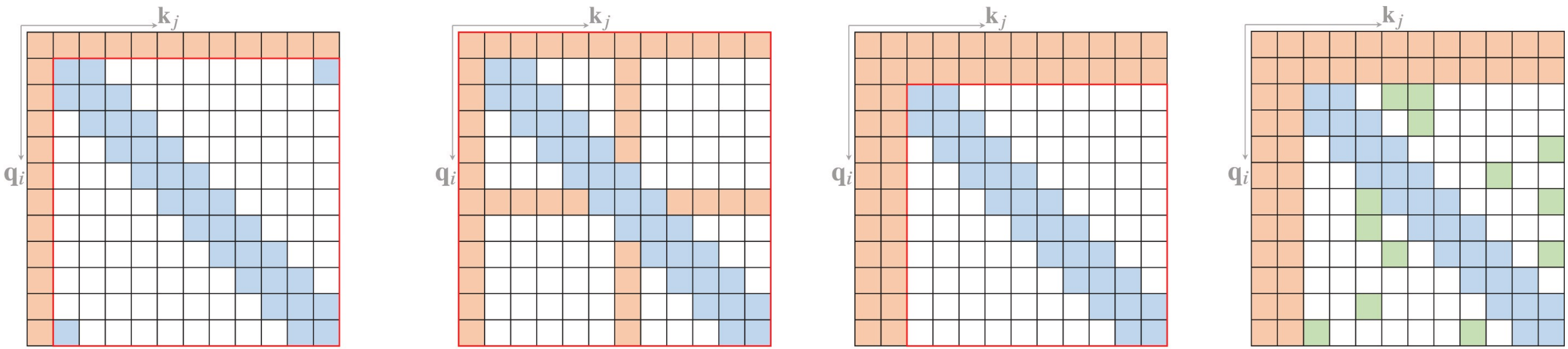
- Block attention
  - input sequence segmented into multiple non-intersection query blocks and each block assigned a local memory block





Position-based Sparse Attention

- COMPOUND attention



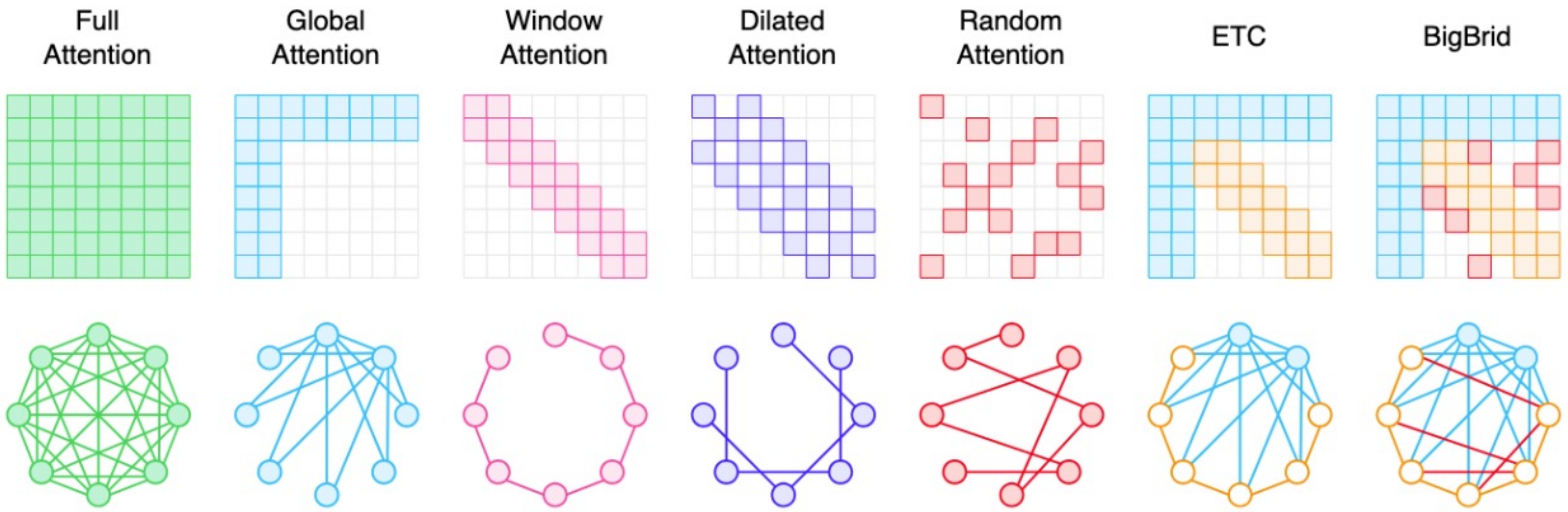
(a) Star-Transformer (b) Longformer (c) ETC (d) BigBird

- BigBird : local + global + random yields dense-equivalent performance
- Increased implementation complexity, more hyperparameters to tune and potential redundancy

https://wikidocs.net/198820

Position-based Sparse Attention

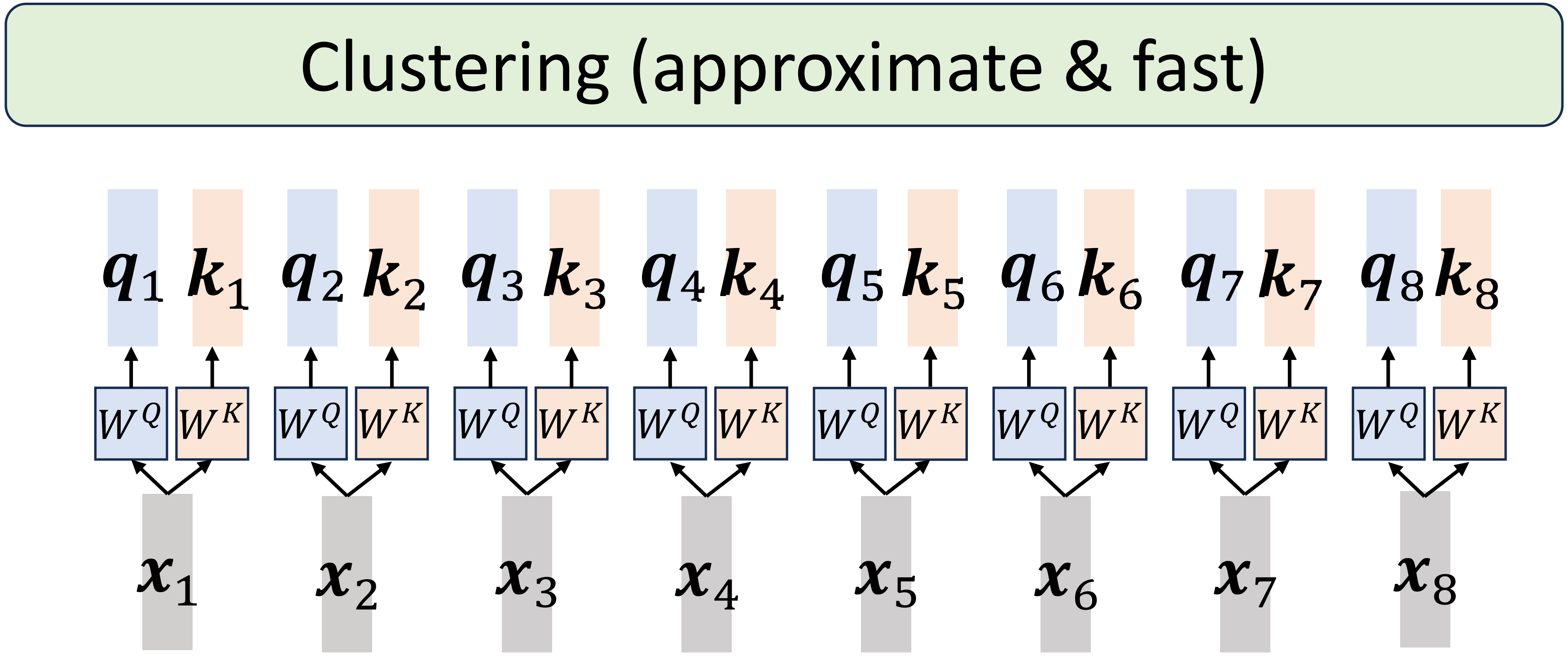
- Representative sparse attentions in a nutshell



Source: Speed Always Wins: A Survey on Efficient Architectures for Large Language Models

Content-based Sparse Attention

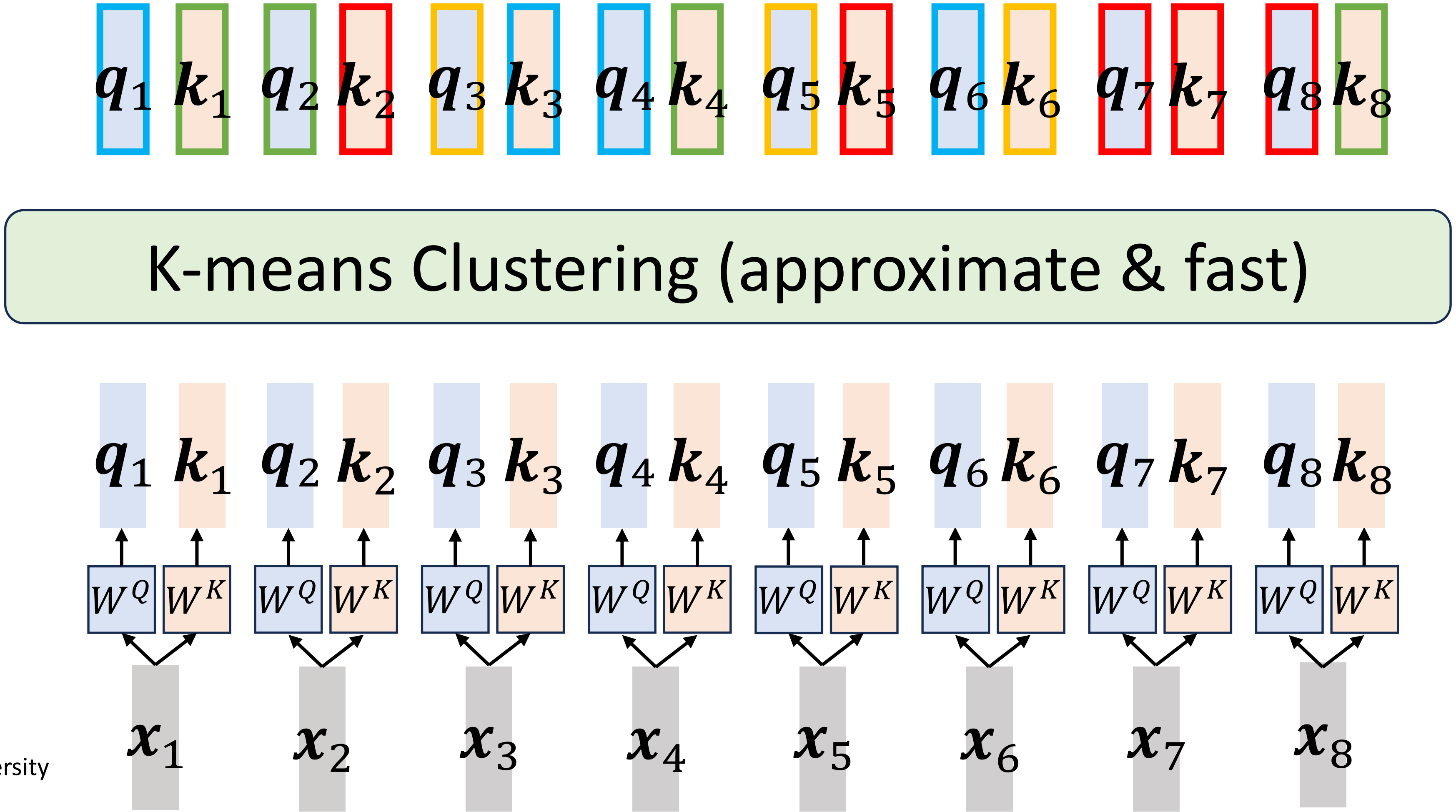
- Routing transformer



Source: Jia-Bin Huang from University of Maryland College Park

Content-based Sparse Attention

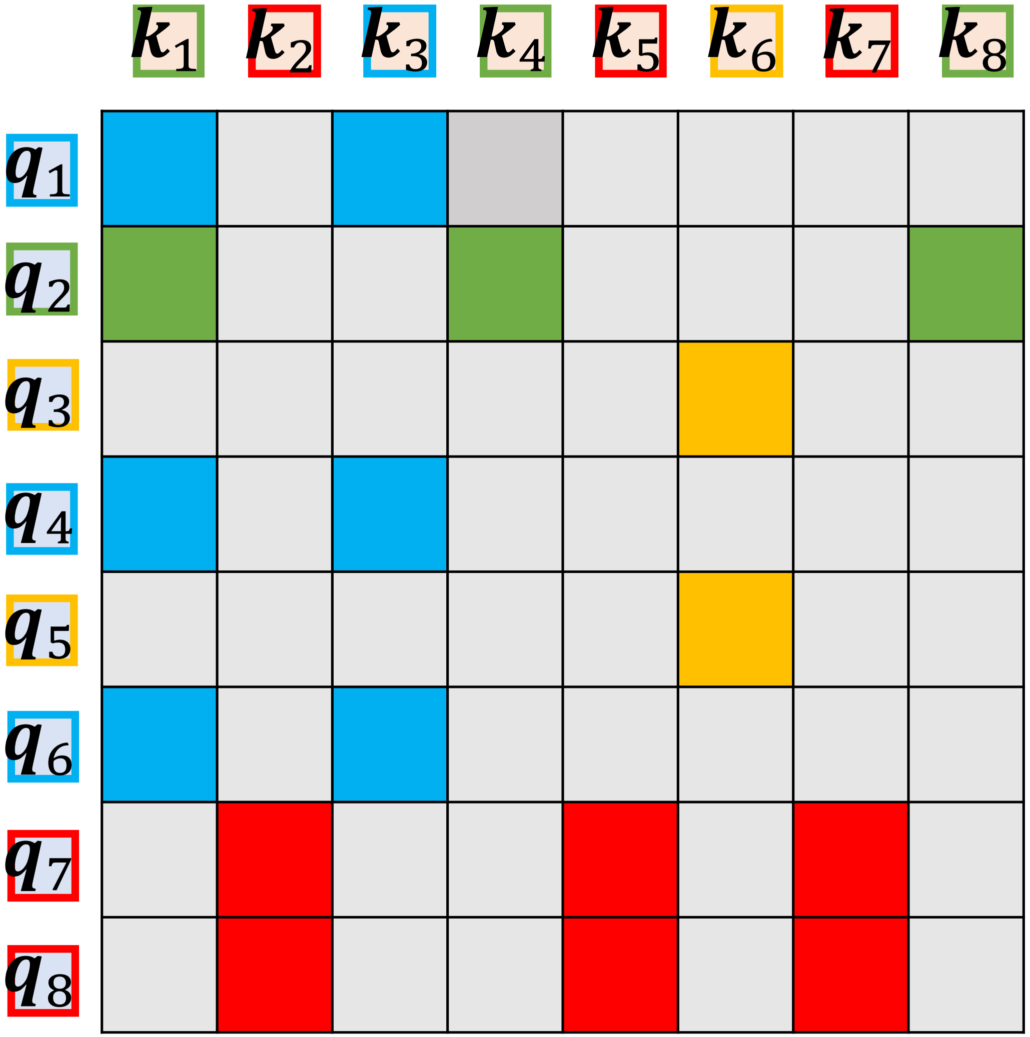
- Routing transformer



Source: Jia-Bin Huang from University of Maryland College Park

Content-based Sparse Attention

- Clusters
  - $\{q1, q4, q6, k1, k3\}$
  - $\{q2, k1, k4, k8\}$
  - $\{q3, q5, k6\}$
  - $\{q7, q8, k2, k5, k7\}$



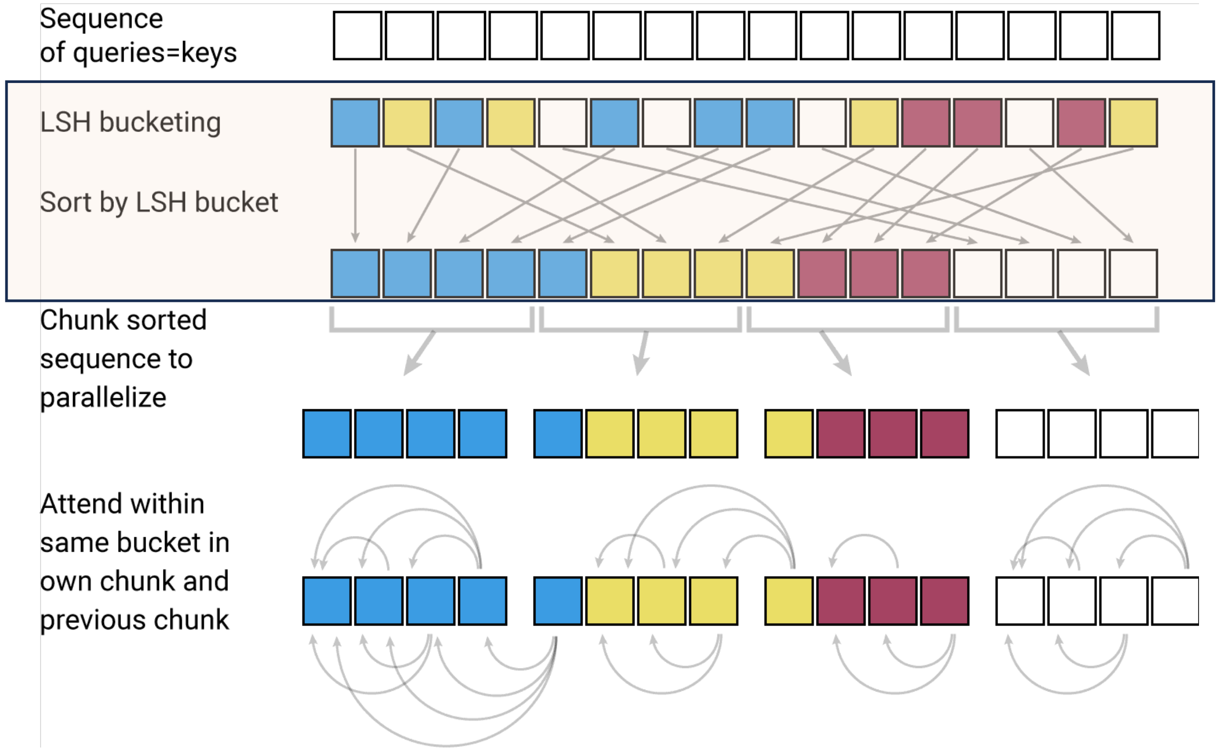
Source: Jia-Bin Huang from University of Maryland College Park

Content-based Sparse Attention

- Routing transformer: summary
  - attention scores in scaled dot-product rely on the similarity between Q and K → approximated by grouping them in a shared projection space
    - needs to be **double checked** by our students manually
  - each query only needs to attend to keys within its assigned cluster (typically $\sqrt{n}$ in size), reducing complexity to $O(n\sqrt{n})$ while preserving relevant long-range dependencies
  - cluster centroids are learned during training, allowing the model to adaptively group based on content

Content-based Sparse Attention

- Reformer: using locality-sensitive hashing (LSH) instead of dot-product attention



- Using **locality-sensitive hashing** (LSH) to select key-value pairs for each query
- Using Reversible Transformer to reduce memory usage during training
- Similar items (queries and keys) falling in the same bucket with high probability
- Processing sequences of length **64K tokens** or more efficiently

Source: Reformer: The Efficient Transformer

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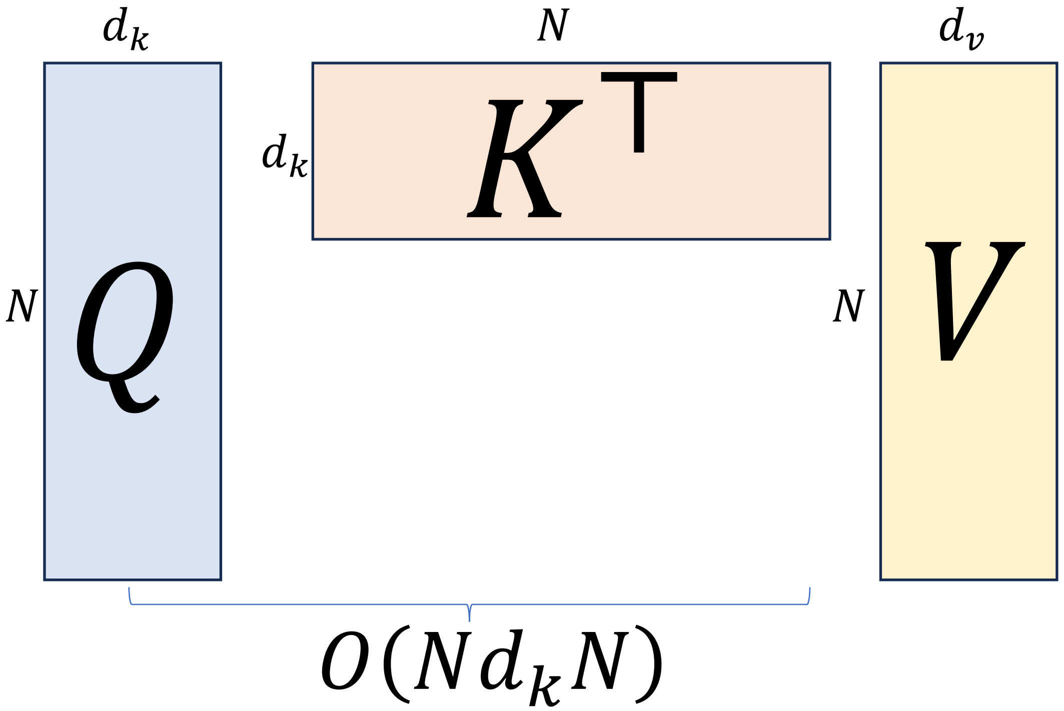
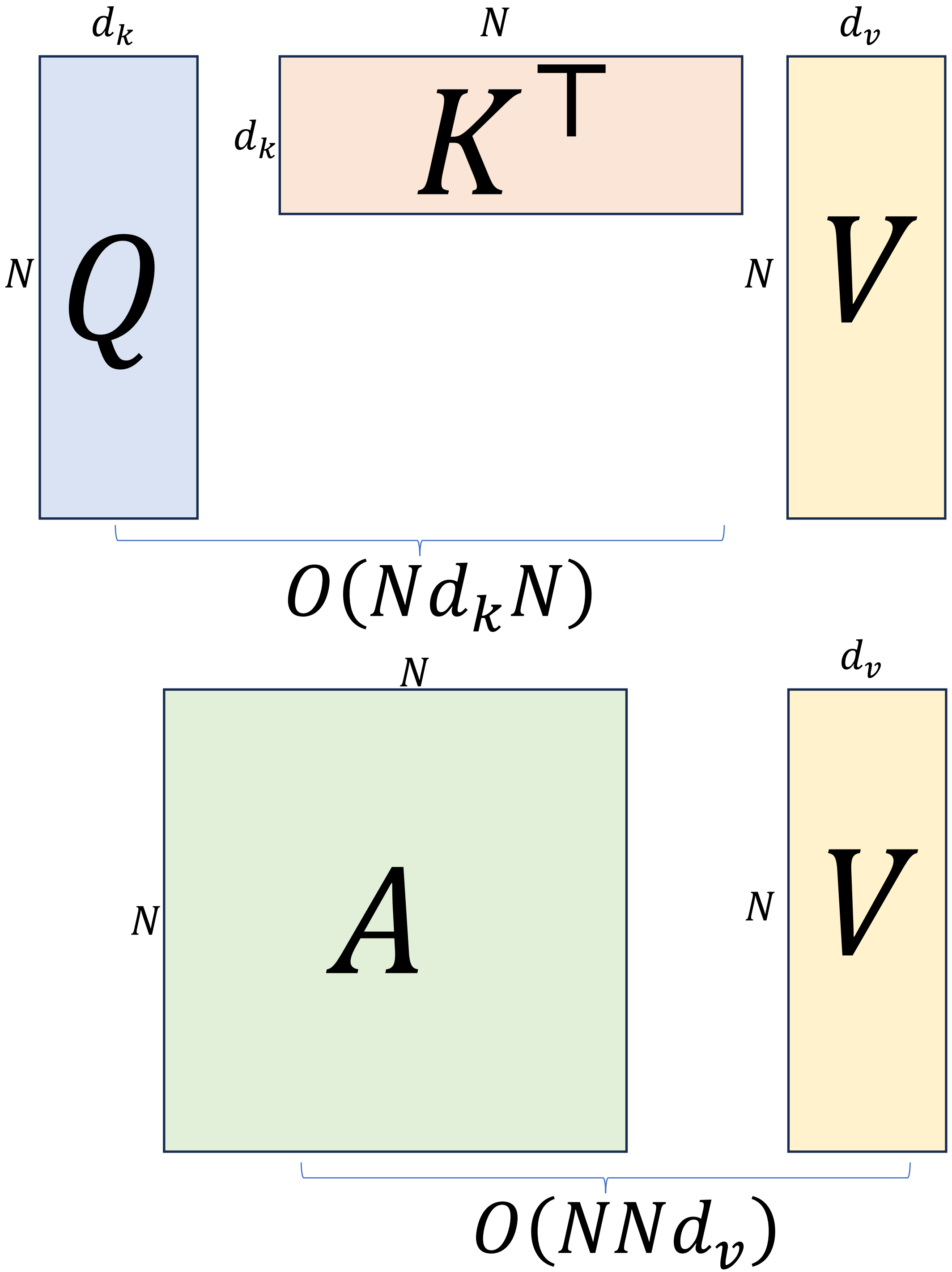


"optimal parenthesization" via dynamic programming

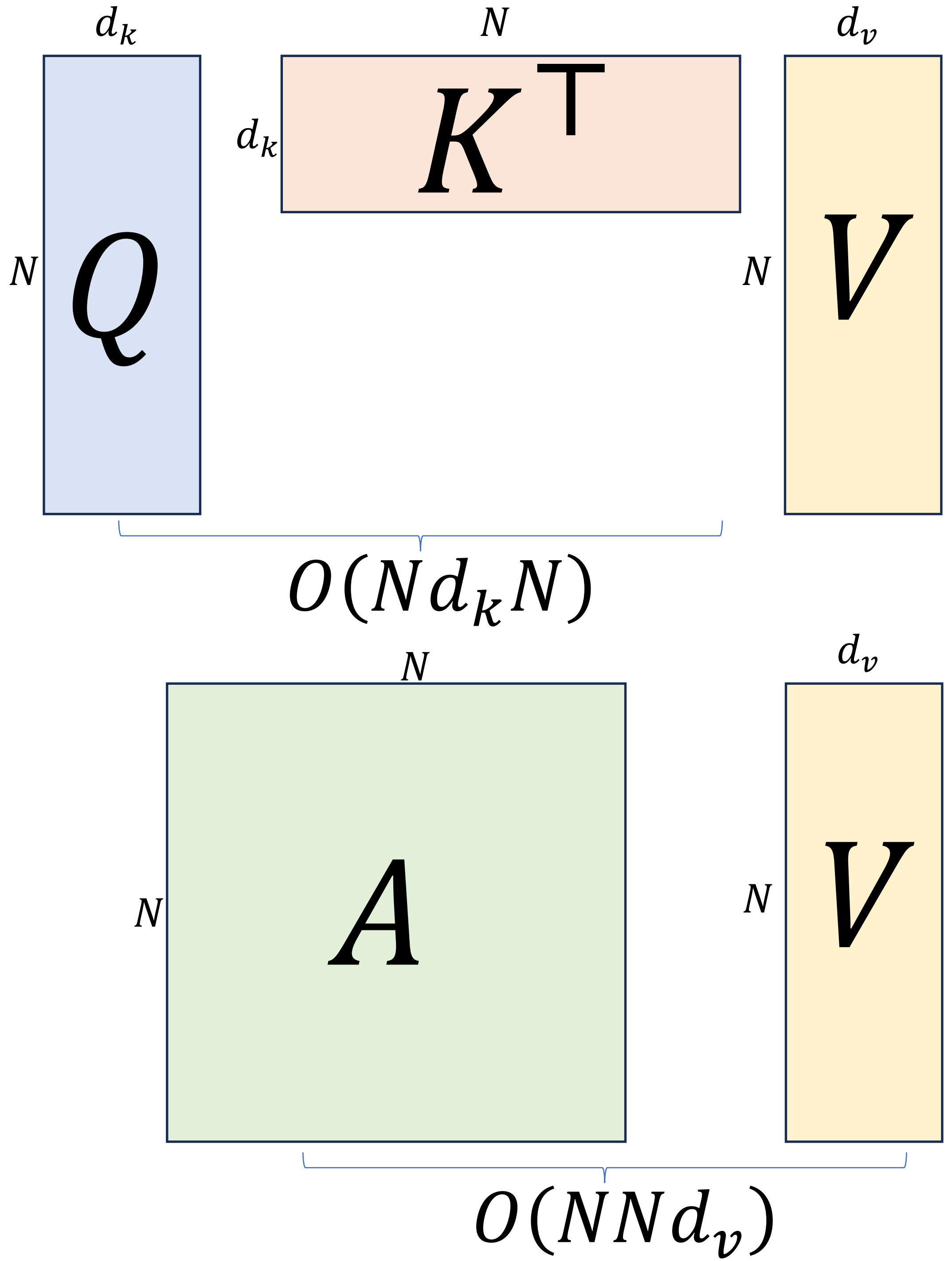
Source: Jia-Bin Huang from University of Maryland College Park



Source: Jia-Bin Huang from University of Maryland College Park

Original

$$O(N^2(d_v+d_k))$$



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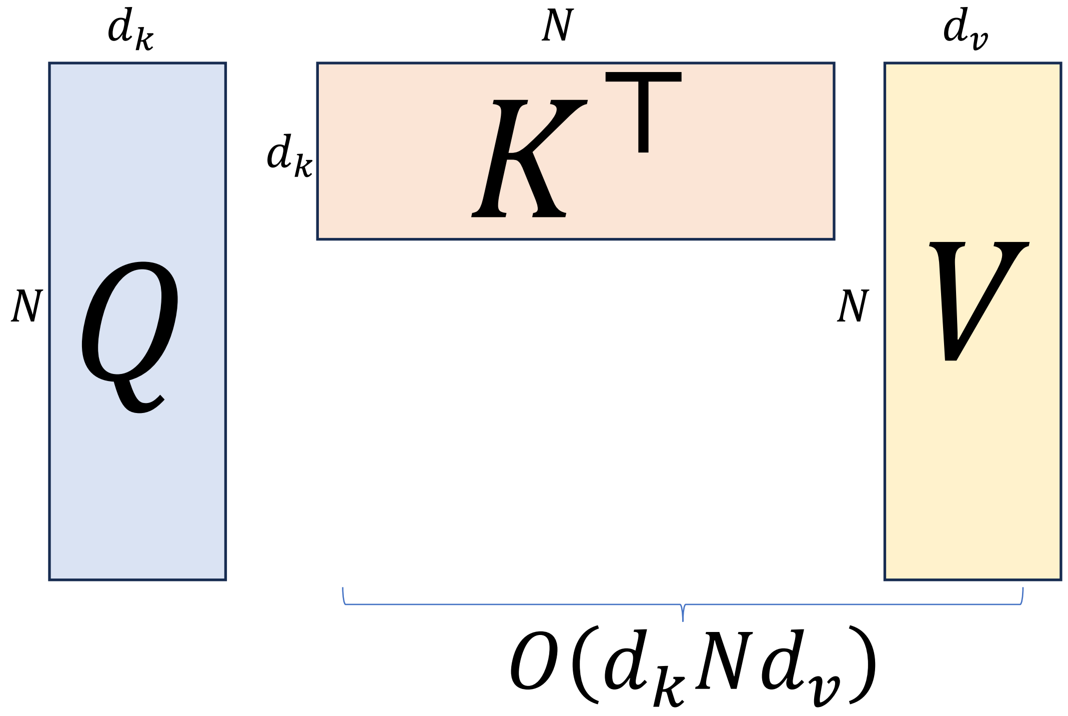

Source: Jia-Bin Huang from University of Maryland College Park

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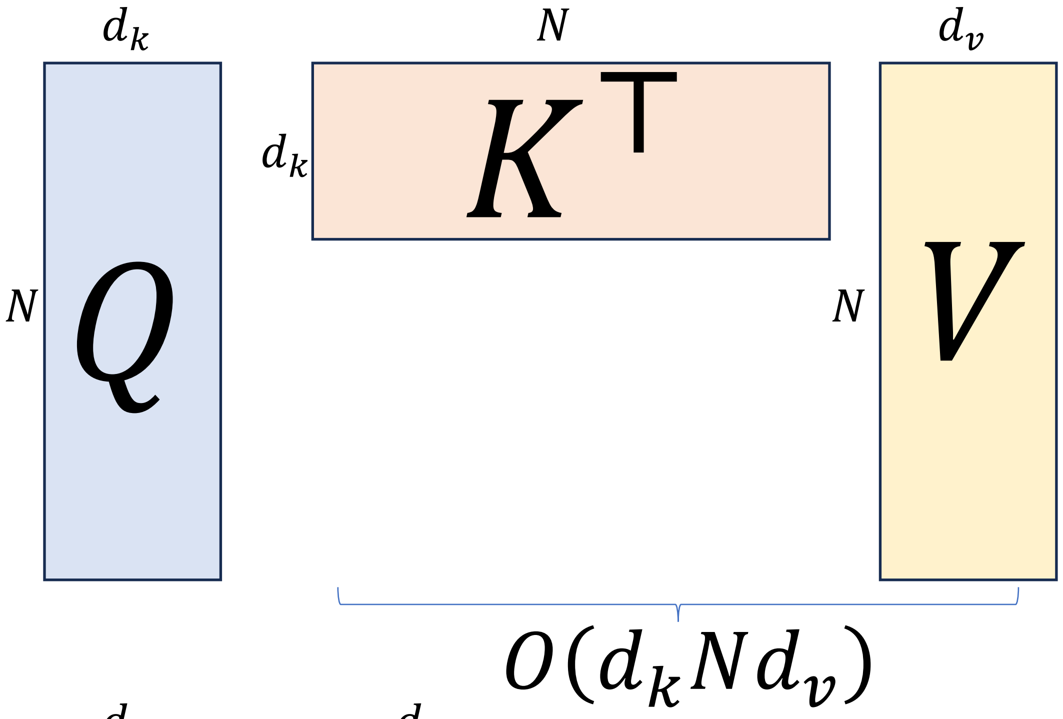
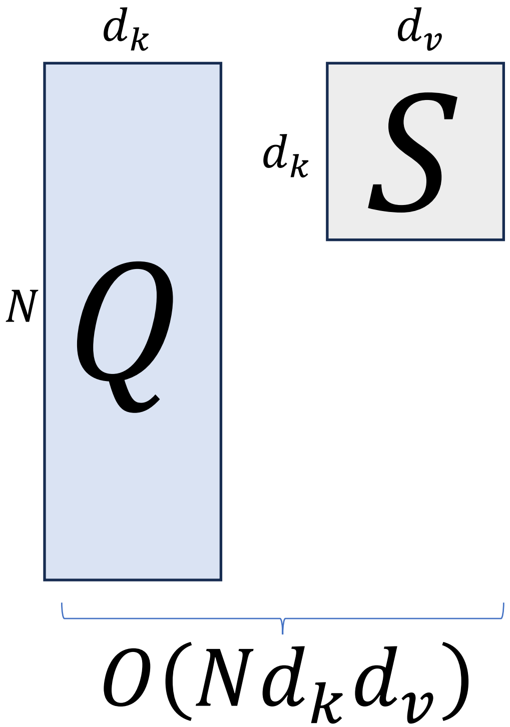


New

$$O(N d_k d_v)$$



Source: Jia-Bin Huang from University of Maryland College Park

Letting $N = 2048$

$d_k = d_v = 64$

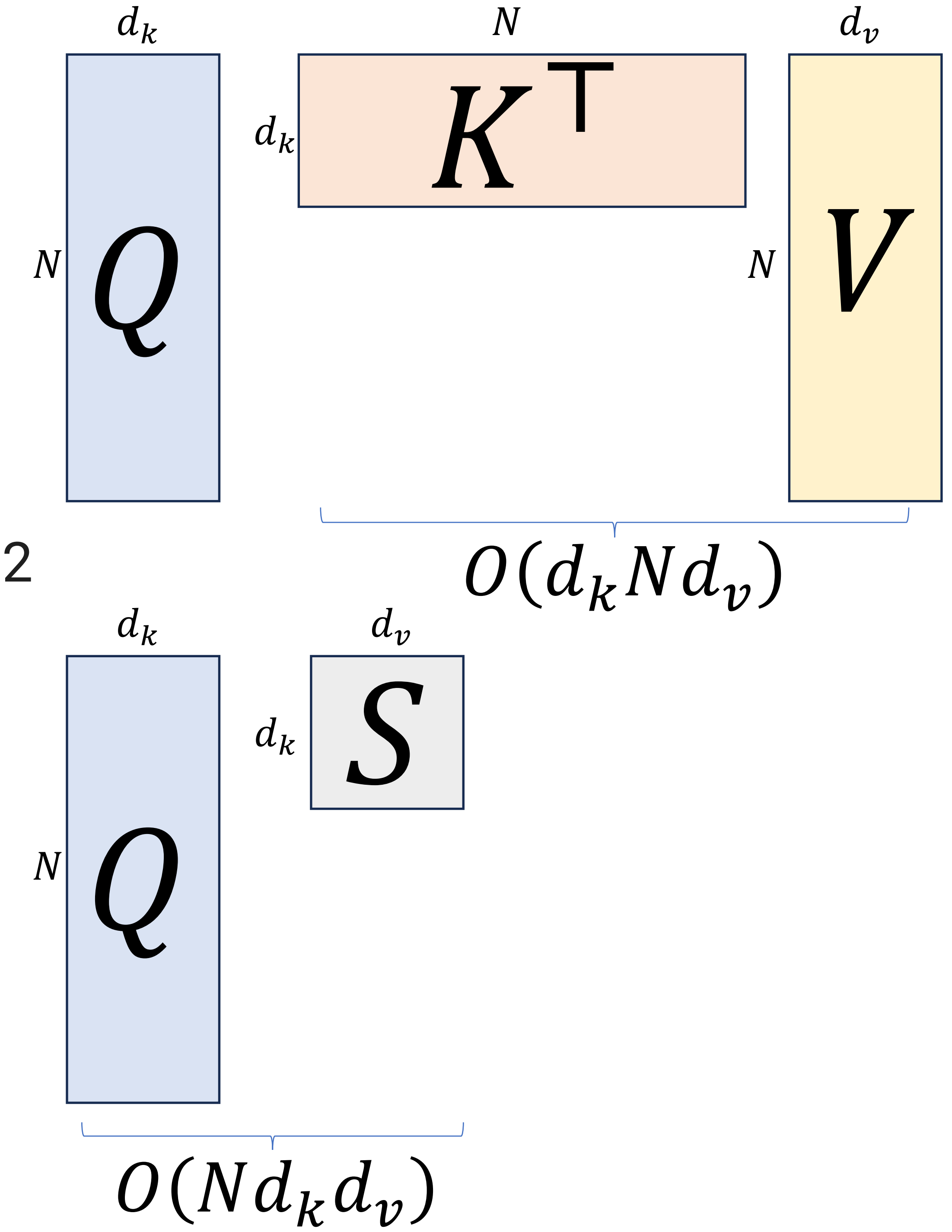
Original

$O(N^2(d_v + d_k))$ = 536,870,912

1.56%!

New

$O(N d_k d_v)$ = 8,388,608



Source: Jia-Bin Huang from University of Maryland College Park

Obstacle

- Softmax prevents this reordered GEMM

$$\text{Softmax}\left(\frac{1}{\sqrt{d_k}}\; {}_{N}\overset{d_k}{Q}\; {}_{d_k}\overset{N}{K^{\top}}\right) {}_{N}\overset{d_v}{V}$$

Source: Jia-Bin Huang from University of Maryland College Park

Linear Attention

- Computing Softmax (Katharopoulos et al., 2020)

$$o_1 = \sum_{i=1}^{N} \alpha_{1,i} v_i = \sum_{i=1}^{N} \frac{\exp(q_1 \cdot k_i)}{\sum_{j=1}^{N} \exp(q_1 \cdot k_j)} v_i$$

- Substituting *exp* by

$$\exp(q \cdot k) \approx \phi(q) \cdot \phi(q)$$

- there exists

Kernel function

$$o_1 \approx \sum_{i=1}^{N} \frac{\phi(q_1) \cdot \phi(k_i)}{\sum_{j=1}^{N} \phi(q_1) \cdot \phi(k_j)} v_i$$

Source: Jia-Bin Huang from University of Maryland College Park

Linear Attention

- Overall Output: complexity order $O(N)$

$$O \approx \phi(Q) \cdot (\phi(K)\, V), \quad O, Q, K, V \in \mathbb{R}^{N \times d}$$

- An example: denominator of o_1

$$\phi(q_1) = \begin{bmatrix} q_1^1 \\ q_1^2 \\ q_1^3 \\ \vdots \\ q_1^d \end{bmatrix} \quad \phi(k_i) = \begin{bmatrix} k_i^1 \\ k_i^2 \\ k_i^3 \\ \vdots \\ k_i^d \end{bmatrix},$$

$$[\; q_1^1 \;\; q_1^2 \;\; q_1^3 \;\; \cdots \;\; q_1^d \;] \begin{bmatrix} \sum_{i=1}^{N} k_i^1 v_i \\ \sum_{i=1}^{N} k_i^2 v_i \\ \sum_{i=1}^{N} k_i^3 v_i \\ \vdots \\ \sum_{i=1}^{N} k_i^d v_i \end{bmatrix}$$

Source: Jia-Bin Huang from University of Maryland College Park

Linear Attention

- Iterative computation to further reduce complexity

$$o_1 \approx \sum_{i=1}^{N} \frac{\phi(q_1) \cdot \phi(k_i)}{\sum_{j=1}^{N} \phi(q_1) \cdot \phi(k_j)} v_i$$

  - where

$$\sum_{i=1}^{N} [\phi(q_1) \cdot \phi(k_i)] v_i = \phi(q_1)^\top \left(\sum_{i=1}^{N} k_i v_i^\top \right) \Rightarrow$$

$$S_t = \sum_{i=1}^{t} k_i v_i^\top \in R^{d \times d}$$

$$S_t = S_{t-1} + k_t v_t^\top$$

$$o_t = \phi(q_t)^\top S_t$$

Source: Jia-Bin Huang from University of Maryland College Park

Linear Attention

- Today's Linear Attention
  - Baby linear attention: Linear Transformer, SANA, CHELA, LightningAttention, etc.
    - **Efficient** in computation, I/O, storage
    - Performance loss compared with ***Softmax***
  - More advanced linear attention:
    - Delta rule
    - Gamma forget
    - Gated attention
  - Qwen3-Next and Kimi-linear, and many others

Inference Optimization: Outline

- Overview
- Attention Computation Optimization
  - Sparse Attention
  - Linear Attention
  - **Flash Attention**
  - Continuous Batching
- KV Cache Optimization
- Speculative Decoding
- Distributed Serving

Flash Attention

- Prior works
  - Reducing # of scaled dot-product, potentially sacrificing attention performance
  - Compute is fast while reading K and V, and writing S back the results are slow
- Challenges
  - I/O cost of loading/storing Q, K and V are non-trival

Algorithm 0 Standard Attention Implementation

Require: Matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times d}$ in HBM.

1: Load $\mathbf{Q}, \mathbf{K}$ by blocks from HBM, compute $\mathbf{S} = \mathbf{Q}\mathbf{K}^\top$, write $\mathbf{S}$ to HBM.
2: Read $\mathbf{S}$ from HBM, compute $\mathbf{P} = \mathrm{softmax}(\mathbf{S})$, write $\mathbf{P}$ to HBM.
3: Load $\mathbf{P}$ and $\mathbf{V}$ by blocks from HBM, compute $\mathbf{O} = \mathbf{P}\mathbf{V}$, write $\mathbf{O}$ to HBM.
4: Return $\mathbf{O}$.

Flash Attention

- From Softmax to Safe Softmax

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^{n} e^{x_j}} \Rightarrow \text{softmax}(x_i) = \frac{e^{x_i - max(x)}}{\sum_{j=1}^{n} e^{x_j - max(x)}}$$

- Online Safe Softmax: not enough space to store entire x
  - Alg1: three-pass algorithm

m_i: $\max_{j=1}^{i}\{x_j\}$ with initial value $m_0 = -\infty$

d_i: $\sum_{j=1}^{i} e^{x_j - \boxed{m_N}}$ is the denominator of Softmax

a_i: is the final Softmax output

for $i \leftarrow 1, N$ **do**

$\quad m_i \leftarrow \max(m_{i-1}, x_i)$

end

for $i \leftarrow 1, N$ **do**

$\quad d_i \leftarrow d_{i-1} + e^{x_i - m_N}$

end

for $i \leftarrow 1, N$ **do**

$\quad a_i \leftarrow \frac{e^{x_i - m_N}}{d_N}$

end

Flash Attention

- Online Safe Softmax
  - d_i can be computed iteratively

$$\boxed{d_i'} = \sum_{j=1}^{\boxed{i}} e^{x_j - \boxed{m_i}}$$

$$= \left(\sum_{j=1}^{i-1} e^{x_j - m_i}\right) + e^{x_i - m_i}$$

$$= \left(\sum_{j=1}^{i-1} e^{x_j - m_{i-1}}\right) e^{m_{i-1} - m_i} + e^{x_i - m_i}$$

$$= d_{i-1}' e^{m_{i-1} - m_i} + e^{x_i - m_i}$$

Algorithm **2-pass online softmax**

for $i \leftarrow 1, N$ **do**

$$m_i \leftarrow \max(m_{i-1}, x_i)$$

$$\boxed{d_i' \leftarrow d_{i-1}' e^{m_{i-1} - m_i} + e^{x_i - m_i}}$$

end

for $i \leftarrow 1, N$ **do**

$$a_i \leftarrow \frac{e^{x_i - m_N}}{d_N'}$$

end

Flash Attention

- Online Safe Softmax
  - One-pass safe softmax algorithm does not exist
  - Safe softmax is only our intermediate result:

$O = \text{Softmax}(QK^T)V$

Notations

$Q[k,:]$: the k-th row vector of Q matrix.
$K^T[:,i]$: the i-th column vector of K^T matrix.
$O[k,:]$: the k-th row of output O matrix.
$V[i,:]$: the i-th row of V matrix.
$\{\boldsymbol{o}_i\}$: $\sum_{j=1}^{i} a_j V[j,:]$, a row vector storing partial aggregation result $A[k,:i] \times V[:i,:]$

Body

for $i \leftarrow 1, N$ **do**

$$\begin{aligned} x_i &\leftarrow Q[k,:]\, K^T[:,i] \\ m_i &\leftarrow \max(m_{i-1}, x_i) \\ d'_i &\leftarrow d'_{i-1} e^{m_{i-1}-m_i} + e^{x_i-m_i} \end{aligned}$$

end

for $i \leftarrow 1, N$ **do**

$$\begin{aligned} a_i &\leftarrow \frac{e^{x_i-m_N}}{d'_N} \\ \boldsymbol{o}_i &\leftarrow \boldsymbol{o}_{i-1} + a_i V[i,:] \end{aligned}$$

end

$$O[k,:] \leftarrow \boldsymbol{o}_N$$

Flash Attention

- One-pass Attention
  - Define an "intermediate" output

$$\boldsymbol{o}'_i := \left(\sum_{j=1}^{i} \frac{e^{x_j - m_i}}{d'_i} V[j,:]\right)$$

$$
\begin{aligned}
&for\,i \rightarrow 1, N \\
&\qquad x_i \leftarrow Q[k,:]K^T[:,i] \\
&\qquad m_i \leftarrow \max(m_{i-1}, x_i) \\
&\qquad d'_i \leftarrow d'_{i-1} e^{m_{i-1}-m_i} + e^{x_i - m_i} \\
&\qquad o'_i = o'_{i-1} \frac{d'_{i-1} e^{m_{i-1}-m_i}}{d'_i} + \frac{e^{x_i - m_i}}{d'_i} V[i,:] \\
&end \\
&O[k,:] \leftarrow o'_N
\end{aligned}
$$

Flash Attention

  - Compute it iteratively

$$
\begin{aligned}
\boldsymbol{o}'_i &= \sum_{j=1}^{i} \frac{e^{x_j - m_i}}{d'_i} V[j,:] \\
&= \left(\sum_{j=1}^{i-1} \frac{e^{x_j - m_i}}{d'_i} V[j,:]\right) + \boxed{\frac{e^{x_i - m_i}}{d'_i} V[i,:]} \\
&= \left(\sum_{j=1}^{i-1} \frac{e^{x_j - m_{i-1}}}{d'_{i-1}} \frac{e^{x_j - m_i}}{e^{x_j - m_{i-1}}} \frac{d'_{i-1}}{d'_i} V[j,:]\right) + \frac{e^{x_i - m_i}}{d'_i} V[i,:] \\
&= \left(\sum_{j=1}^{i-1} \frac{e^{x_j - m_{i-1}}}{d'_{i-1}} V[j,:]\right) \frac{d'_{i-1}}{d'_i} e^{m_{i-1} - m_i} + \frac{e^{x_i - m_i}}{d'_i} V[i,:] \\
&= \boldsymbol{o}'_{i-1} \frac{d'_{i-1} e^{m_{i-1}-m_i}}{d'_i} + \frac{e^{x_i - m_i}}{d'_i} V[i,:]
\end{aligned}
$$

Flash Attention

Tiling: loading and computing at the block granularity

- Pseudocode of Flash Attention

Algorithm 1 FlashAttention

Require: Matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times d}$ in HBM, on-chip SRAM of size M.

Set block sizes $B_c = \left\lceil \frac{M}{4d} \right\rceil, B_r = \min\left(\left\lceil \frac{M}{4d} \right\rceil, d\right)$.

Initialize $\mathbf{O} = (0)_{N \times d} \in \mathbb{R}^{N \times d}, \ell = (0)_N \in \mathbb{R}^N, m = (-\infty)_N \in \mathbb{R}^N$ in HBM.

Divide $\mathbf{Q}$ into $T_r = \left\lceil \frac{N}{B_r} \right\rceil$ blocks $\mathbf{Q}_1, \dots, \mathbf{Q}_{T_r}$ of size $B_r \times d$ each, and divide $\mathbf{K}, \mathbf{V}$ in to $T_c = \left\lceil \frac{N}{B_c} \right\rceil$ blocks $\mathbf{K}_1, \dots, \mathbf{K}_{T_c}$ and $\mathbf{V}_1, \dots, \mathbf{V}_{T_c}$, of size $B_c \times d$ each.

Divide $\mathbf{O}$ into T_r blocks $\mathbf{O}_i, \dots, \mathbf{O}_{T_r}$ of size $B_r \times d$ each, divide ℓ into T_r blocks $\ell_i, \dots, \ell_{T_r}$ of size B_r each, divide m into T_r blocks $m_1, \dots, m_{T_r}$ of size B_r each.

for $1 \le j \le T_c$ **do** — Outer Loop on Key/Value

  Load $\mathbf{K}_j, \mathbf{V}_j$ from HBM to on-chip SRAM.

  for $1 \le i \le T_r$ **do** — Inner Loop on Key/Value

    Load $\mathbf{Q}_i, \mathbf{O}_i, \ell_i, m_i$ from HBM to on-chip SRAM.

    On chip, compute $\mathbf{S}_{ij} = \mathbf{Q}_i \mathbf{K}_j^T \in \mathbb{R}^{B_r \times B_c}$.

    On chip, compute $\tilde{m}_{ij} = \mathrm{rowmax}(\mathbf{S}_{ij}) \in \mathbb{R}^{B_r}$, $\tilde{\mathbf{P}}_{ij} = \exp(\mathbf{S}_{ij} - \tilde{m}_{ij}) \in \mathbb{R}^{B_r \times B_c}$ (pointwise), $\tilde{\ell}_{ij} = \mathrm{rowsum}(\tilde{\mathbf{P}}_{ij}) \in \mathbb{R}^{B_r}$.

    On chip, compute $m_i^{\text{new}} = \max(m_i, \tilde{m}_{ij}) \in \mathbb{R}^{B_r}$, $\ell_i^{\text{new}} = e^{m_i - m_i^{\text{new}}} \ell_i + e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{\ell}_{ij} \in \mathbb{R}^{B_r}$.

    Write $\mathbf{O}_i \leftarrow \mathrm{diag}(\ell_i^{\text{new}})^{-1} (\mathrm{diag}(\ell_i) e^{m_i - m_i^{\text{new}}} \mathbf{O}_i + e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{\mathbf{P}}_{ij} \mathbf{V}_j)$ to HBM.

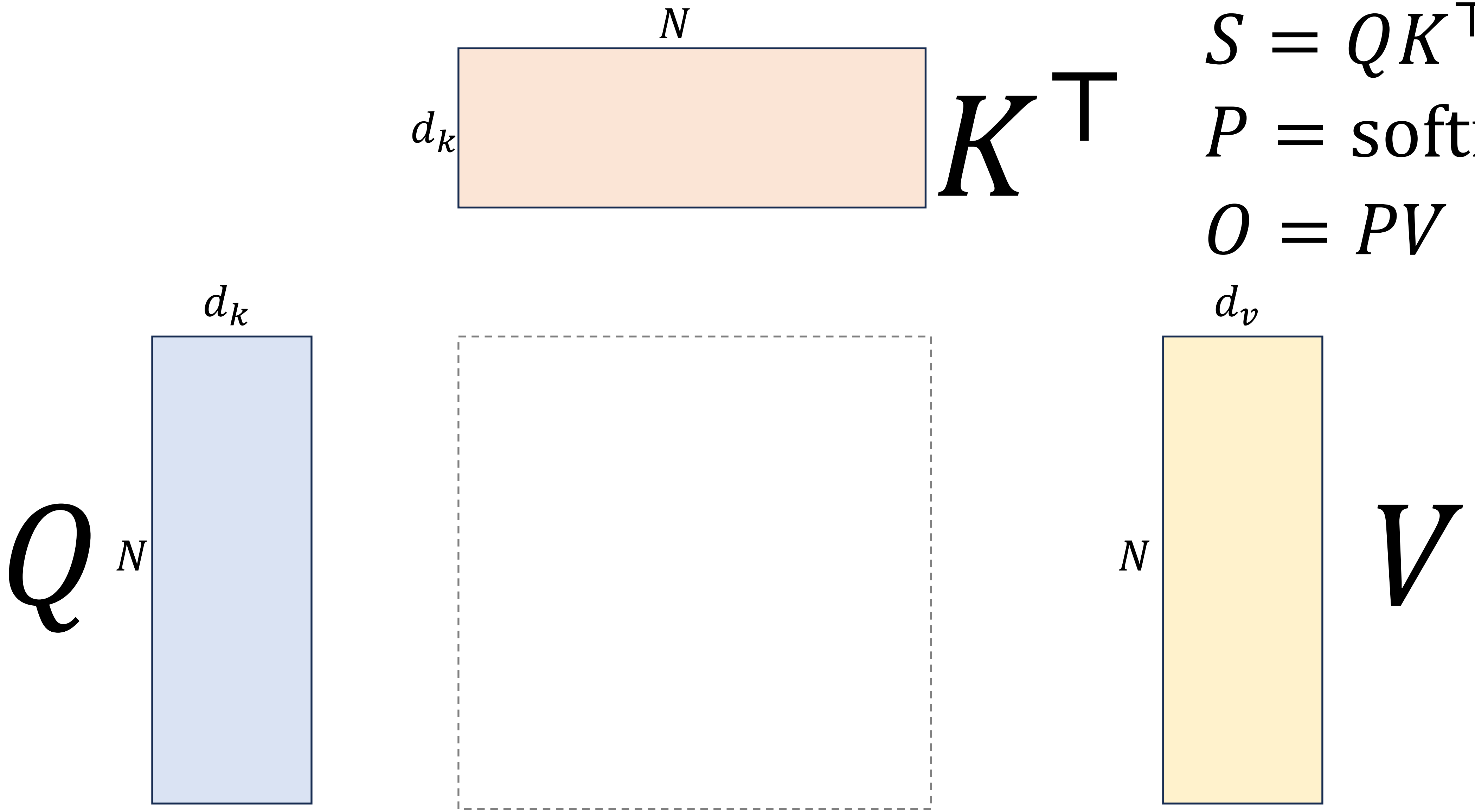
    Write $\ell_i \leftarrow \ell_i^{\text{new}}$, $m_i \leftarrow m_i^{\text{new}}$ to HBM.

  end for

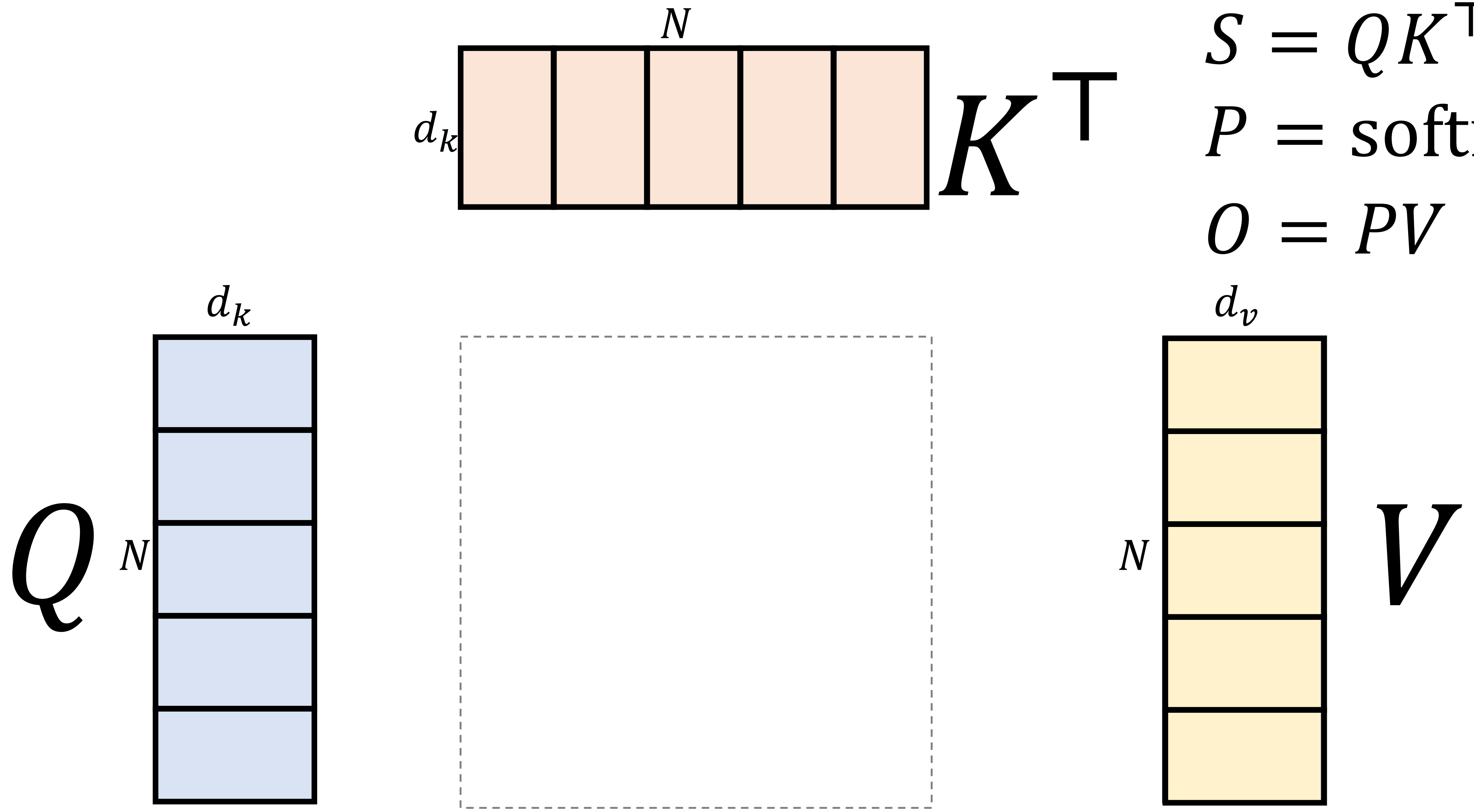
end for

Return $\mathbf{O}$.

$$S = QK^\top$$
$$P = \mathrm{softmax}(S)$$
$$O = PV$$



Source: Jia-Bin Huang from University of Maryland College Park



$$S = QK^\top$$
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Source: Jia-Bin Huang from University of Maryland College Park

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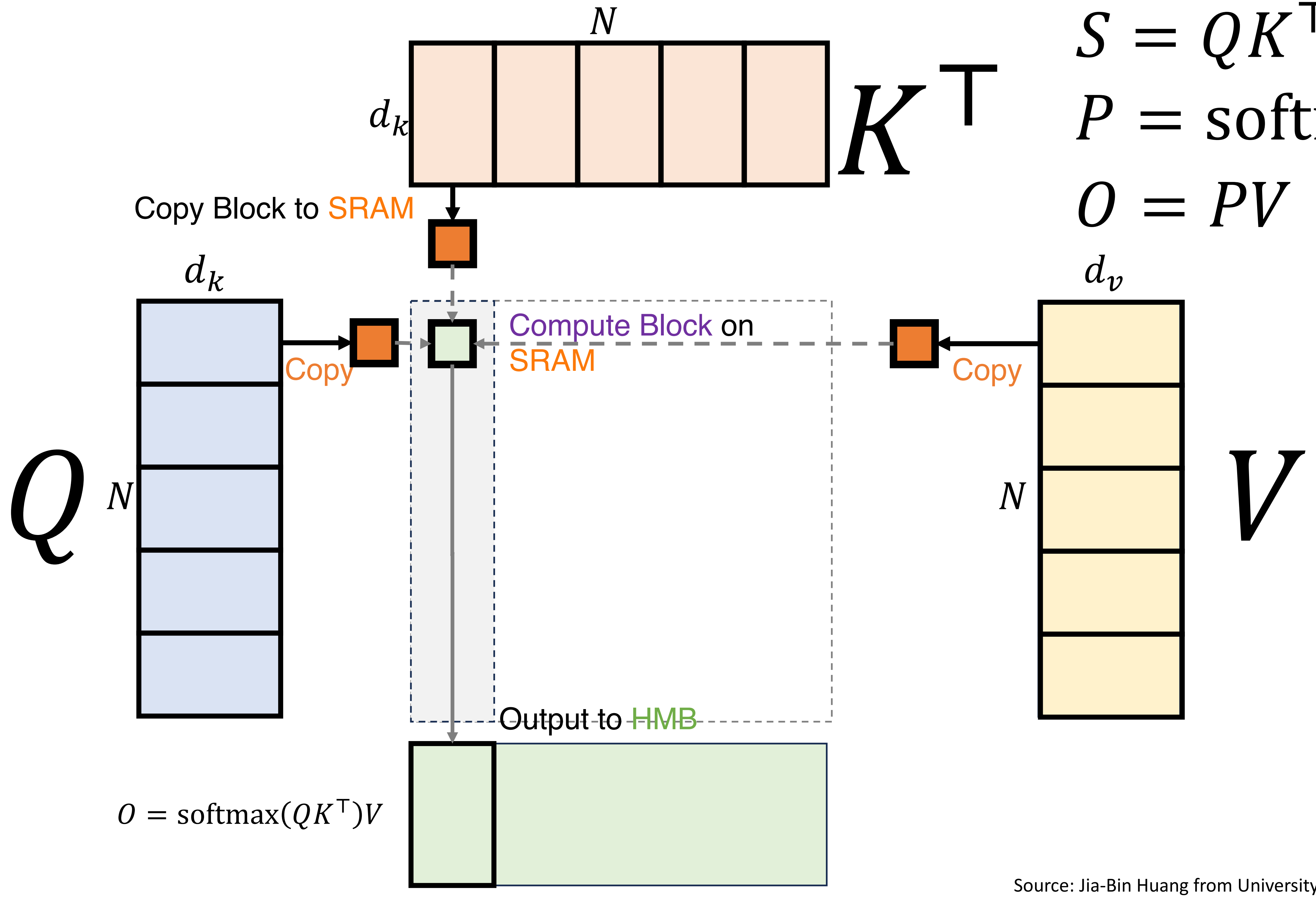
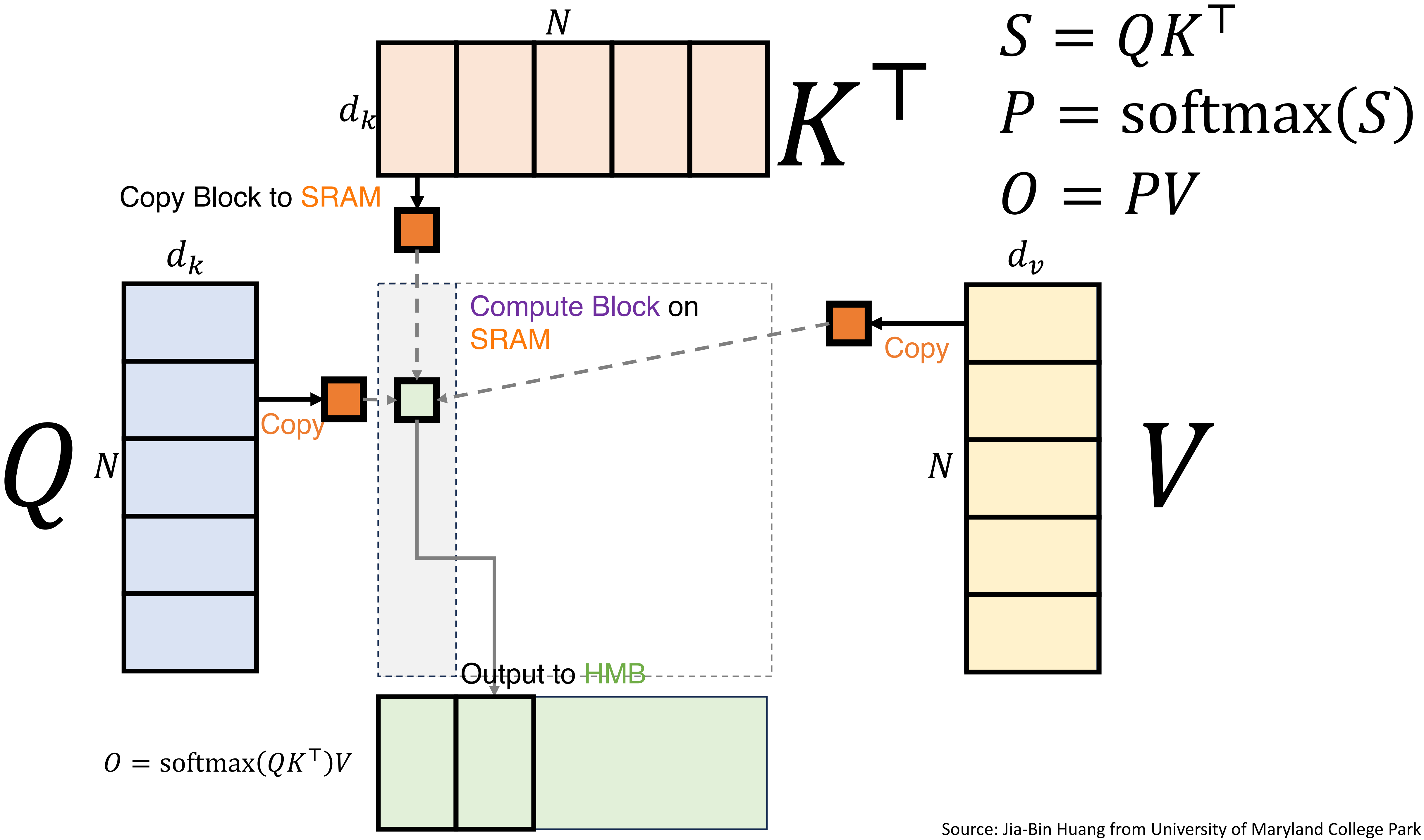
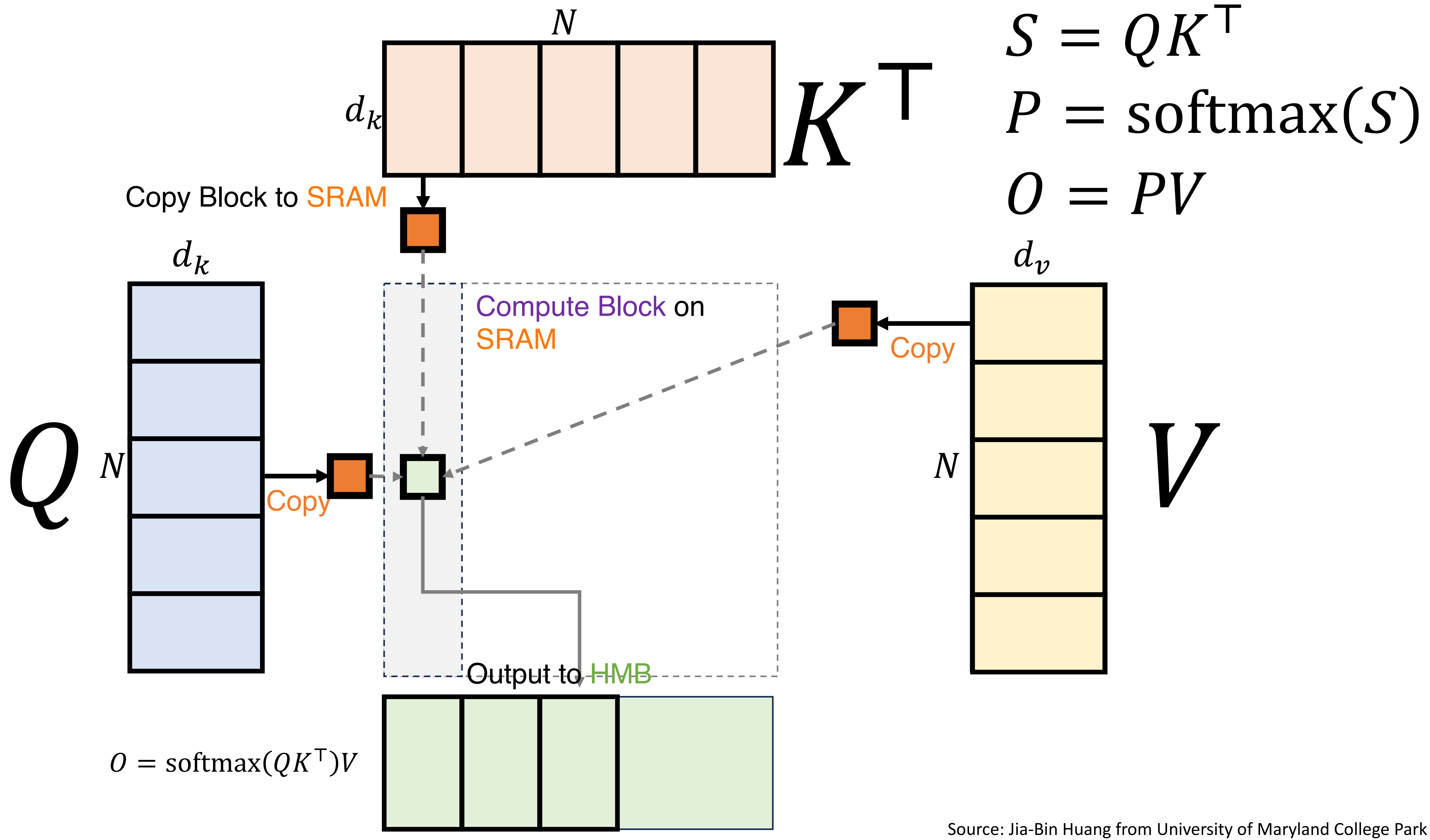


Source: Jia-Bin Huang from University of Maryland College Park



Source: Jia-Bin Huang from University of Maryland College Park



Source: Jia-Bin Huang from University of Maryland College Park

$$S = QK^{\top}$$
$$P = \mathrm{softmax}(S)$$
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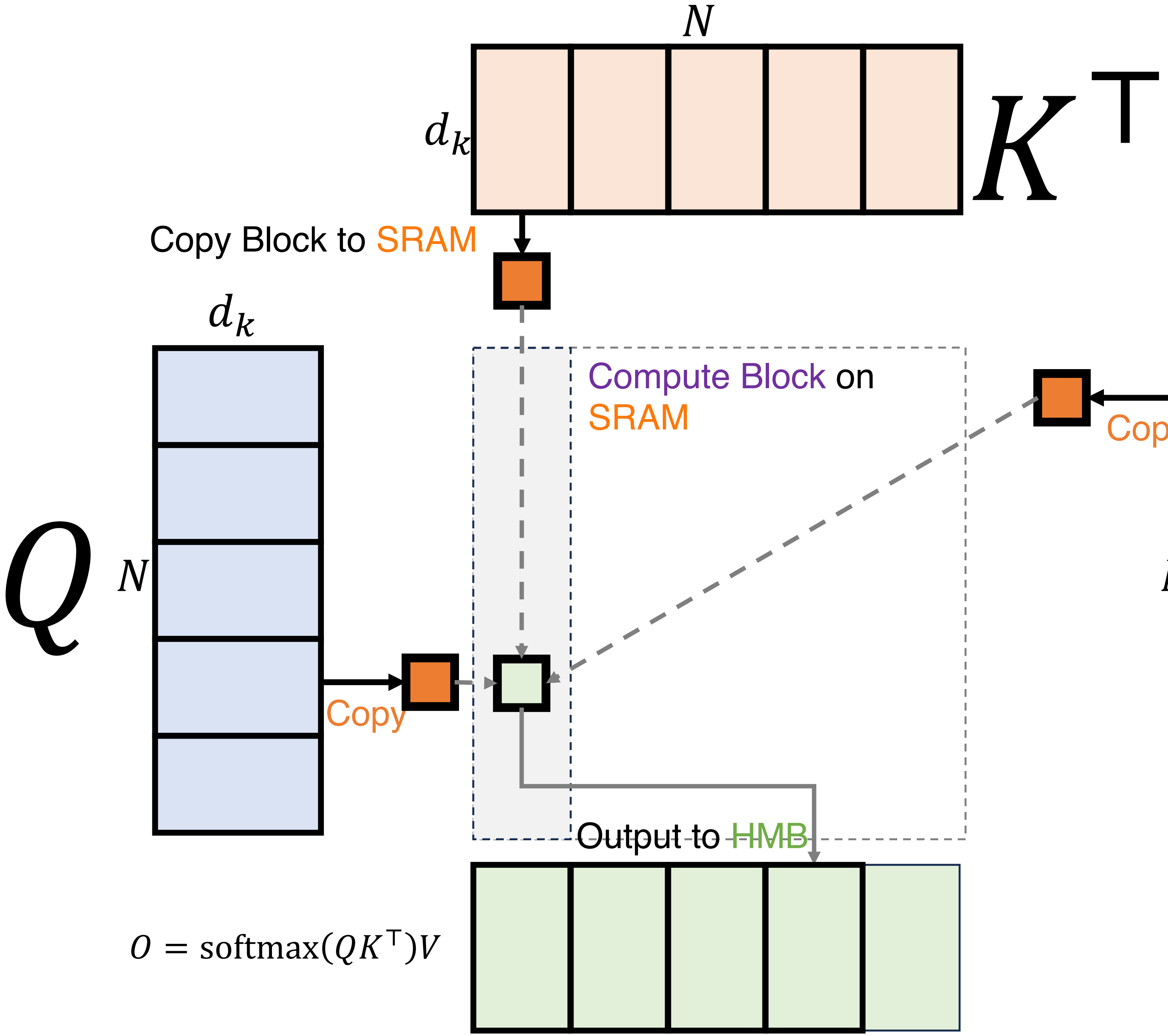

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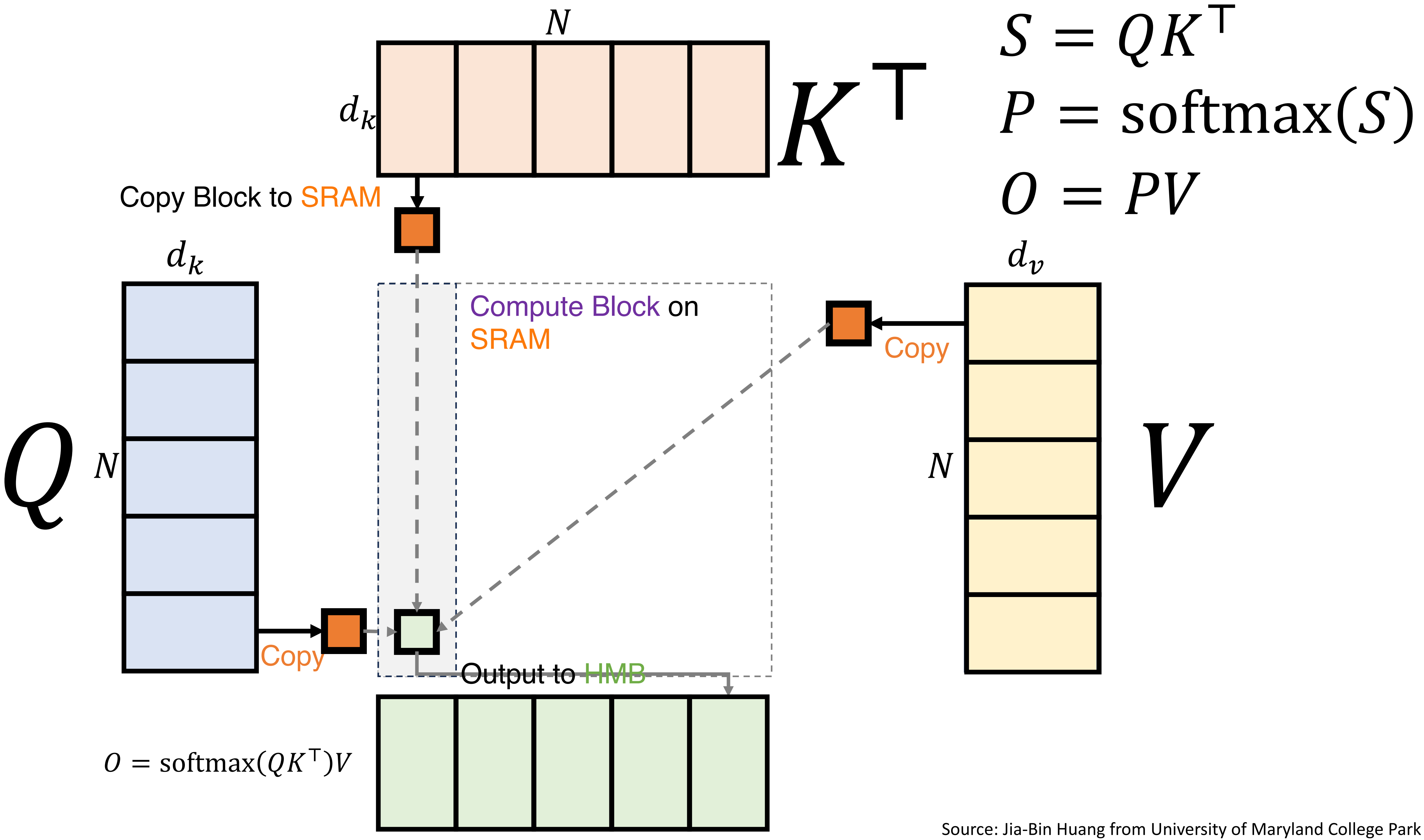
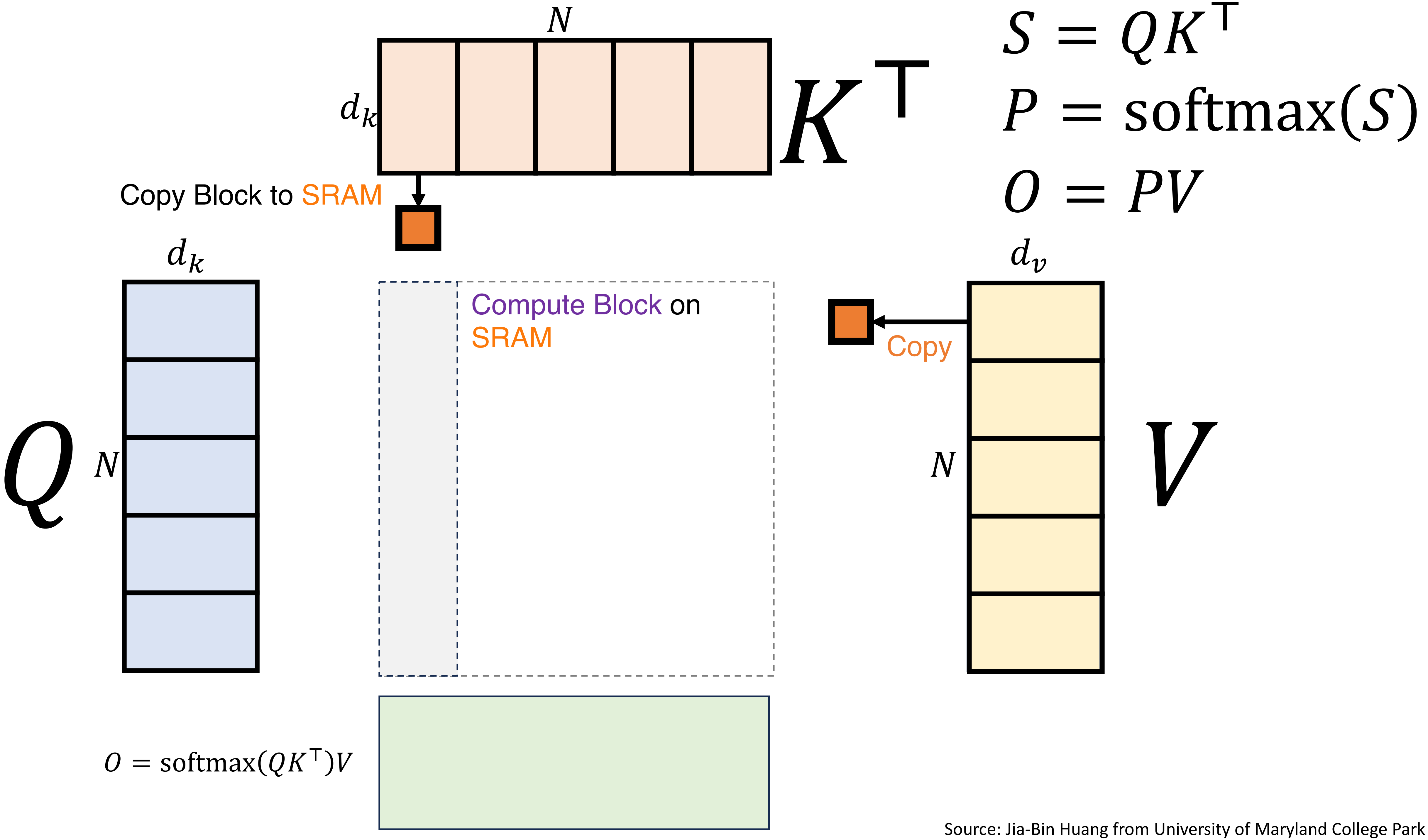


Source: Jia-Bin Huang from University of Maryland College Park



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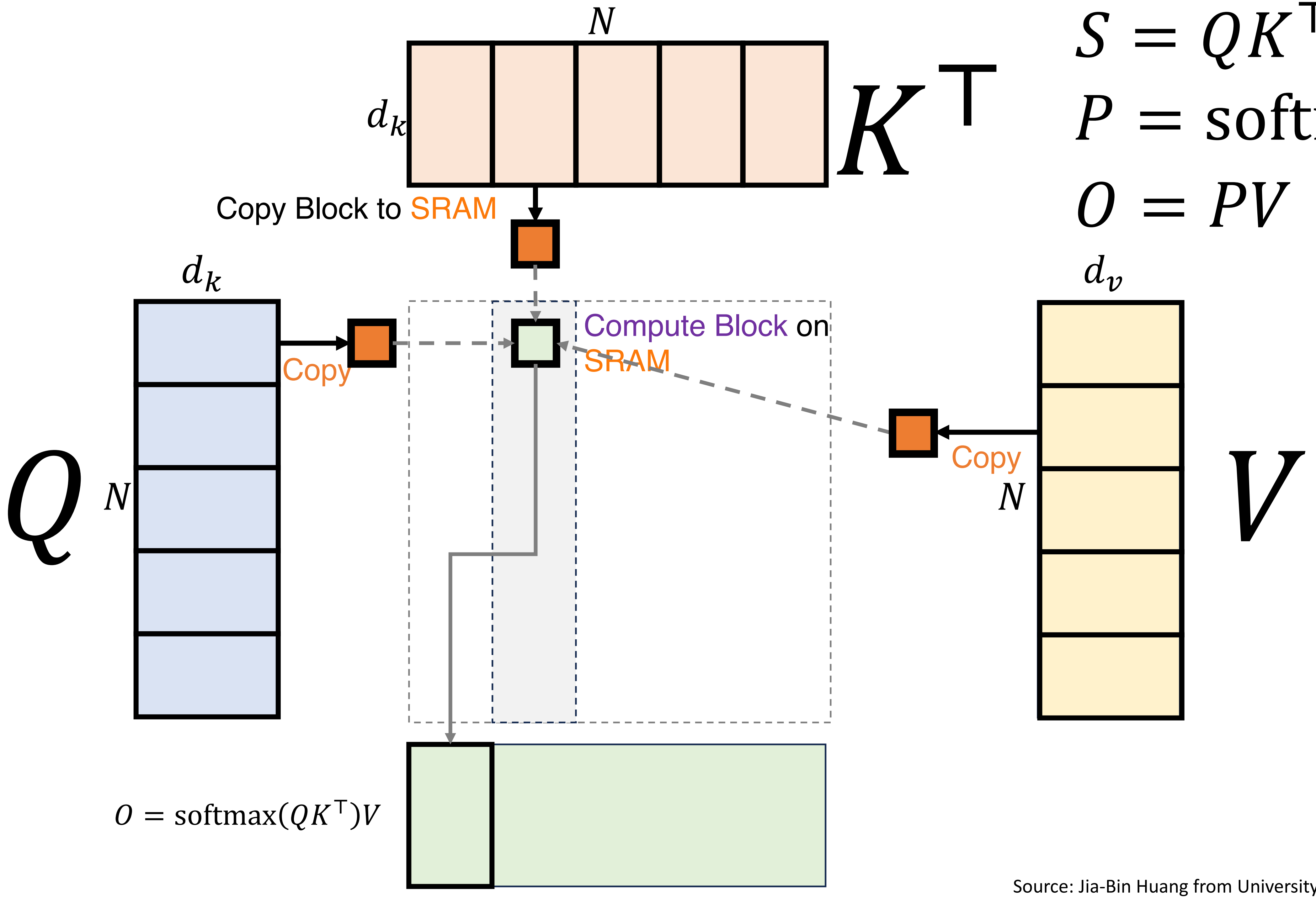

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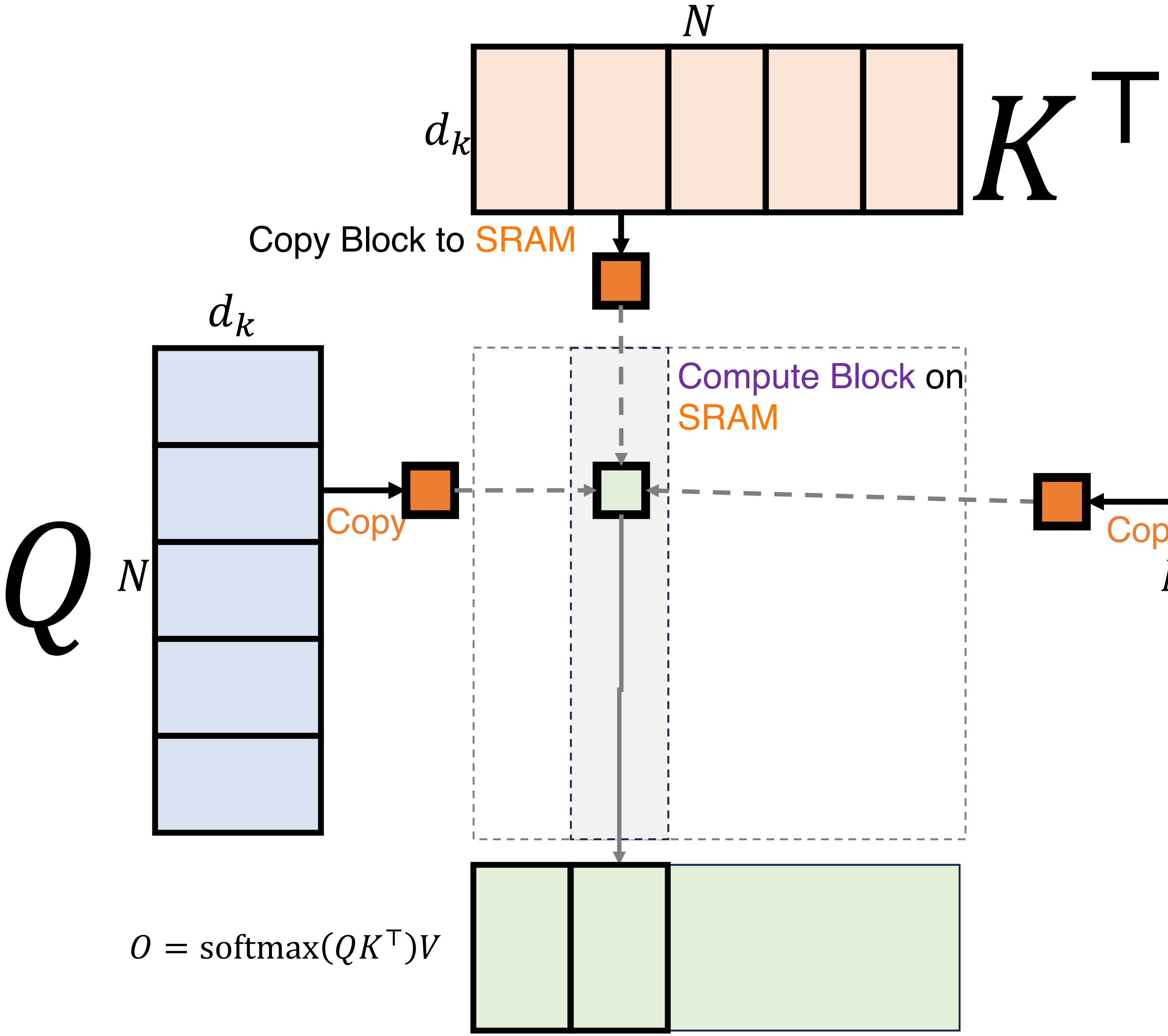

$$S = QK^\top$$
$$P = \text{softmax}(S)$$
$$O = PV$$

Source: Jia-Bin Huang from University of Maryland College Park

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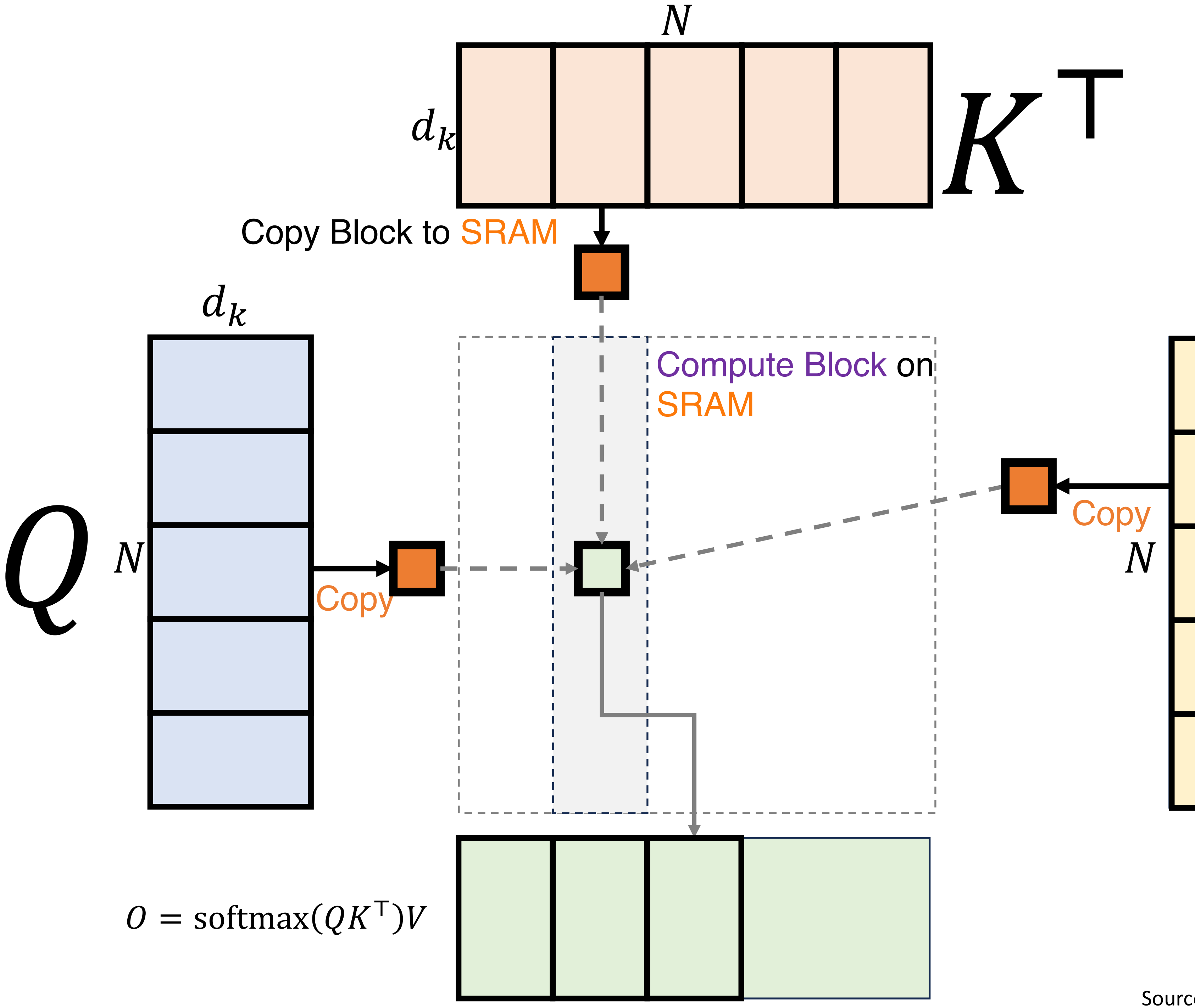

Source: Jia-Bin Huang from University of Maryland College Park

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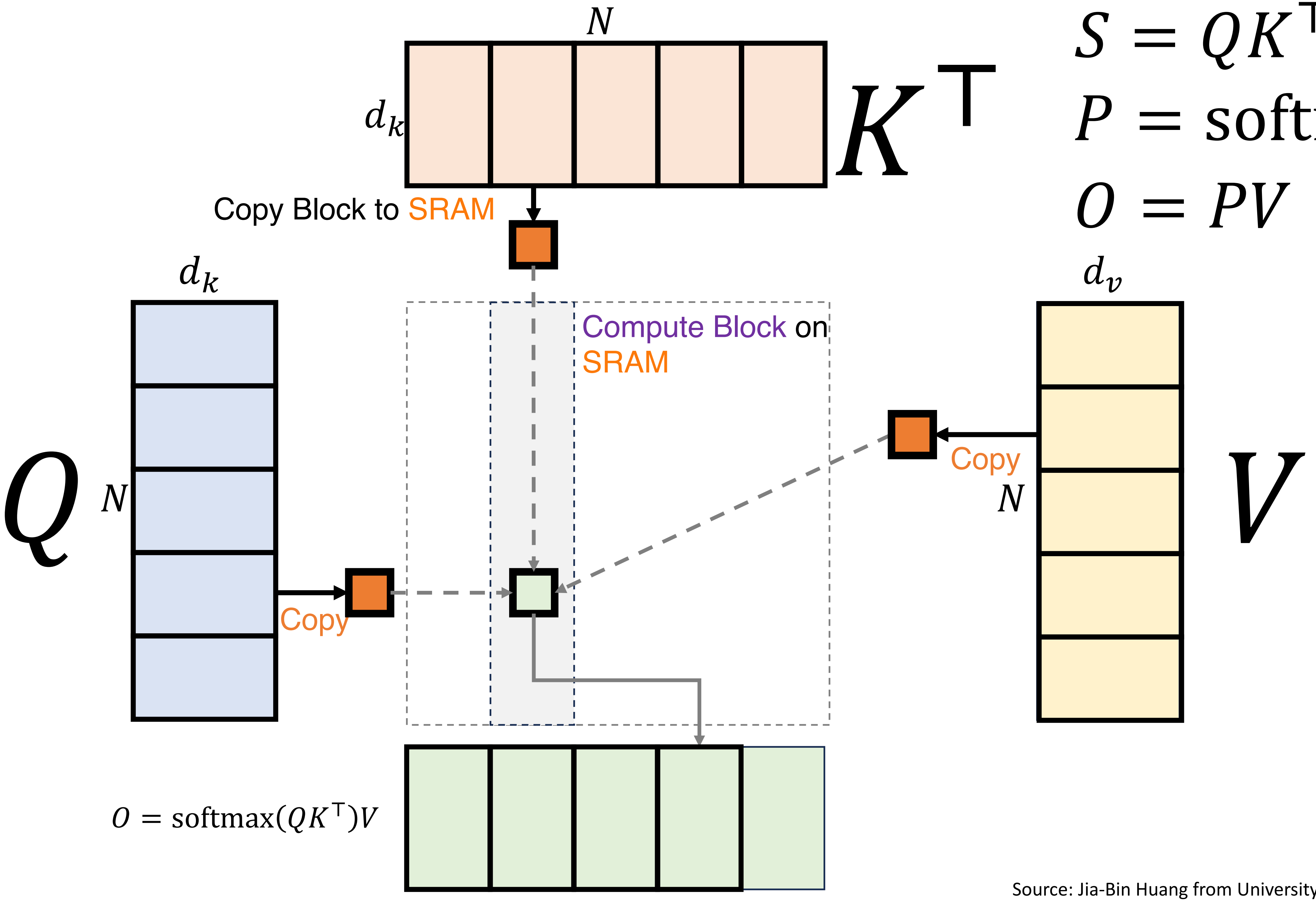
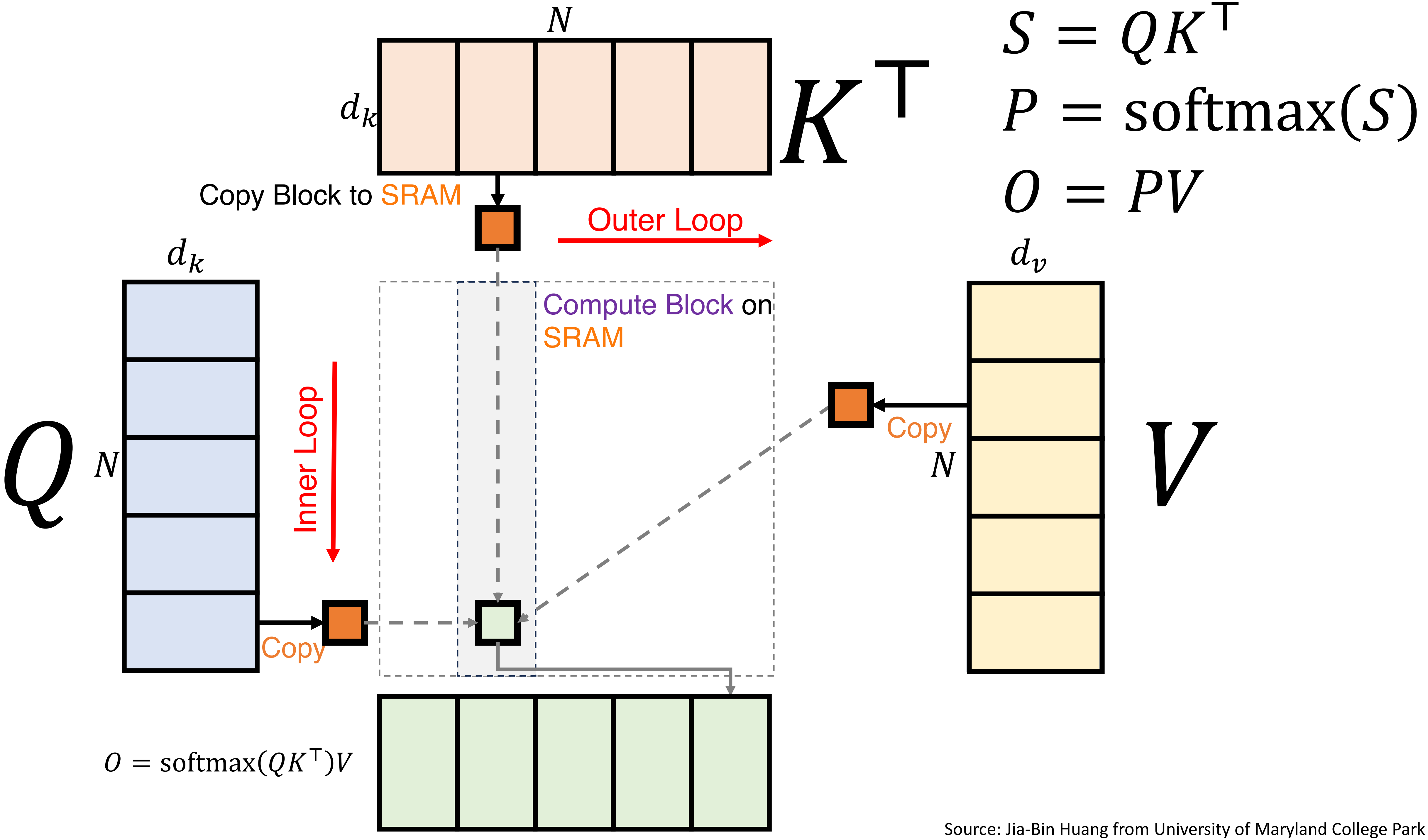


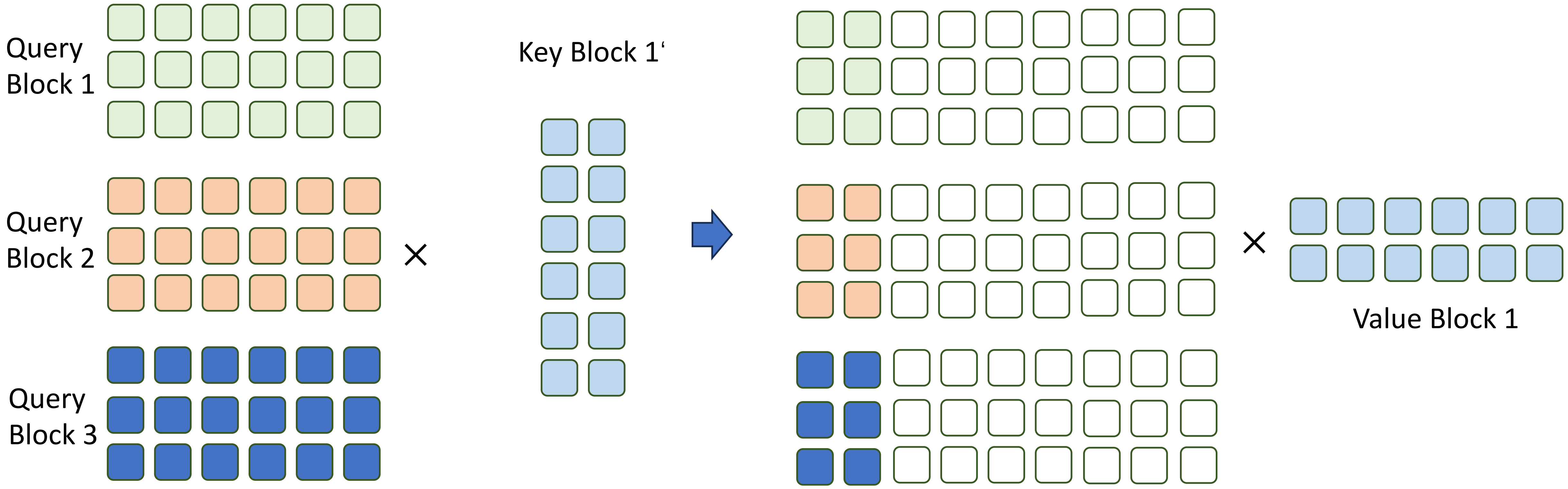
Source: Jia-Bin Huang from University of Maryland College Park



Source: Jia-Bin Huang from University of Maryland College Park

Flash Attention

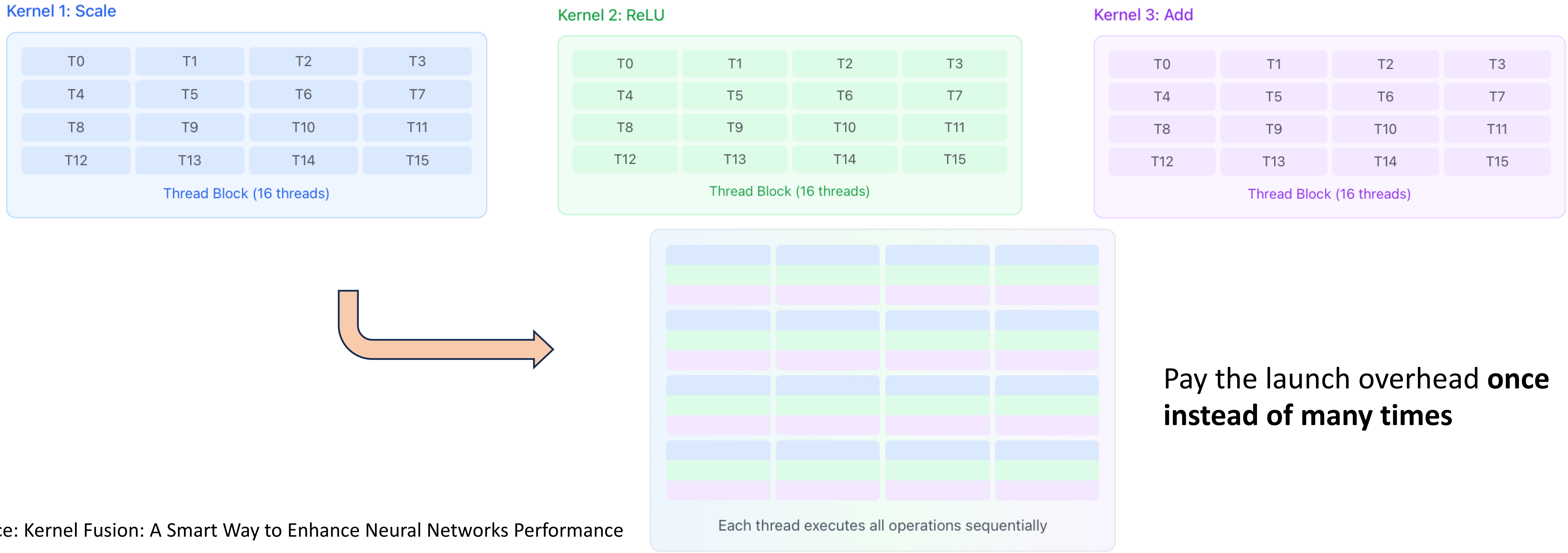
- Flash Attention (Tiling)



Source: Jia-Bin Huang from University of Maryland College Park

Flash Attention

- Kernel Fusion
  - Multiple separate operations or kernels (small, independent programs that run on hardware like GPUs) are combined or "fused" into a single, more efficient kernel.

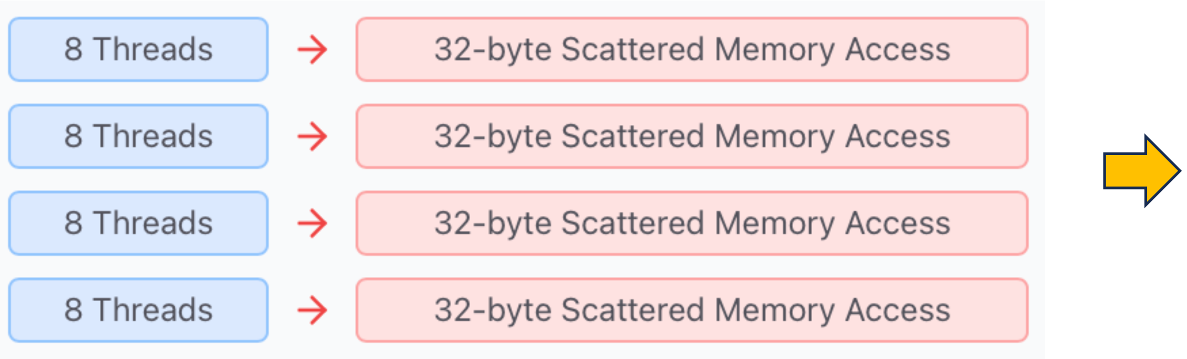


Pay the launch overhead **once instead of many times**

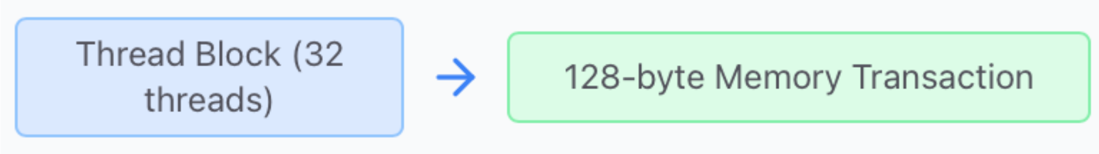
Source: Kernel Fusion: A Smart Way to Enhance Neural Networks Performance

Flash Attention

- Kernel Fusion
    - Minimizing data movement between slow global memory and faster on-chip memory (like SRAM or registers), avoids storing intermediate results



Multiple smaller memory transactions

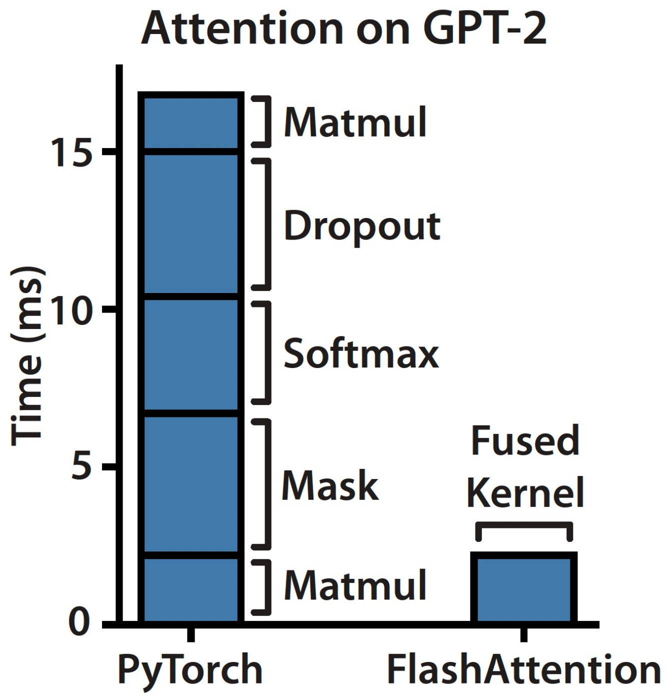


Single memory transaction for 32 consecutive elements

Source: Kernel Fusion: A Smart Way to Enhance Neural Networks Performance

Flash Attention

- Kernel Fusion
  - FlashAttention kernel fusion combining multiple (block-wise) steps of the attention computation—such as scaling, masking, softmax normalization, and matrix multiplications—into custom fused CUDA kernels



6 - 7 × improvement

Flash Attention

- Complexity Analysis
  - GPT2 Medium (seq. length 1024, head dim. 64, 16 heads, batch size 64) on A100

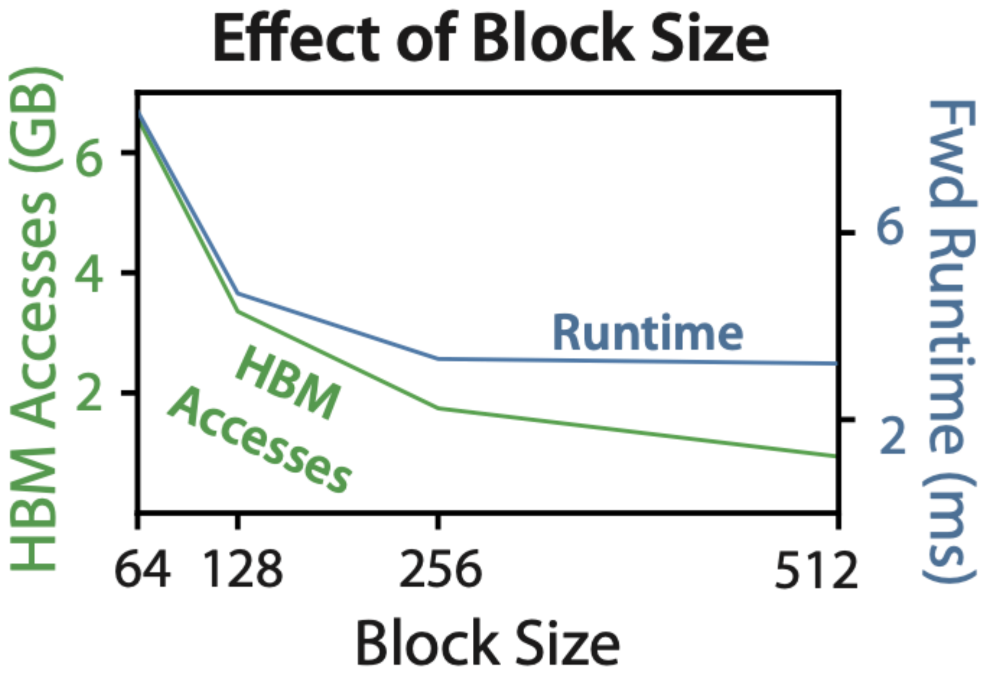
Attention	Standard	FlashAttention
GFLOPs	66.6	75.2
HBM R/W (GB)	40.3	4.4
Runtime (ms)	41.7	7.3

  - Computational complexity: Flash Attention is slightly higher
  - Memory complexity: Flash Attention is significantly better $O(Nd)$
  - I/O complexity
    - Standard Attention $O(N^2d)$ versus Flash Attention $O(N^2d^2M^{-1})$
      - Outer loop runs $T_c = N/B_c = 4d/M$ where M is SRAM size of a SM, and reads K and V once
      - Inner loop reads Q, O, and write O back to HBM
      - Total I/O complexity: $O(Nd)\times T_c = O(N^2d^2M^{-1})$

Flash Attention

- Flash Attention in practice
  - Impact of hyper parameters:

    Larger block size reduces the number of data loading, thus improving the forward runtime

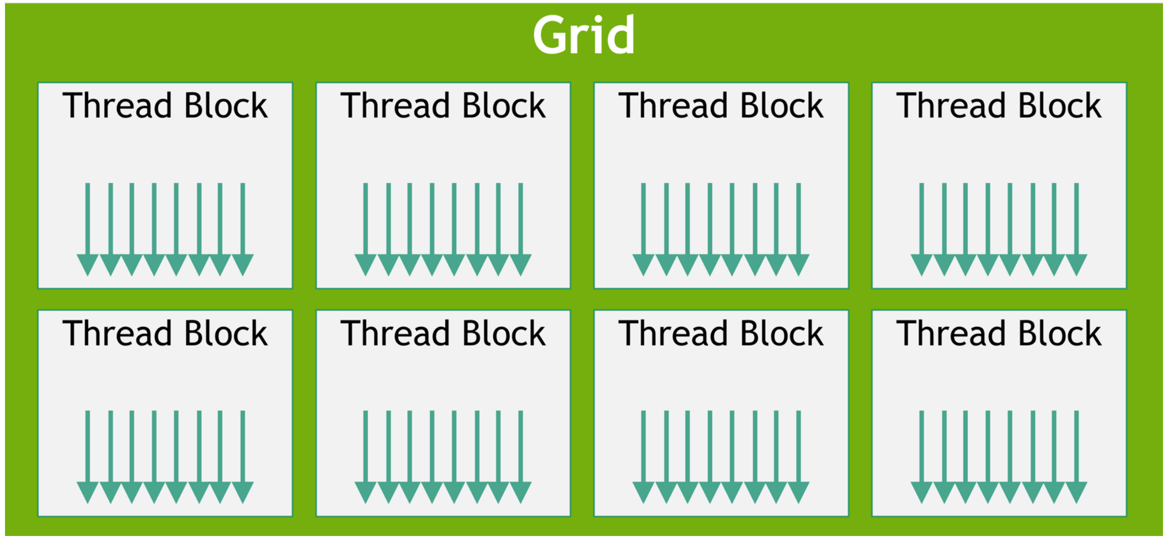


Forward runtime of FlashAttention

  - As hidden dimension becomes large, the block size shrinks
    - $O(N^2 d^2 M^{-1})$ is quadratic to the hidden dimension d, while M is constrained by the hardware

Flash Attention

- Flash Attention
  - Underutilized streaming multi-processors for **small batch size** but **long sequences**

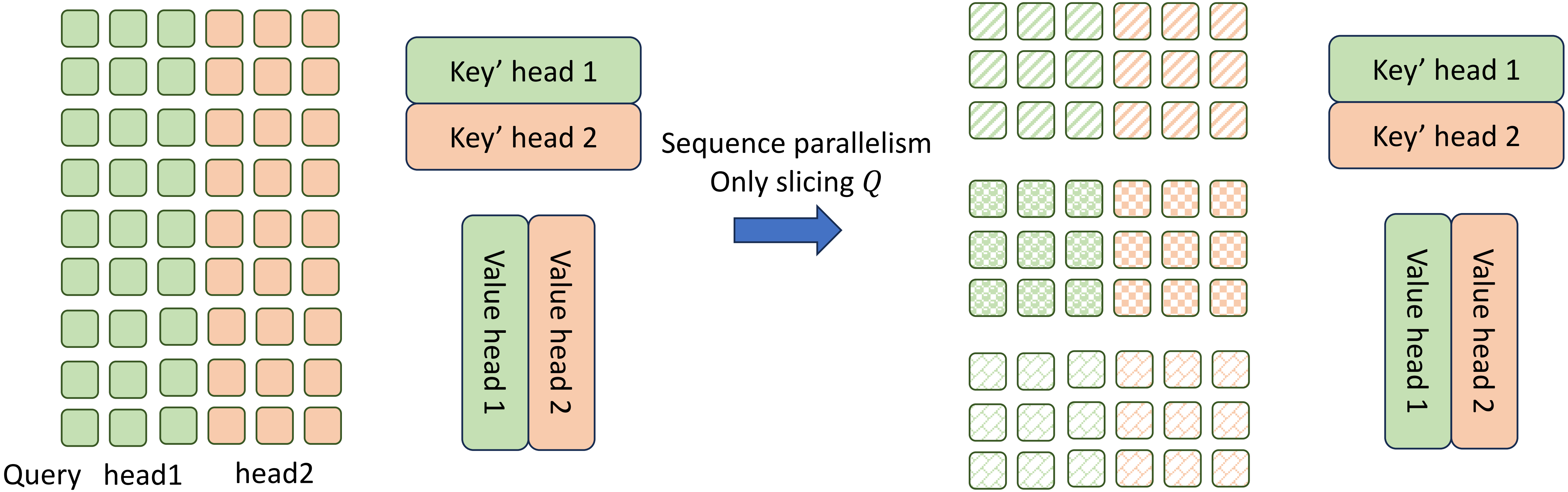


Grid of Thread Blocks

- GPU grid
  - Flash Attention enables Parallel computing on **batches and heads**
  - The grid consists of *batch_size* × *head_number* thread blocks
    - Batch_size = 2 and *head_number = 16*, meaning that only 16 out of 108 SMs are used on A100

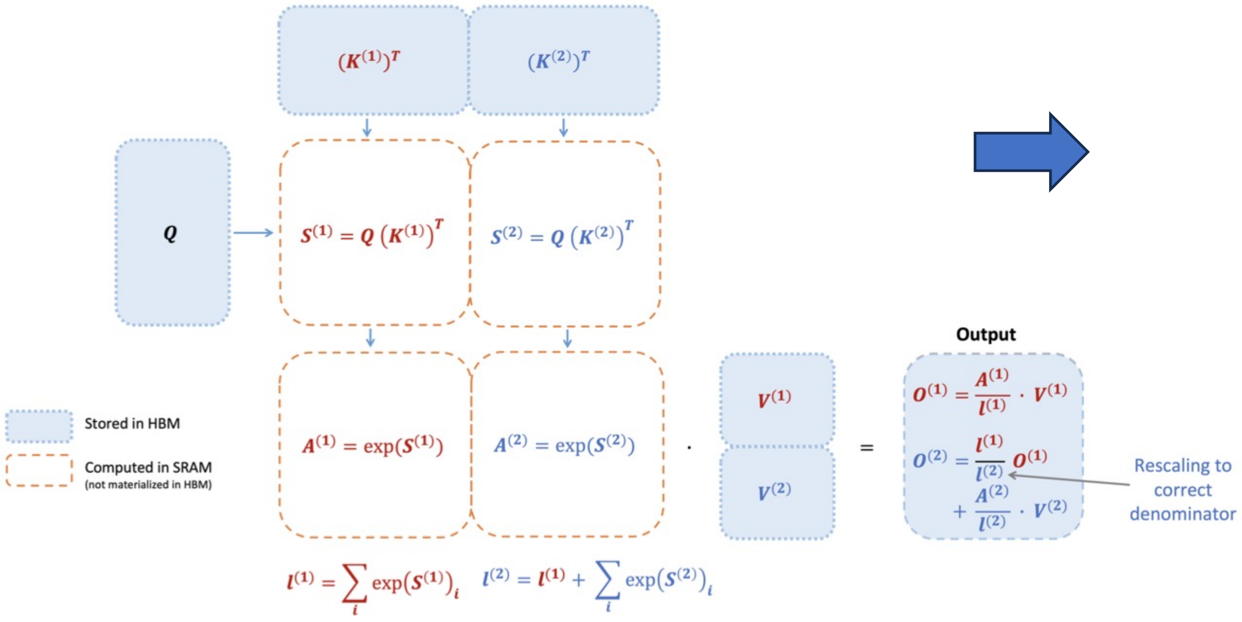
Flash Attention

- Flash Attention v2
  - Making FlashAttention sequence parallel
    - The shape of a grid: grid(num_m_block, batch_size, num_heads)



Flash Attention

- Flash Attention v2: Loop Reordering



$$m^{(1)} = \mathrm{rowmax}(\mathbf{S}^{(1)}) \in \mathbb{R}^{B_r}$$

$$\ell^{(1)} = \mathrm{rowsum}(e^{\mathbf{S}^{(1)}-m^{(1)}}) \in \mathbb{R}^{B_r}$$

$$\tilde{\mathbf{O}}^{(1)} = e^{\mathbf{S}^{(1)}-m^{(1)}}\mathbf{V}^{(1)} \in \mathbb{R}^{B_r \times d}$$

$$m^{(2)} = \max(m^{(1)}, \mathrm{rowmax}(\mathbf{S}^{(2)})) = m$$

$$\ell^{(2)} = e^{m^{(1)}-m^{(2)}}\ell^{(1)} + \mathrm{rowsum}(e^{\mathbf{S}^{(2)}-m^{(2)}}) = \mathrm{rowsum}(e^{\mathbf{S}^{(1)}-m}) + \mathrm{rowsum}(e^{\mathbf{S}^{(2)}-m}) = \ell$$

$$\tilde{\mathbf{P}}^{(2)} = \mathrm{diag}(\ell^{(2)})^{-1}e^{\mathbf{S}^{(2)}-m^{(2)}}$$

$$\tilde{\mathbf{O}}^{(2)} = \mathrm{diag}(e^{m^{(1)}-m^{(2)}})^{-1}\tilde{\mathbf{O}}^{(1)} + e^{\mathbf{S}^{(2)}-m^{(2)}}\mathbf{V}^{(2)} = e^{s^{(1)}-m}\mathbf{V}^{(1)} + e^{s^{(2)}-m}\mathbf{V}^{(2)}$$

$$\mathbf{O}^{(2)} = \mathrm{diag}(\ell^{(2)})^{-1}\tilde{\mathbf{O}}^{(2)} = \mathbf{O}.$$

FlashAttention forward pass: key K is partitioned into two blocks and value V is also partitioned into two blocks

Source: Flash Attention v2

Flash Attention

- FlashAttention v2
  - Algorithm: fewer *non-**matmul*** FLOPs
    - GPU is highly optimized on *matmul* (e.g. Tensor Cores)
      - NVIDIA A100 GPU: 312 TFLOPs/s of FP16/BF16 *matmul*, only 19.5 TFLOPs/s of *non-matmul* FP32

$$
\begin{aligned}
S_{ij} &\leftarrow Q_i K_j^T \\
m_{ij} &\leftarrow \mathrm{rowmax}(S_{ij}) \\
m_i^{new} &\leftarrow \max(m_i, m_{ij}) \\
P_{ij} &\leftarrow \exp(S_{ij} - m_i^{new}) \\
l_{ij} &\leftarrow \mathrm{rowsum}(P_{ij}) \\
l_i^{new} &\leftarrow \exp(m_i - m_i^{new}) l_i + l_{ij} \\
O_i^{new} &\leftarrow \boxed{(l_i^{new})^{-1}} \boxed{(l_i} \exp(m_i - m_i^{new}) O_i + P_{ij} V_j)
\end{aligned}
$$

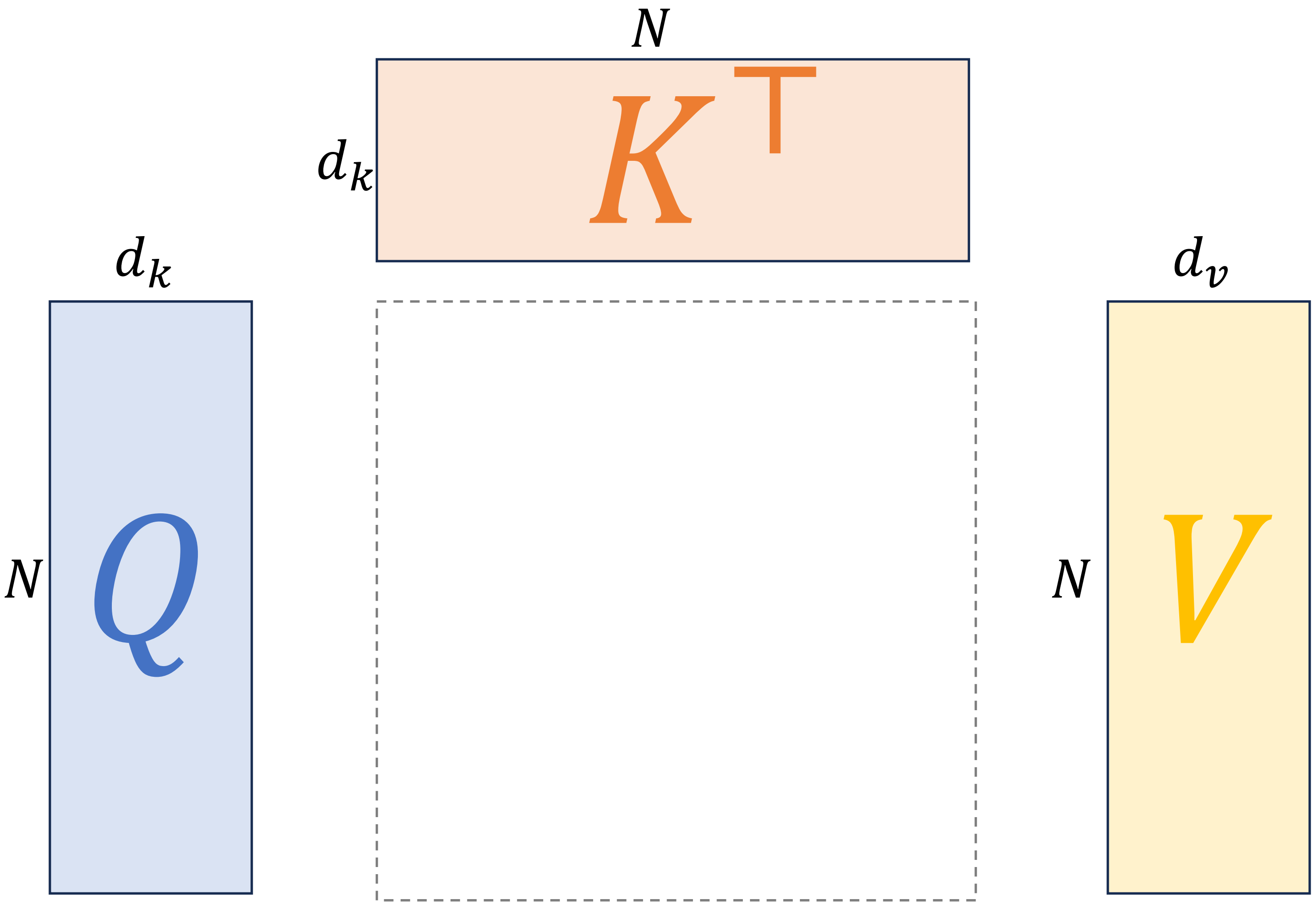
Scaling at next round

Scaling at this round in FlashAttention v1

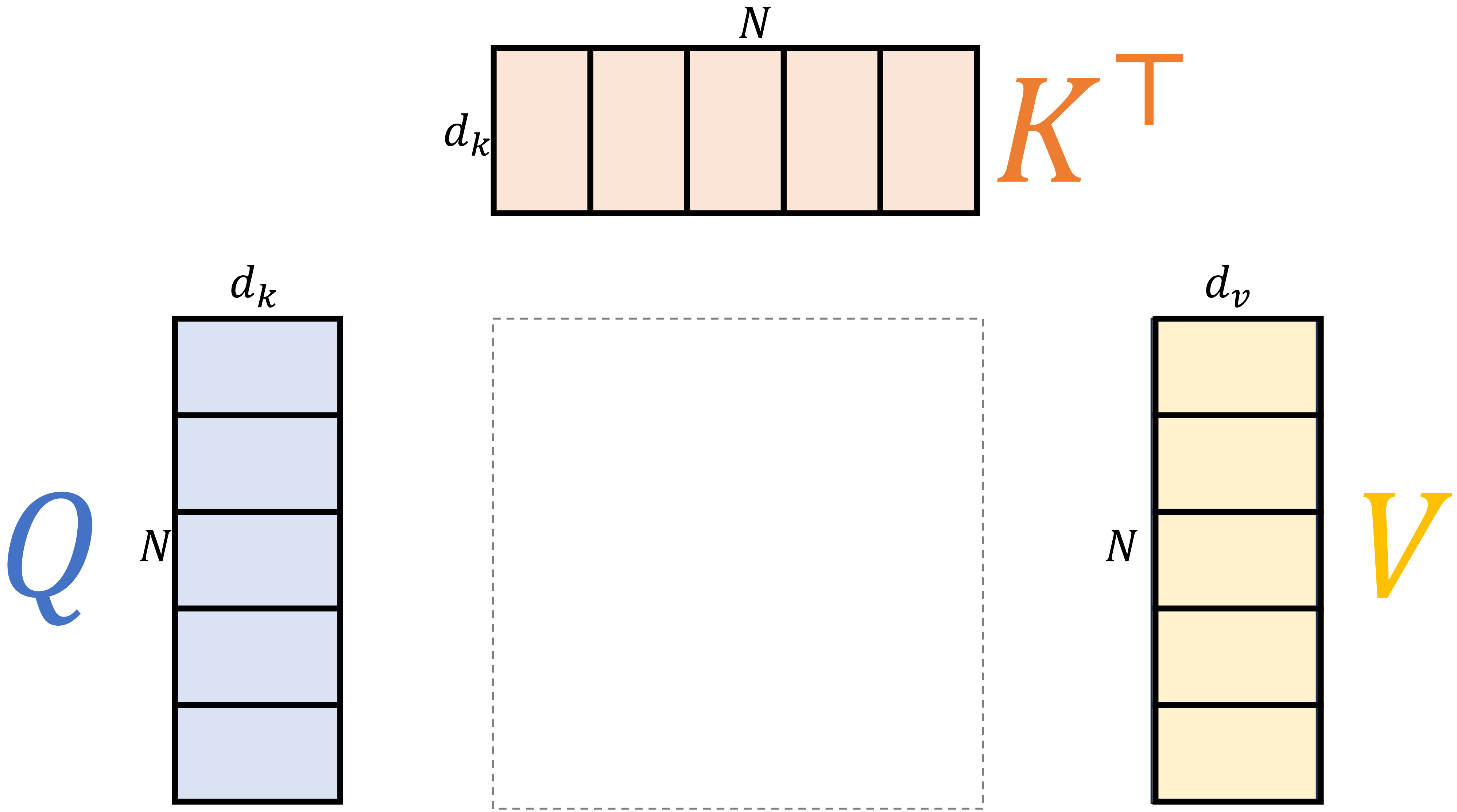
$$
\begin{aligned}
\tilde{O}_i^{new} &\leftarrow \exp(m_i - m_i^{new}) \tilde{O}_i + P_{ij} V_j \\
O_i^{last} &\leftarrow \boxed{(l_i^{last})^{-1}} \tilde{O}_i^{last}
\end{aligned}
$$

Only scaling the final result in FlashAttention v2

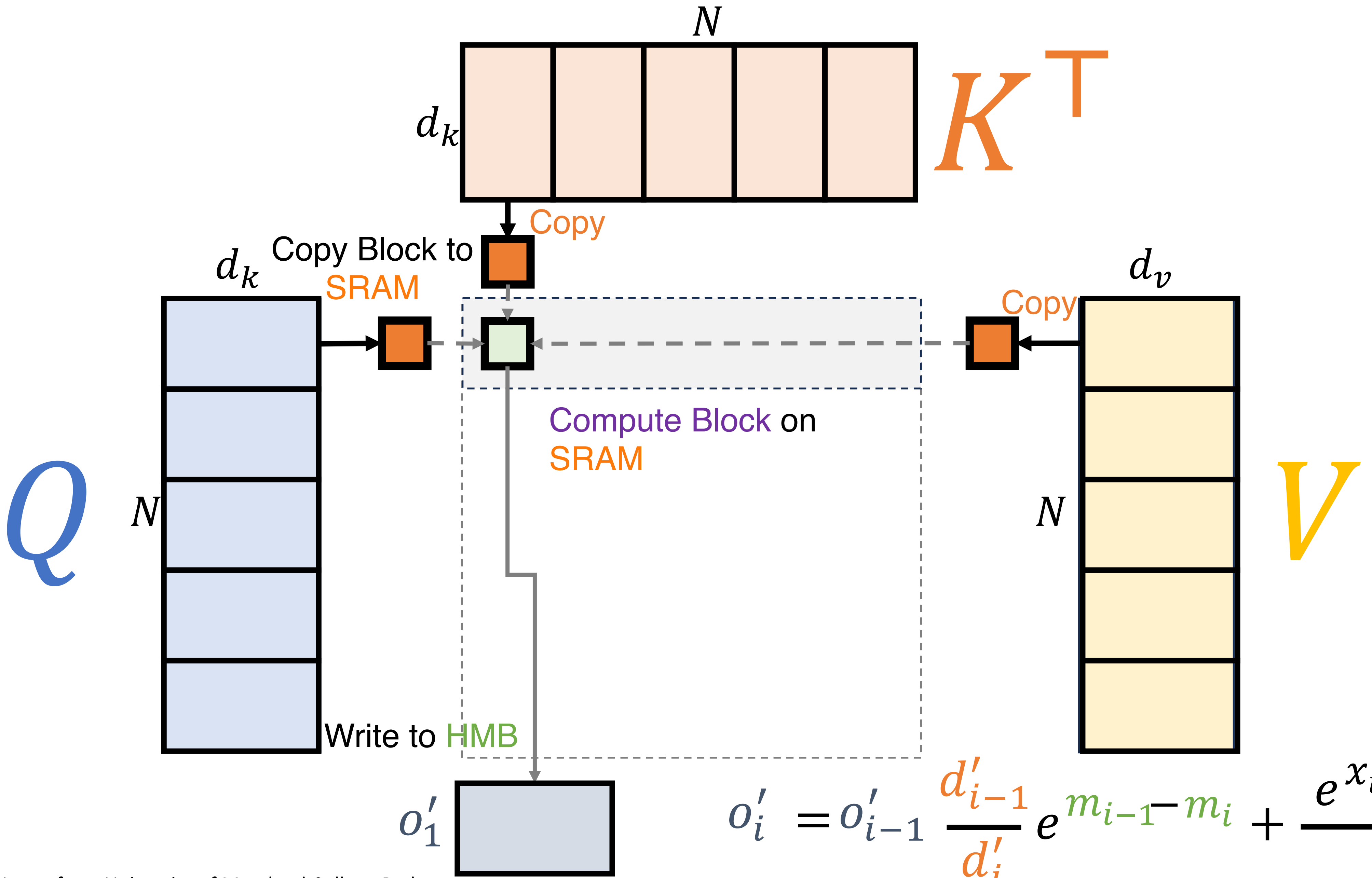
FlashAttention v2: reduce the number of rescaling ops, as well as bound-checking and causal masking operations



Source: Jia-Bin Huang from University of Maryland College Park

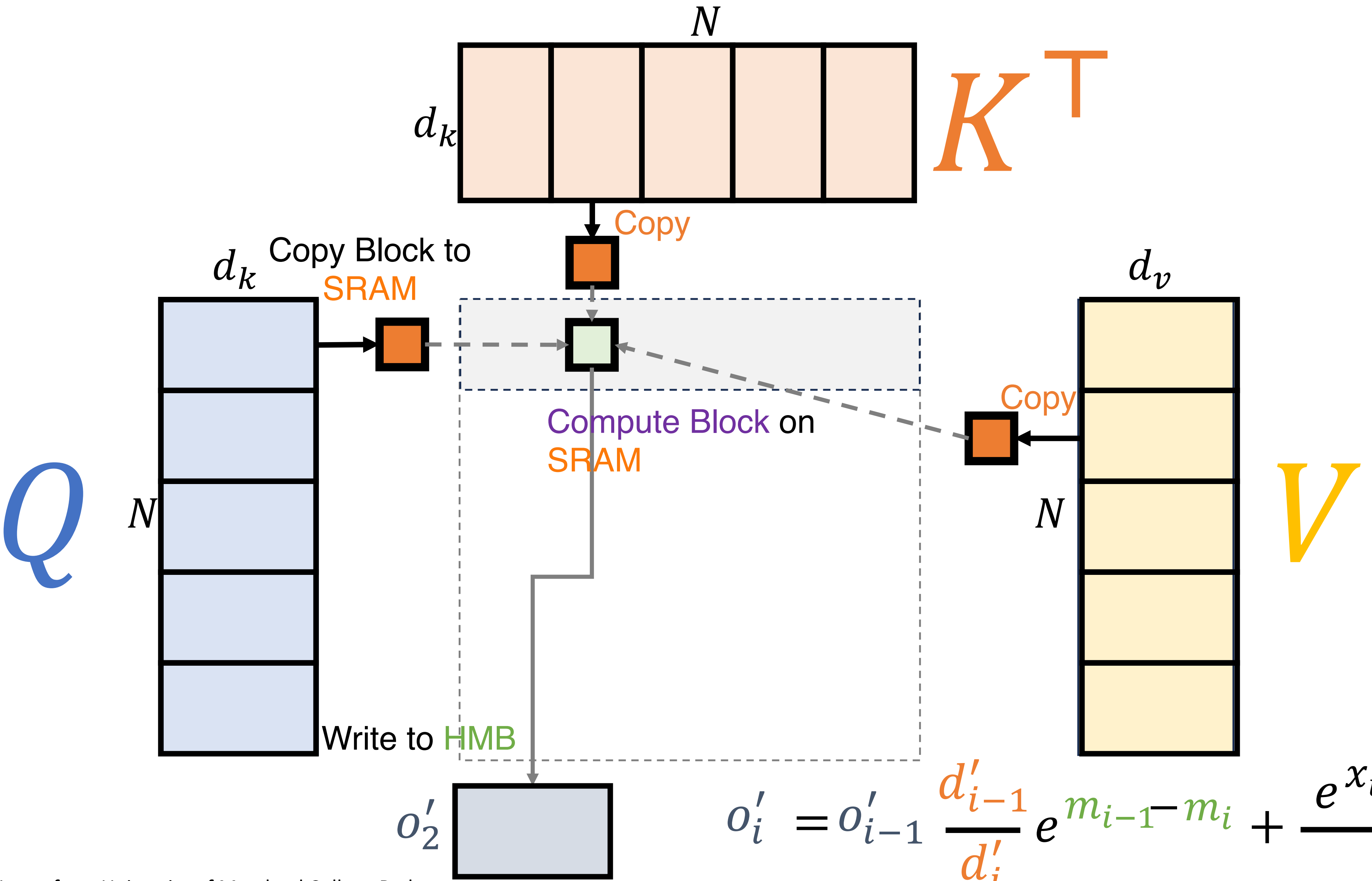


Source: Jia-Bin Huang from University of Maryland College Park



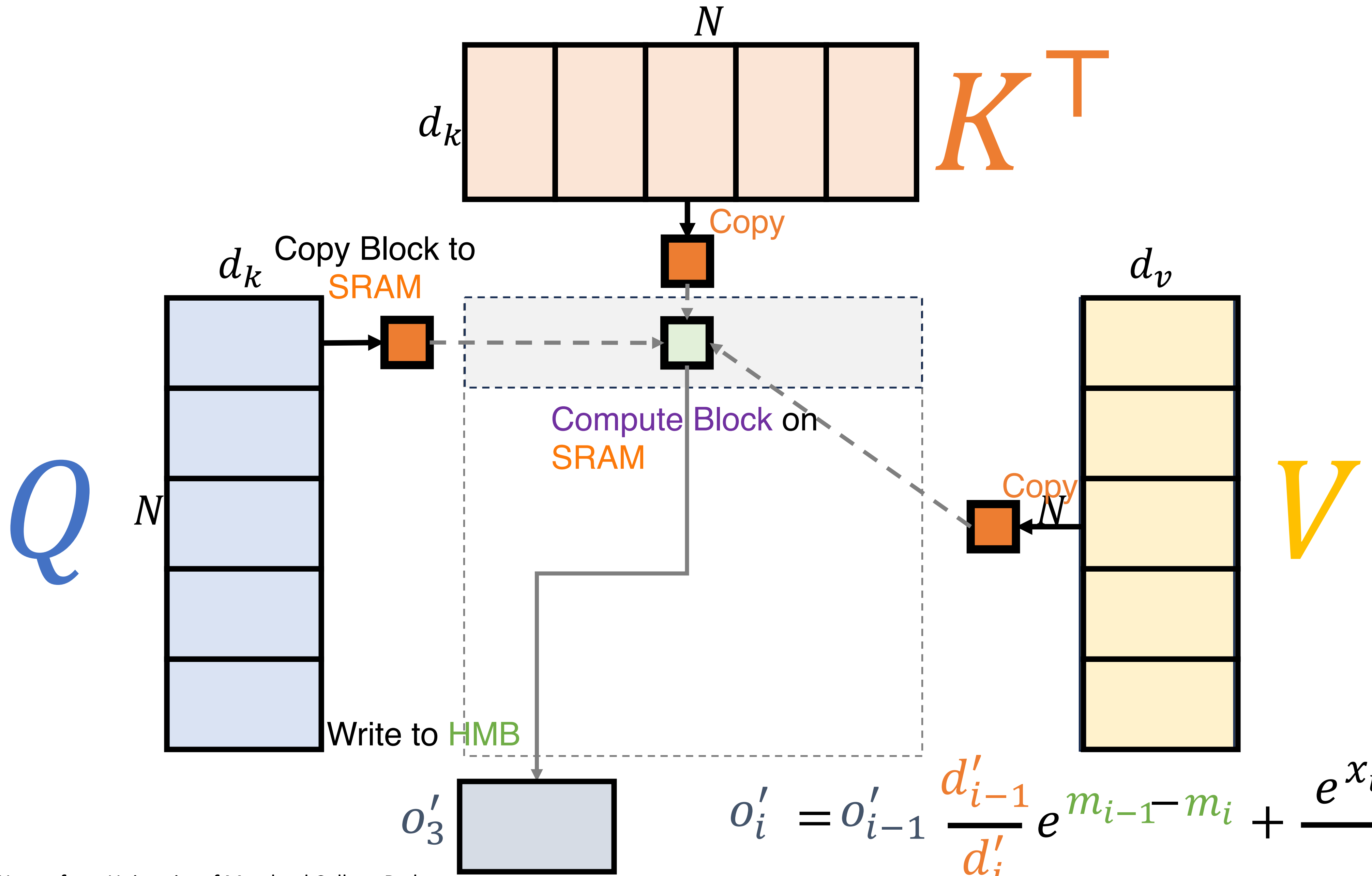
$$o_i' = o_{i-1}' \frac{d_{i-1}'}{d_i'} e^{m_{i-1}-m_i} + \frac{e^{x_i - m_i}}{d_i'} V_i$$

Source: Jia-Bin Huang from University of Maryland College Park



$$o'_i = o'_{i-1} \frac{d'_{i-1}}{d'_i} e^{m_{i-1} - m_i} + \frac{e^{x_i - m_i}}{d'_i} V_i$$

Source: Jia-Bin Huang from University of Maryland College Park



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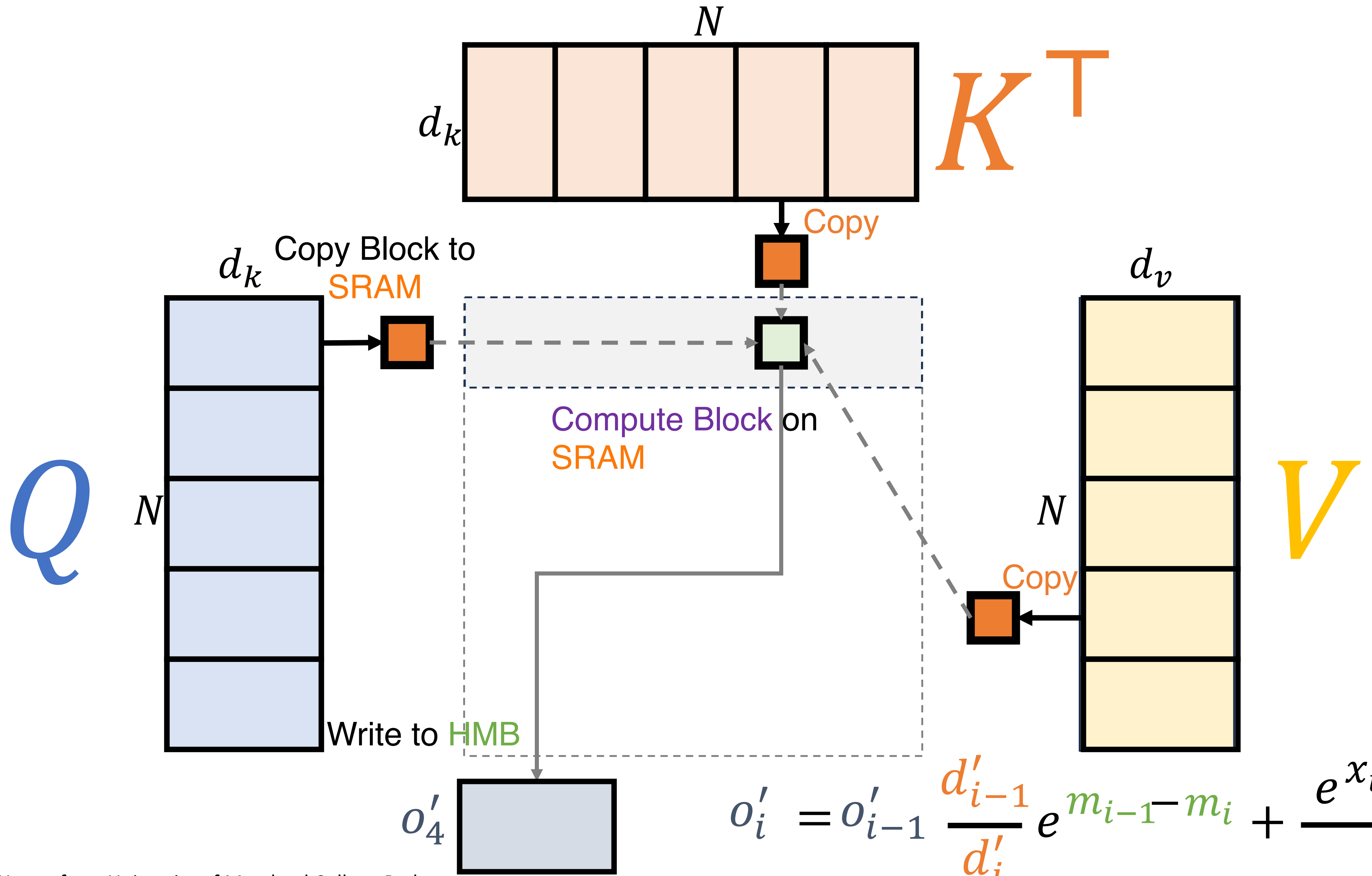

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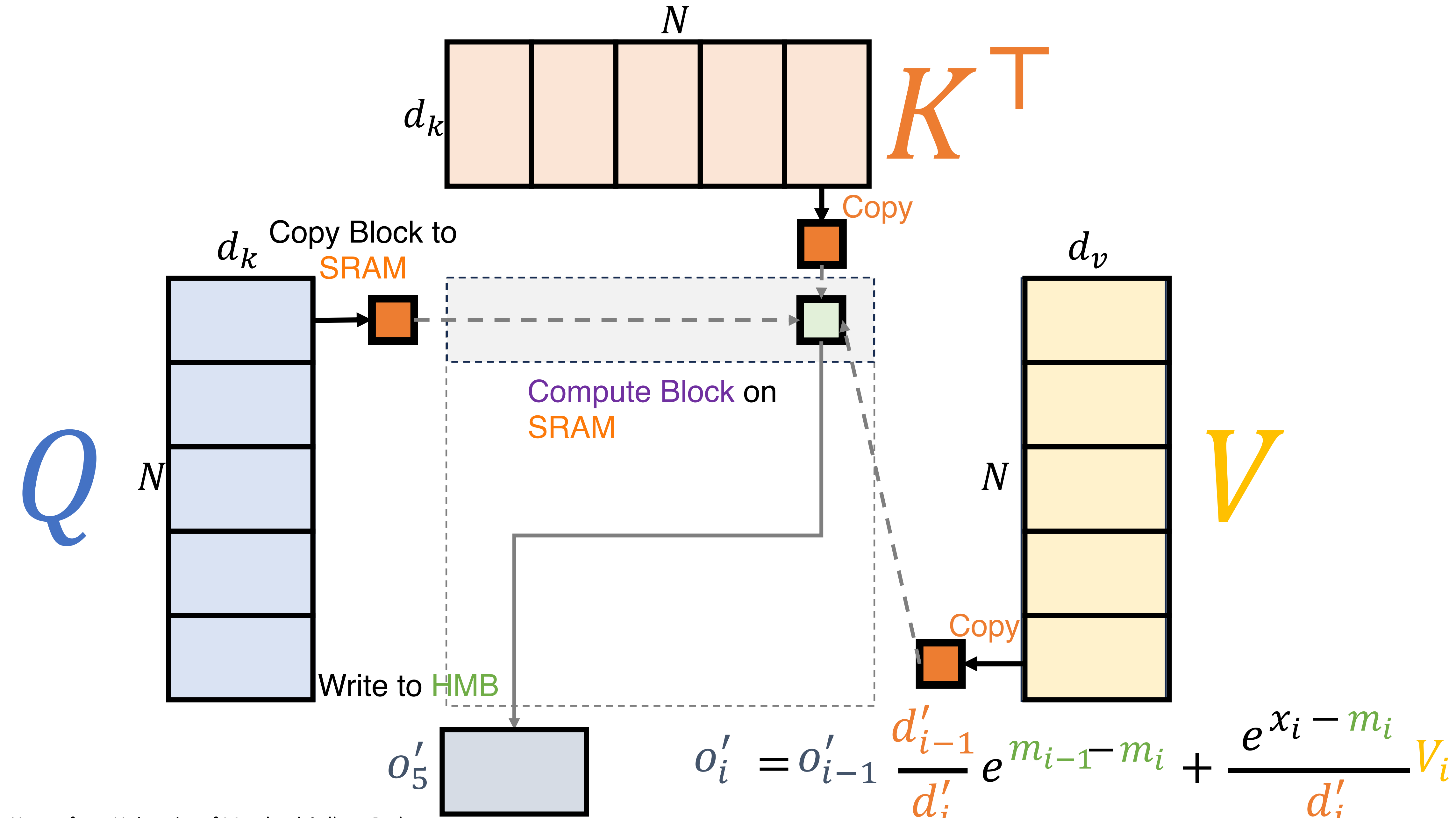

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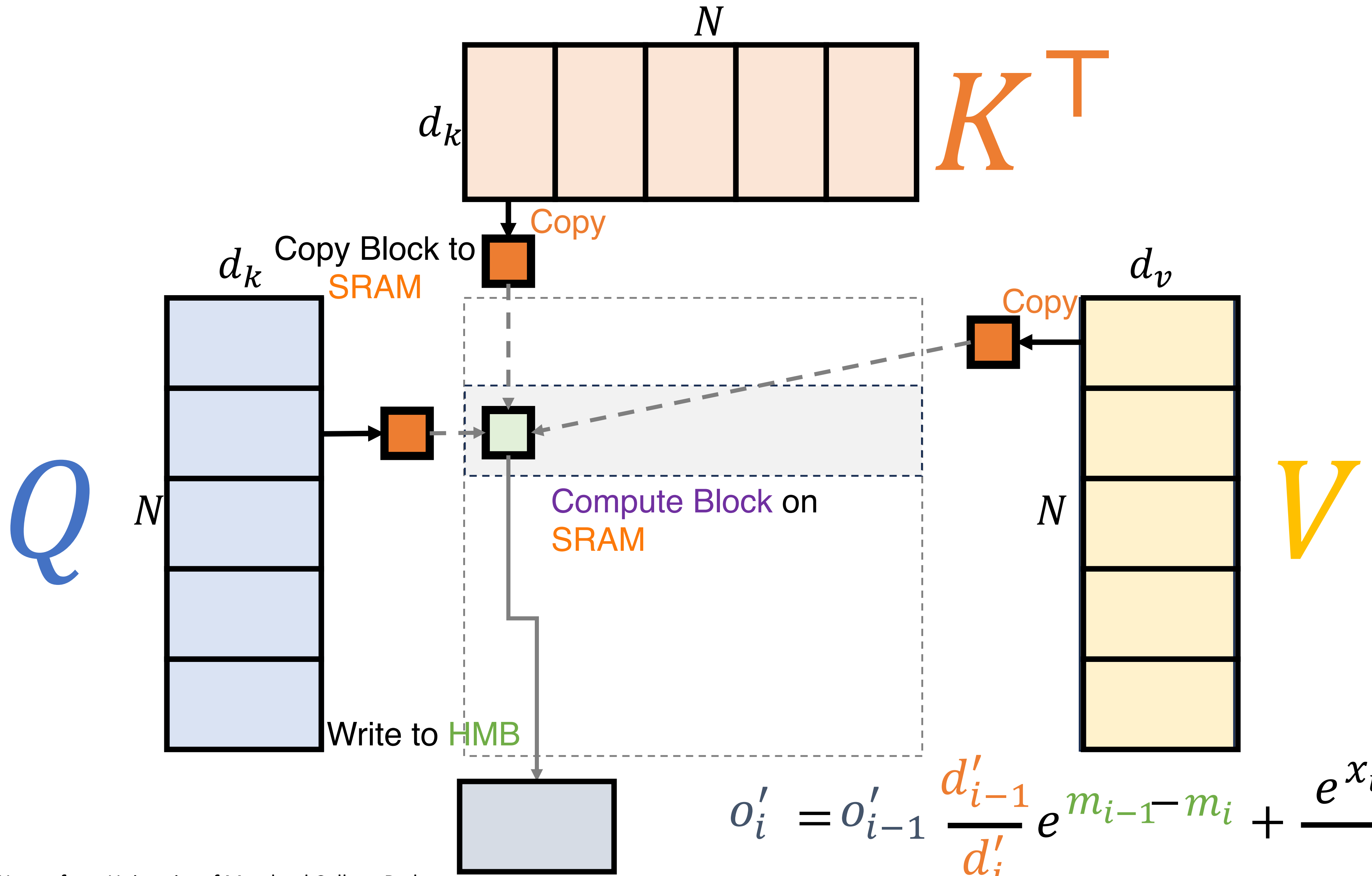

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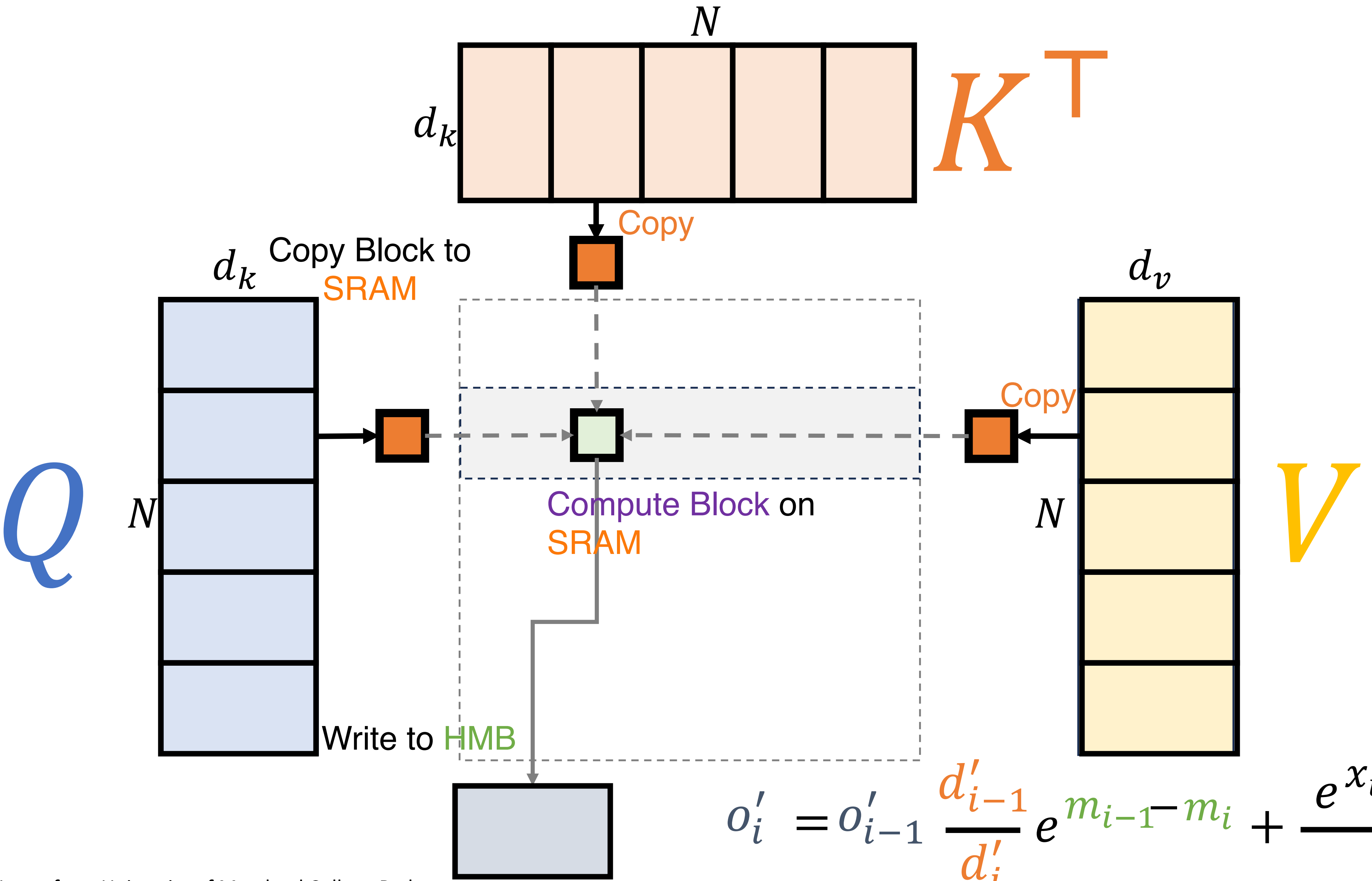

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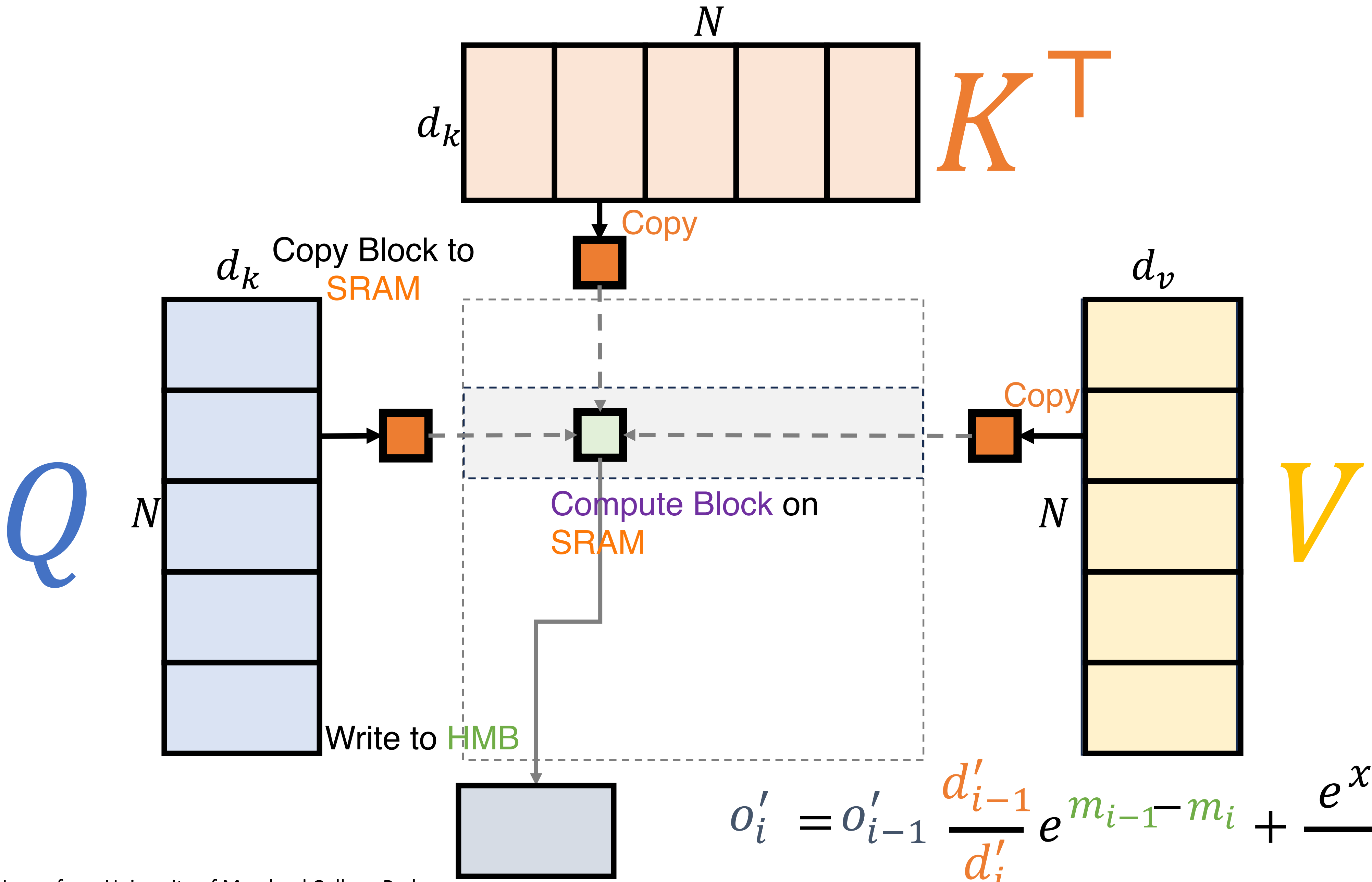

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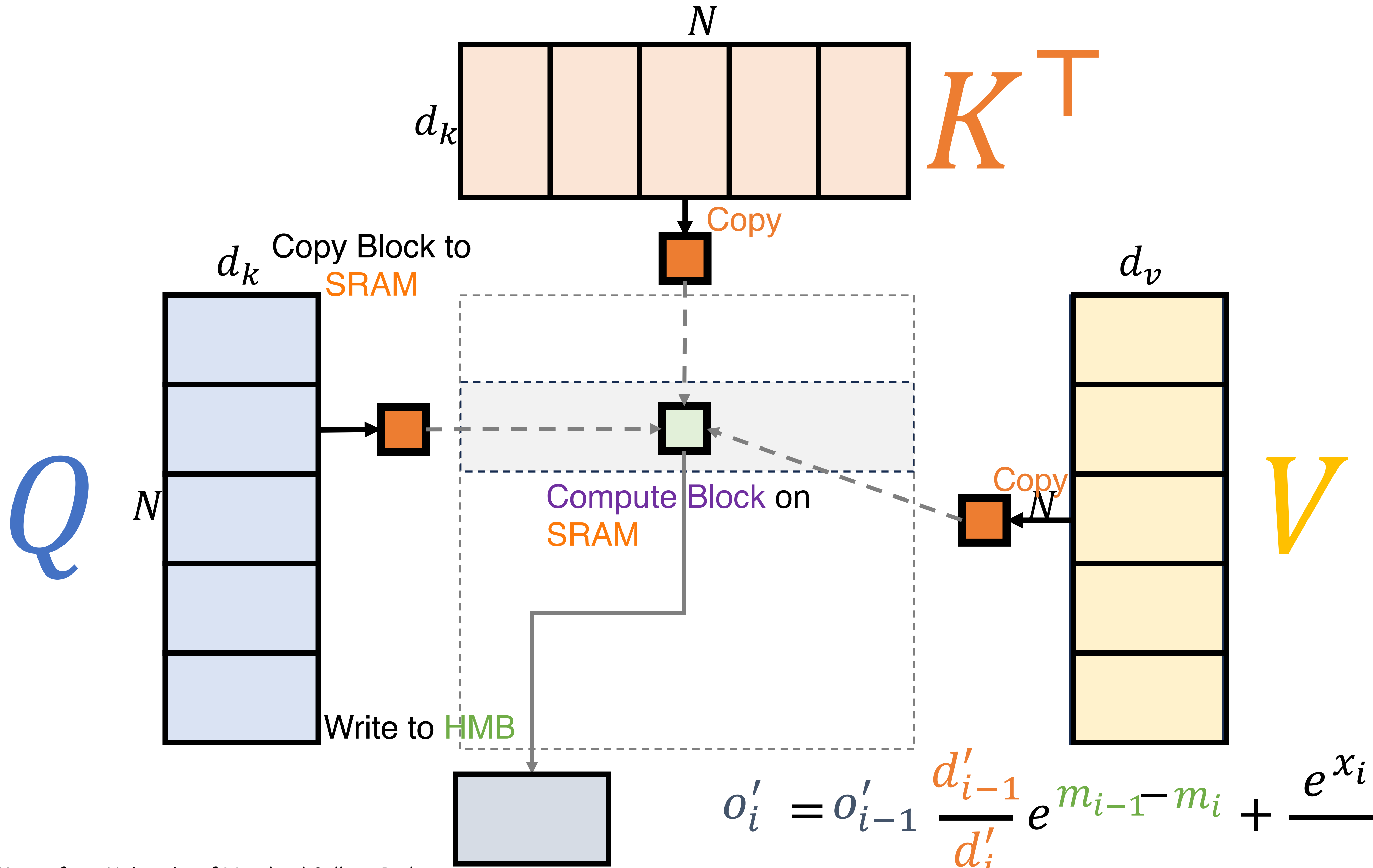

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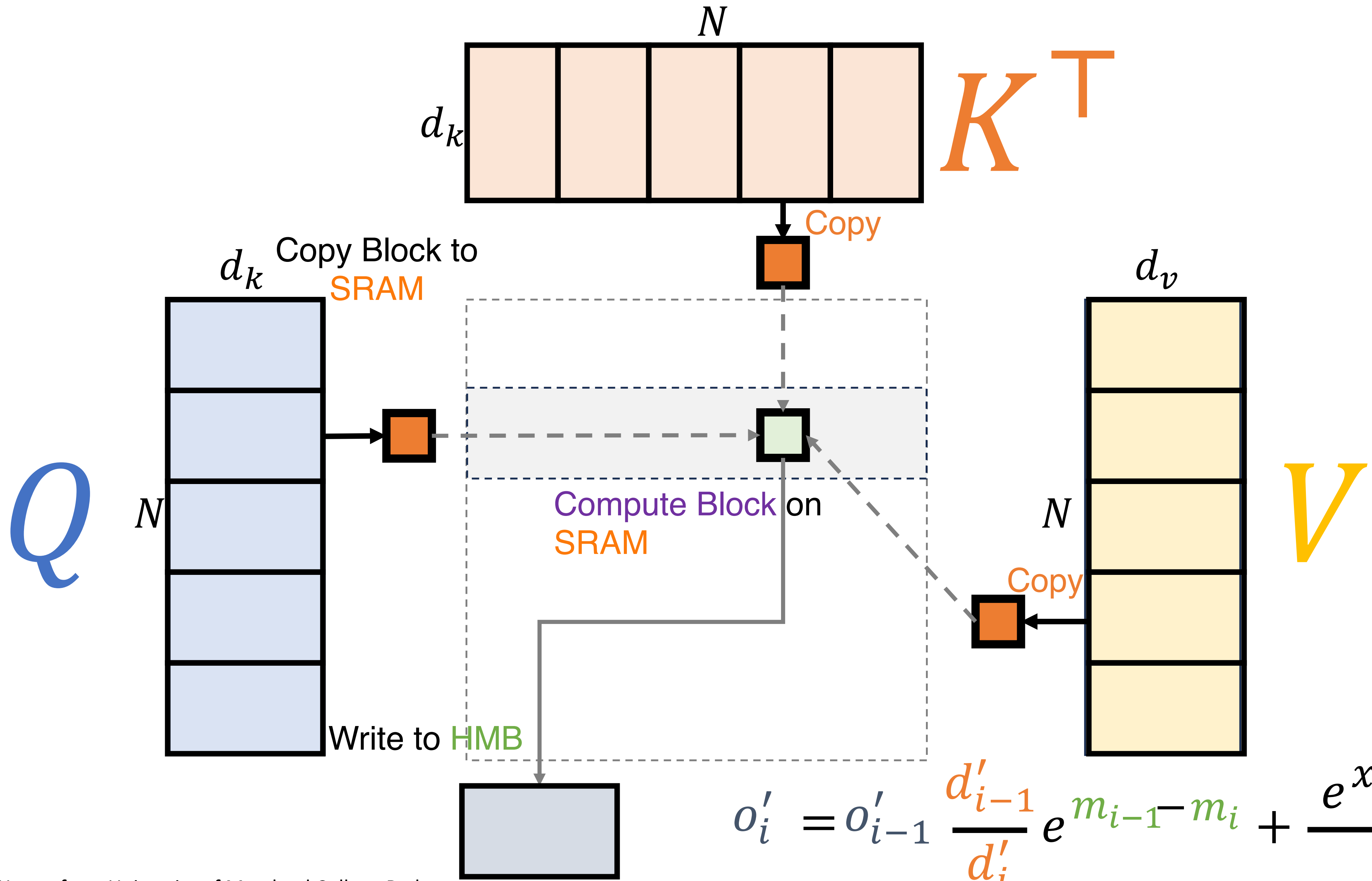

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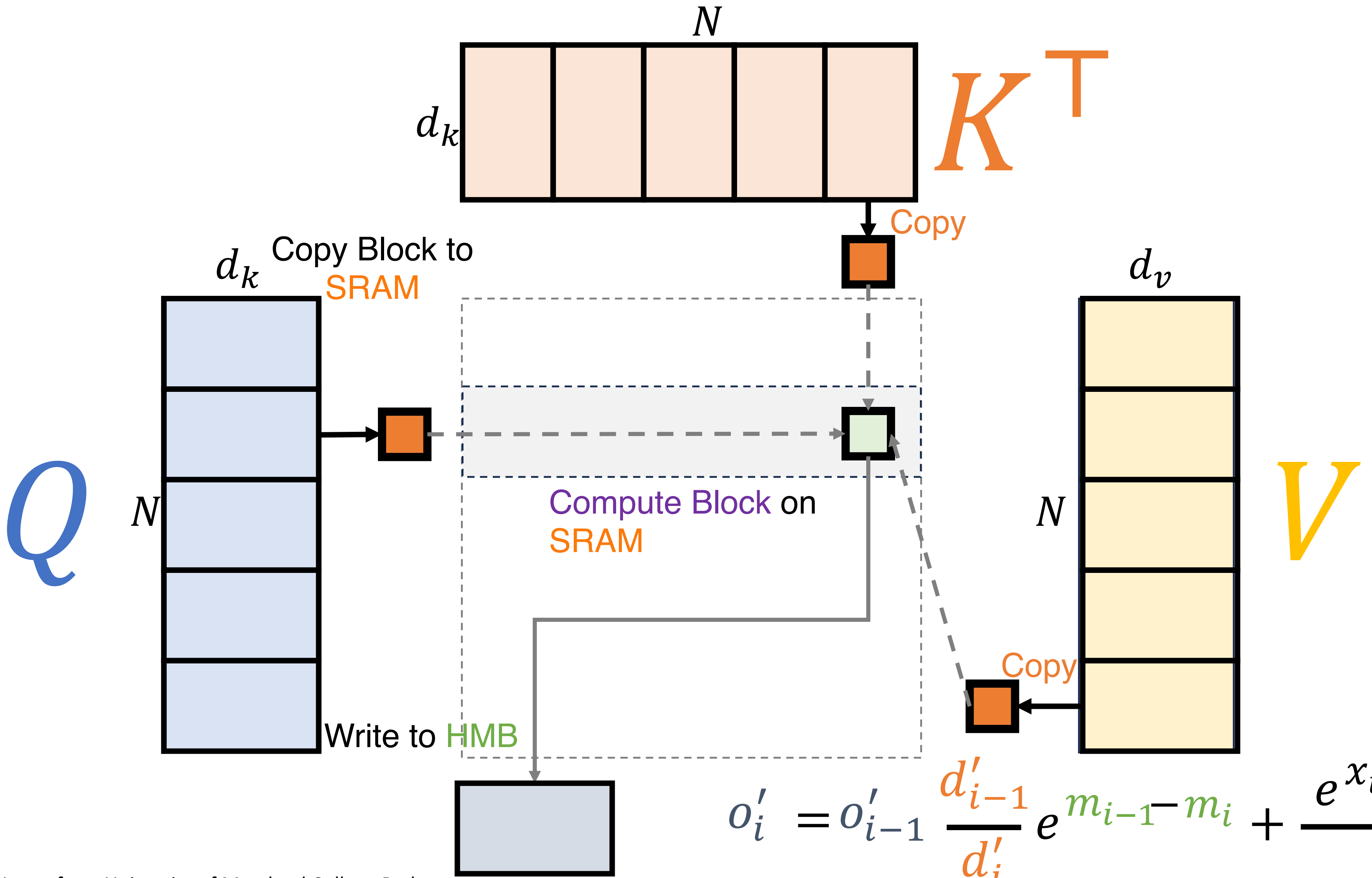
Source: Jia-Bin Huang from University of Maryland College Park



$$o'_i = o'_{i-1} \frac{d'_{i-1}}{d'_i} e^{m_{i-1}-m_i} + \frac{e^{x_i - m_i}}{d'_i} V_i$$

Source: Jia-Bin Huang from University of Maryland College Park

Flash Attention

Algorithm 1 FlashAttention-2 forward pass

Require: Matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times d}$ in HBM, block sizes B_c, B_r.

Divide $\mathbf{Q}$ into $T_r = \left\lceil \frac{N}{B_r} \right\rceil$ blocks $\mathbf{Q}_1, \dots, \mathbf{Q}_{T_r}$ of size $B_r \times d$ each, and divide $\mathbf{K}, \mathbf{V}$ in to $T_c = \left\lceil \frac{N}{B_c} \right\rceil$ blocks $\mathbf{K}_1, \dots, \mathbf{K}_{T_c}$ and $\mathbf{V}_1, \dots, \mathbf{V}_{T_c}$, of size $B_c \times d$ each.

Divide the output $\mathbf{O} \in \mathbb{R}^{N \times d}$ into T_r blocks $\mathbf{O}_i, \dots, \mathbf{O}_{T_r}$ of size $B_r \times d$ each, and divide the logsumexp L into T_r blocks $L_i, \dots, L_{T_r}$ of size B_r each.

Outerloop: traversing Q submatrices first;

Innerloop: traversing K and V submatrices

for $1 \le i \le T_r$ **do**

  Load $\mathbf{Q}_i$ from HBM to on-chip SRAM.

  On chip, initialize $\mathbf{O}_i^{(0)} = (0)_{B_r \times d} \in \mathbb{R}^{B_r \times d}$, $\ell_i^{(0)} = (0)_{B_r} \in \mathbb{R}^{B_r}$, $m_i^{(0)} = (-\infty)_{B_r} \in \mathbb{R}^{B_r}$.

  for $1 \le j \le T_c$ **do**

    Load $\mathbf{K}_j, \mathbf{V}_j$ from HBM to on-chip SRAM.

    On chip, compute $\mathbf{S}_i^{(j)} = \mathbf{Q}_i \mathbf{K}_j^T \in \mathbb{R}^{B_r \times B_c}$.

    On chip, compute $m_i^{(j)} = \max(m_i^{(j-1)}, \mathrm{rowmax}(\mathbf{S}_i^{(j)})) \in \mathbb{R}^{B_r}$, $\tilde{\mathbf{P}}_i^{(j)} = \exp(\mathbf{S}_i^{(j)} - m_i^{(j)}) \in \mathbb{R}^{B_r \times B_c}$ (pointwise), $\ell_i^{(j)} = e^{m_i^{j-1} - m_i^{(j)}} \ell_i^{(j-1)} + \mathrm{rowsum}(\tilde{\mathbf{P}}_i^{(j)}) \in \mathbb{R}^{B_r}$.

    On chip, compute $\mathbf{O}_i^{(j)} = \mathrm{diag}(e^{m_i^{(j-1)} - m_i^{(j)}})^{-1} \mathbf{O}_i^{(j-1)} + \tilde{\mathbf{P}}_i^{(j)} \mathbf{V}_j$.

  end for

  On chip, compute $\mathbf{O}_i = \mathrm{diag}(\ell_i^{(T_c)})^{-1} \mathbf{O}_i^{(T_c)}$.

  On chip, compute $L_i = m_i^{(T_c)} + \log(\ell_i^{(T_c)})$.

  Write $\mathbf{O}_i$ to HBM as the i-th block of $\mathbf{O}$.

  Write L_i to HBM as the i-th block of L.

end for

Return the output $\mathbf{O}$ and the logsumexp L.

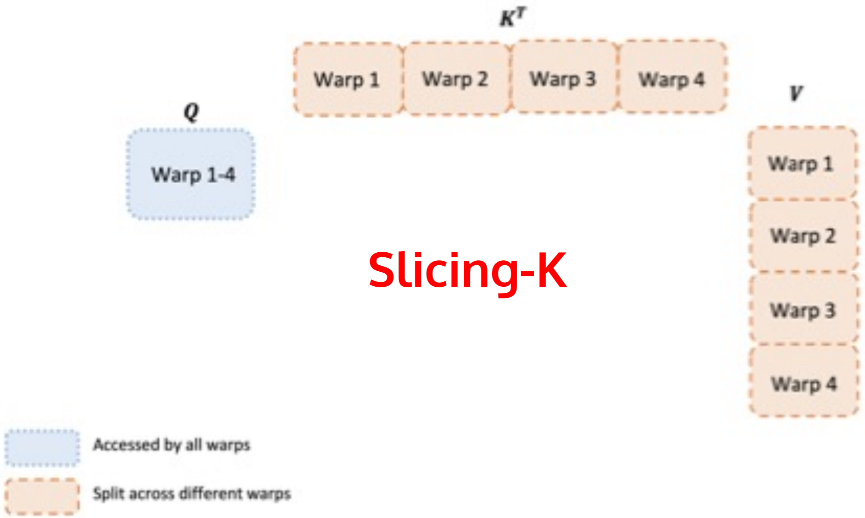
Write back to HBM after the inner-loop: reducing HBM read/write compared with v1

Flash Attention v2: Loop Reordering

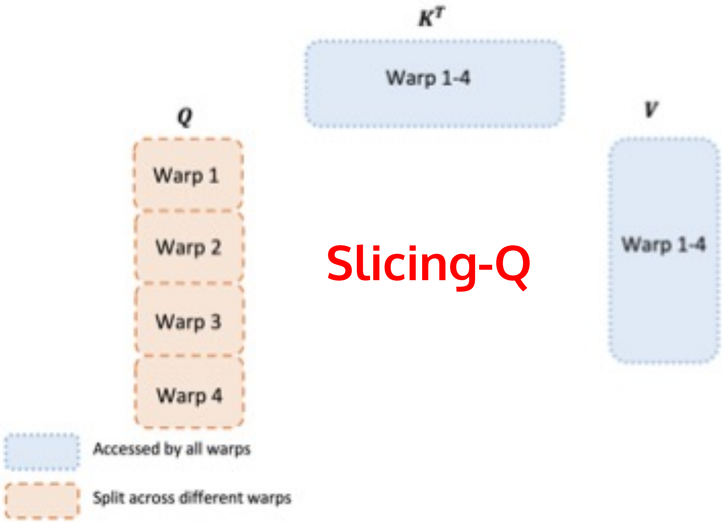
- Maintaining the same order of I/O complexity: $O(N^2 d^2 M^{-1})$
- Reducing read/write of O_i: write O_i to HBM after traversing all K/V blocks at the inner loop

Flash Attention

- FlashAttention v2
  - Better Work Partitioning
    - A thread block is partitioned into multiple warps (32 threads per warp, 4~8 warps per block)



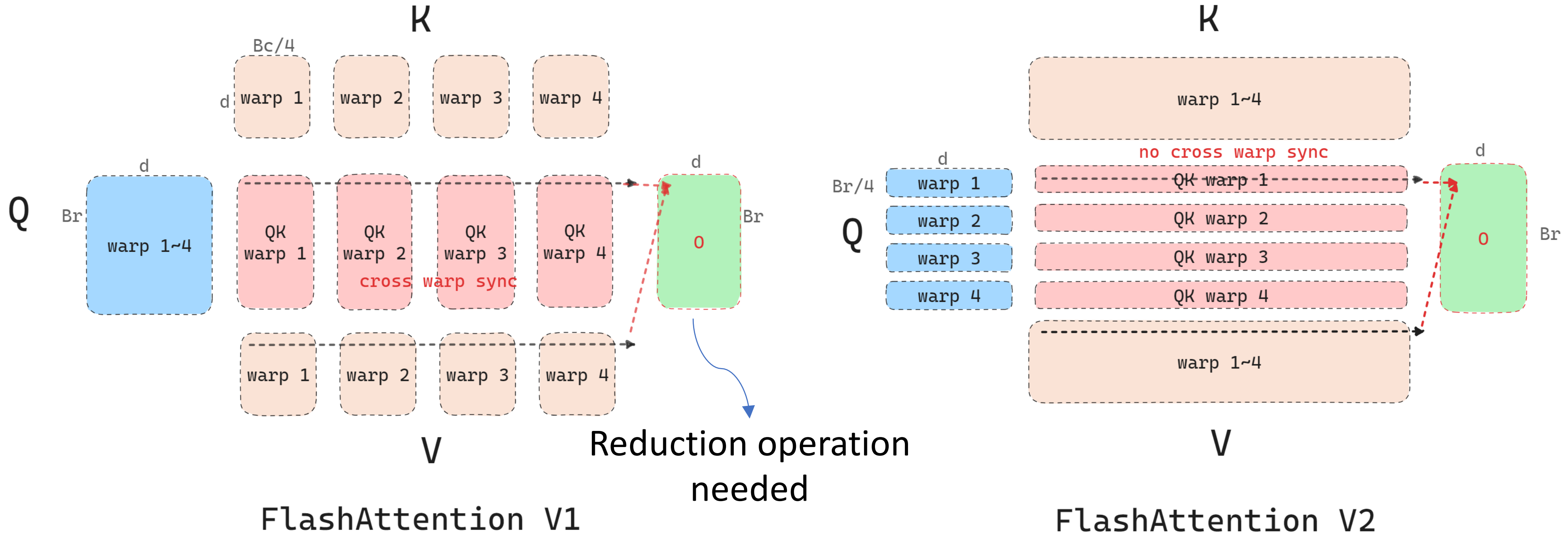
- Slicing K/V at each Warp while maining complete Q
- QK^T has four partitions, and $(QK^T)V$ needs REDUCTION (write to shared memory, synchronize, and add-up intermediate results)



- Slicing Q while keeping K/V accessible by all Warps
- Each warp performs matrix multiply to obtain a slice of QK^T, and no intra-block communication

Flash Attention

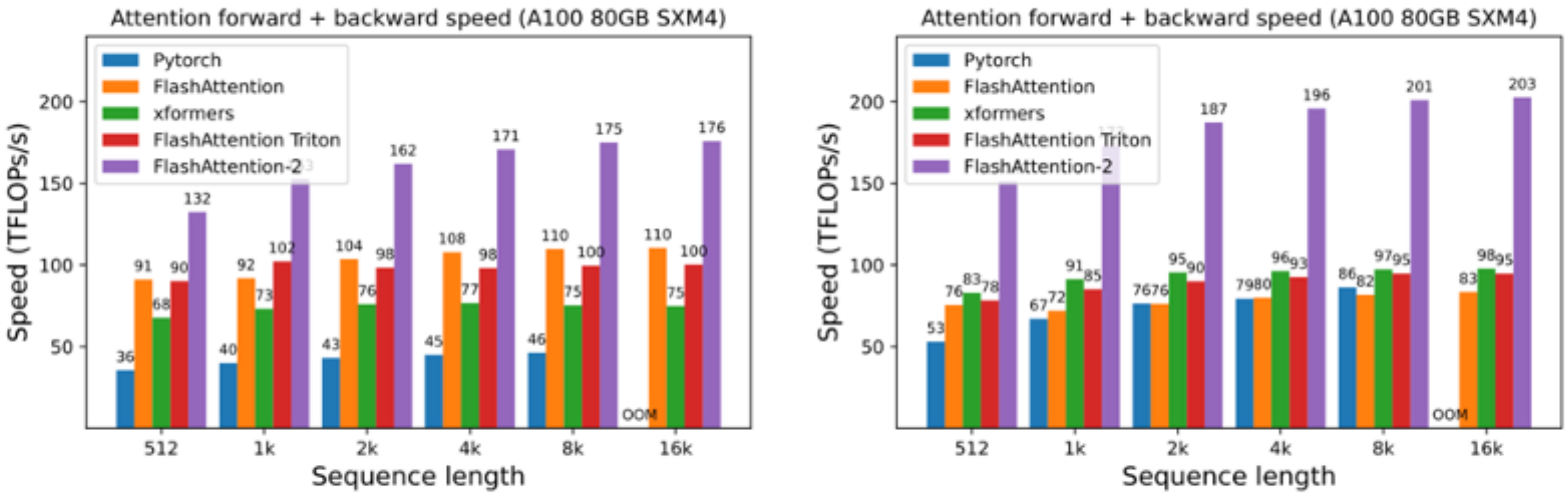
- FlashAttention v2
  - Better Work Partitioning
    - A thread block is partitioned into multiple warps (32 threads per warp, 4~8 warps per block)



By Zhihu blogger: Antinomi

Flash Attention

- FlashAttention v2
  - Performance speedup

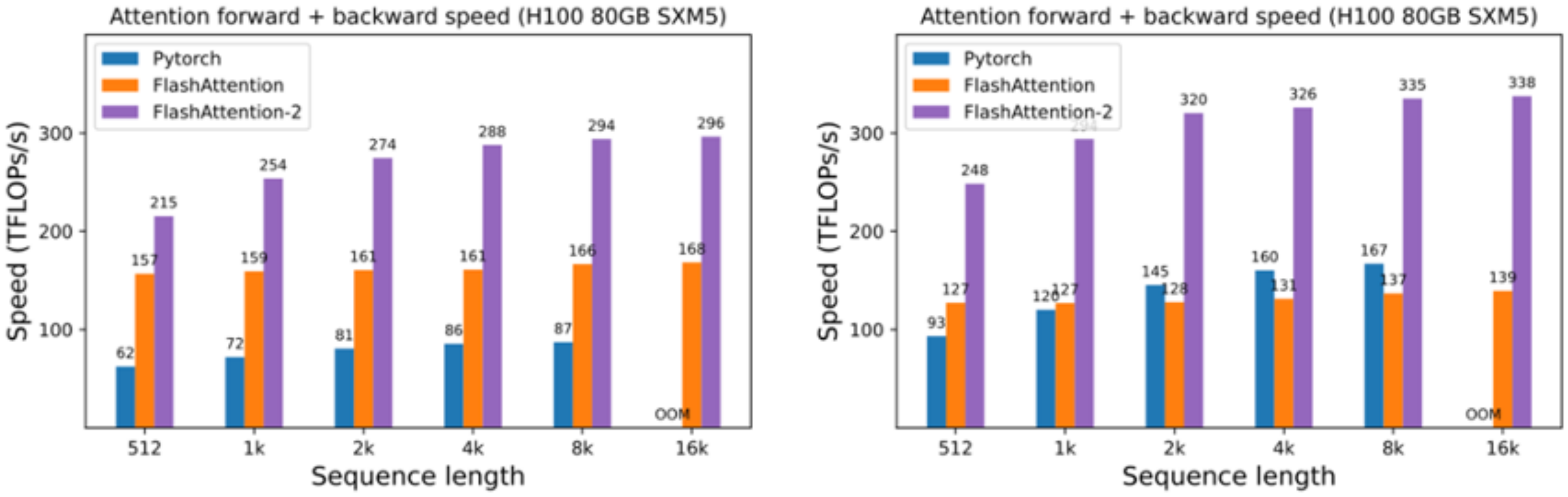


(a) Without causal mask, head dimension 64 (b) Without causal mask, head dimension 128

Attention forward + backward speed on A100 GPU

Flash Attention

- FlashAttention v2
  - Performance speedup



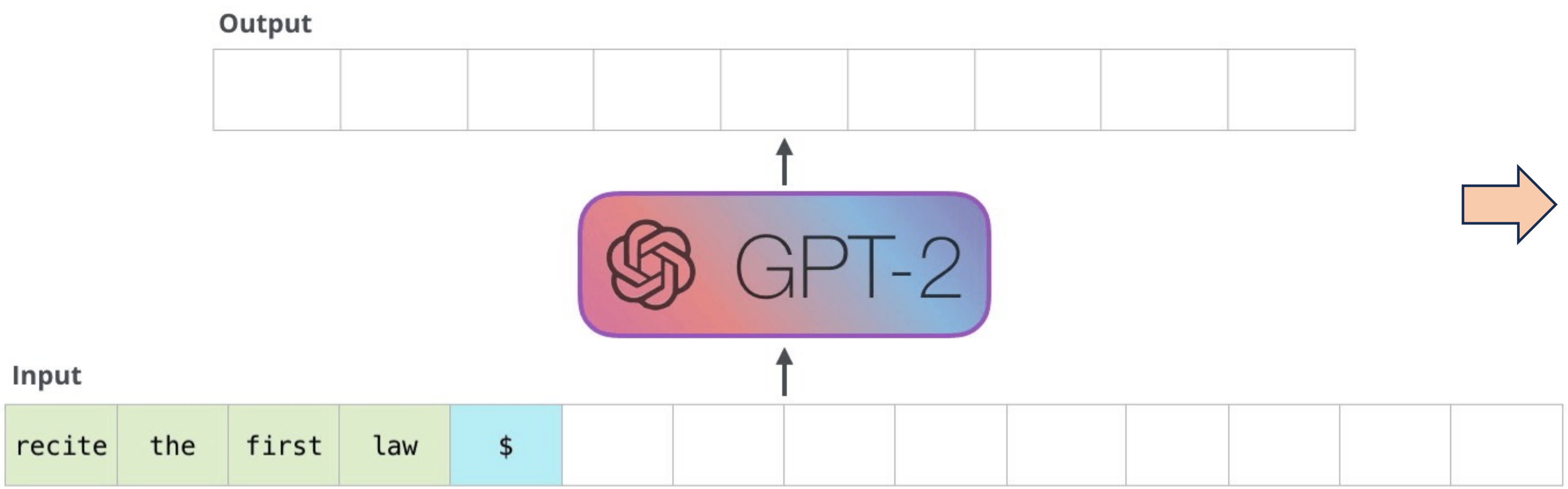
(a) Without causal mask, head dimension 64

(b) Without causal mask, head dimension 128

Attention forward + backward speed on H100 GPU

Flash Attention

- Flash Decoding
  - New challenges in autoregressive decoding



- Key (K) and Value (V) keep growing to be extraordinarily large
  - Out of Memory error
  - Swapping to CPU memory which is very slow
  - Truncating the input causes poor performance

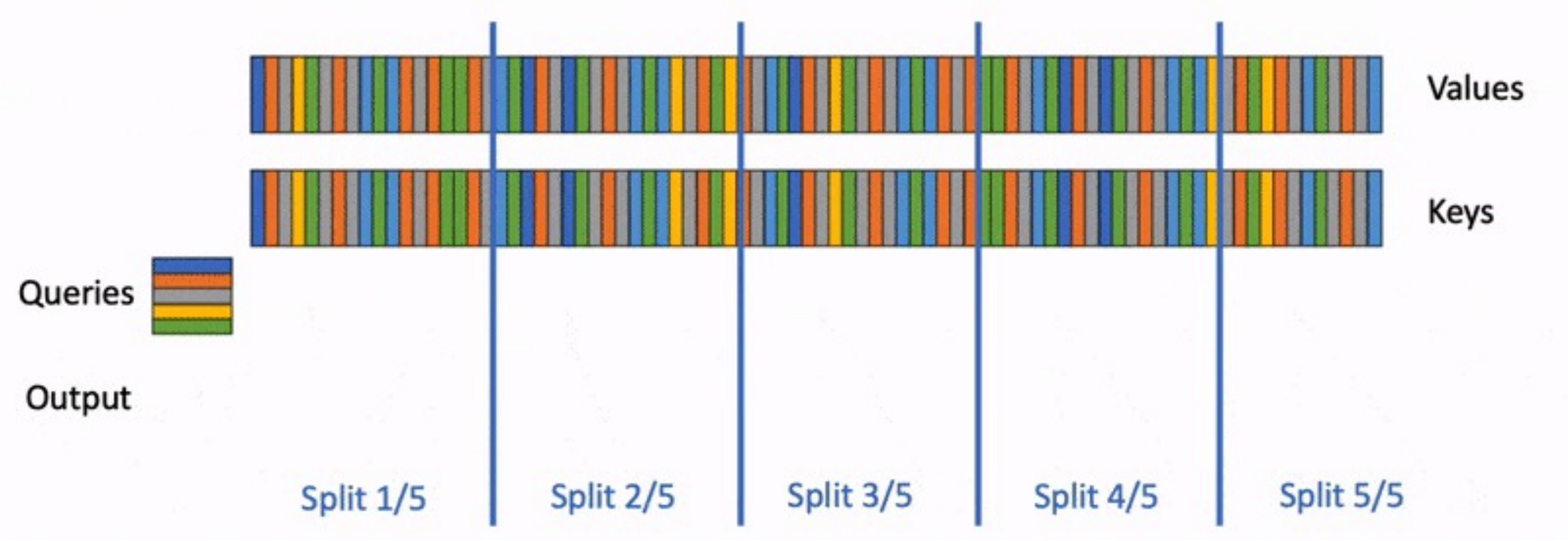
Query (Q) length is always 1, i.e. the newly generated token

Flash Attention

- Flash Decoding
  - Low efficiency due to small Q (i.e. sequence length of 1 in decoding)

Flash Attention

- Flash Decoding
  - Sharding Key and Value matrices (e.g. 5 pieces) in order to create many more thread blocks



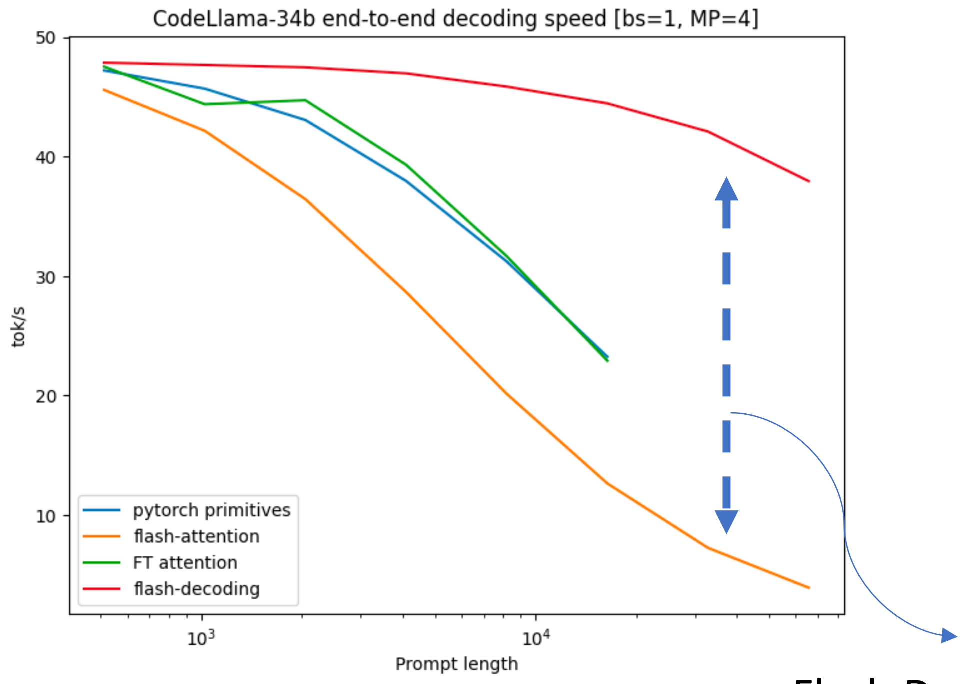
Thread block 1 | Thread block 2 | Thread block 3 | Thread block 4 | Thread block 5

Flash Attention

- Flash Decoding
  - Computing attention scores with K/V splits using FlashAttention v1/v2
    - Only obtaining local ***max*** and local ***sum***
    - Output O not properly scaled
  - Launching an independent ***REDUCE*** kernel
    - Obtaining global ***max*** and global ***sum***
    - Rescaling local O to obtain the final output

Flash Decoding

- Flash Decoding

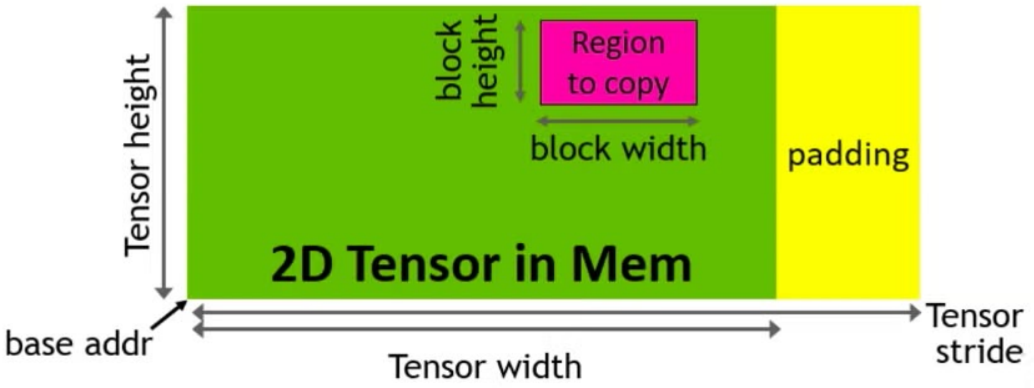


Setting \ Algorithm	PyTorch Eager	Flash-Attention v2.0.9	Flash-Decoding
B=256, seqlen=256	3058.6	390.5	63.4
B=128, seqlen=512	3151.4	366.3	67.7
B=64, seqlen=1024	3160.4	364.8	77.7
B=32, seqlen=2048	3158.3	352	58.5
B=16, seqlen=4096	3157	401.7	57
B=8, seqlen=8192	3173.1	529.2	56.4
B=4, seqlen=16384	3223	582.7	58.2
B=2, seqlen=32768	3224.1	1156.1	60.3
B=1, seqlen=65536	1335.6	2300.6	64.4
B=1, seqlen=131072	2664	4592.2	106.6

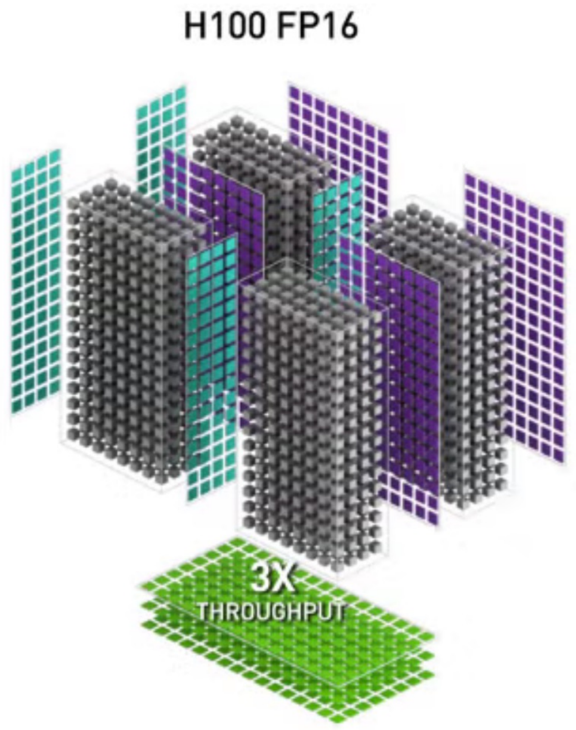
Flash Decoding versus Flash Attention

Flash Attention v3

- Hardware improvement (only for your reference)
  - TMA (Tensor Memory Accelerator)



Warpgroup Matrix Multiply-Accumulate



  - Asynchrony
    - Overlap overall computation and data movement via warp-specialization and interleave block-wise ***matmul*** and ***softmax*** operations

Source: FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision

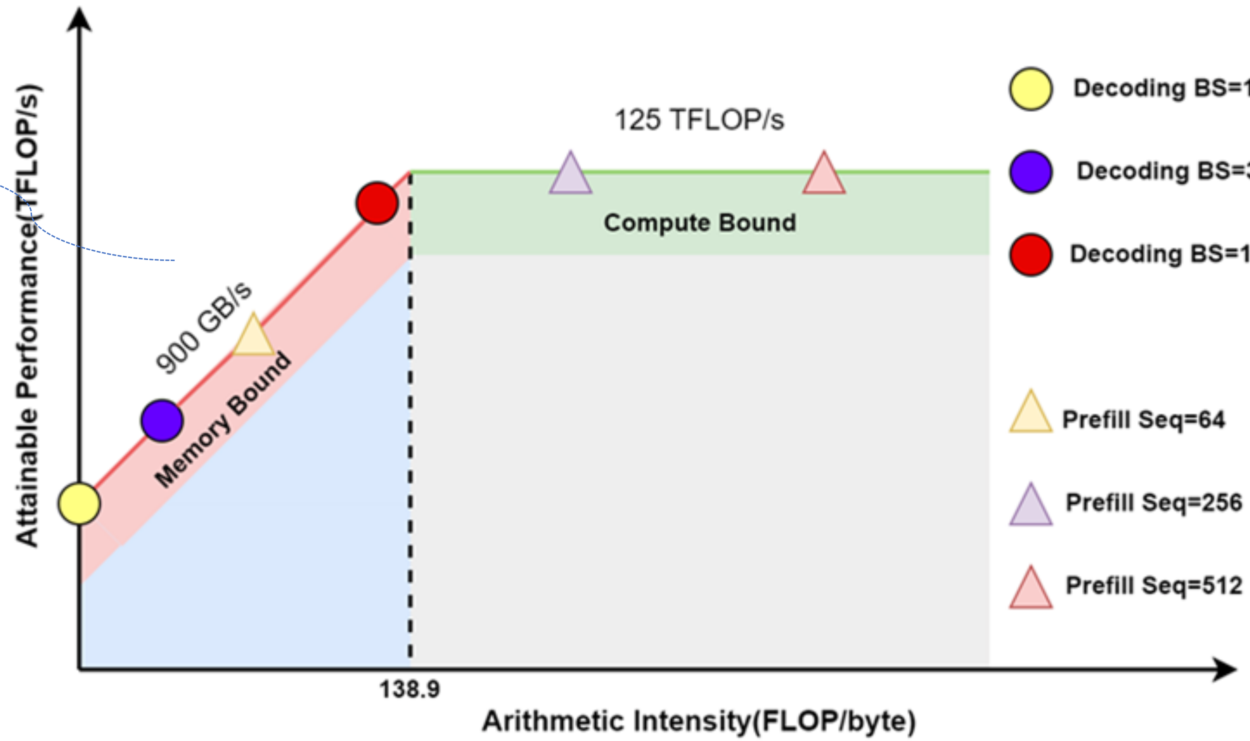
Inference Optimization: Outline

- Overview
- Attention Computation Optimization
  - Sparse Attention
  - Linear Attention
  - Flash Attention
- **Continuous Batching**
- KV Cache Optimization
- Speculative Decoding
- Distributed Serving

Batching to Meet Compute-Bound

- Why batching multiple requests?
  - Generating tokens for a large number of prompts to match the compute bound

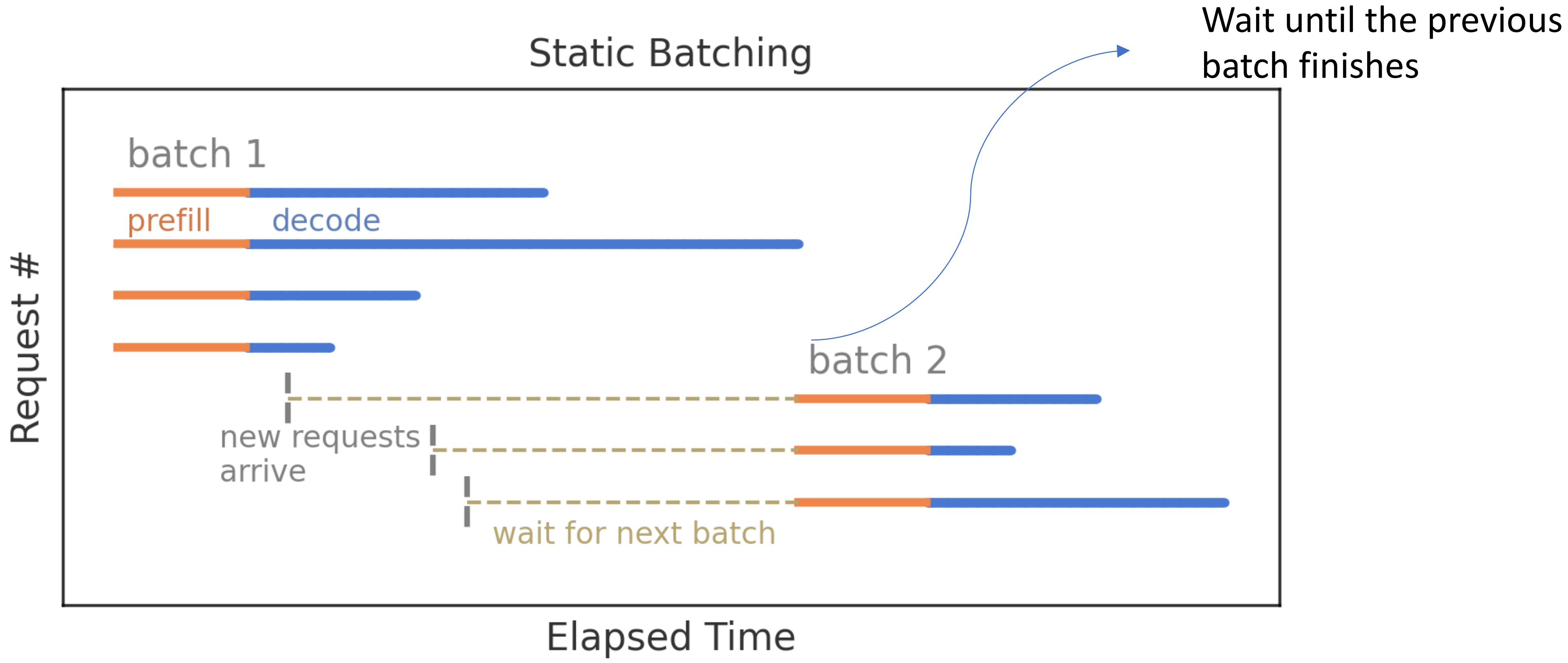
Wasted FLOPS



Roofline model of NVIDIA V100 GPU

Static Batching

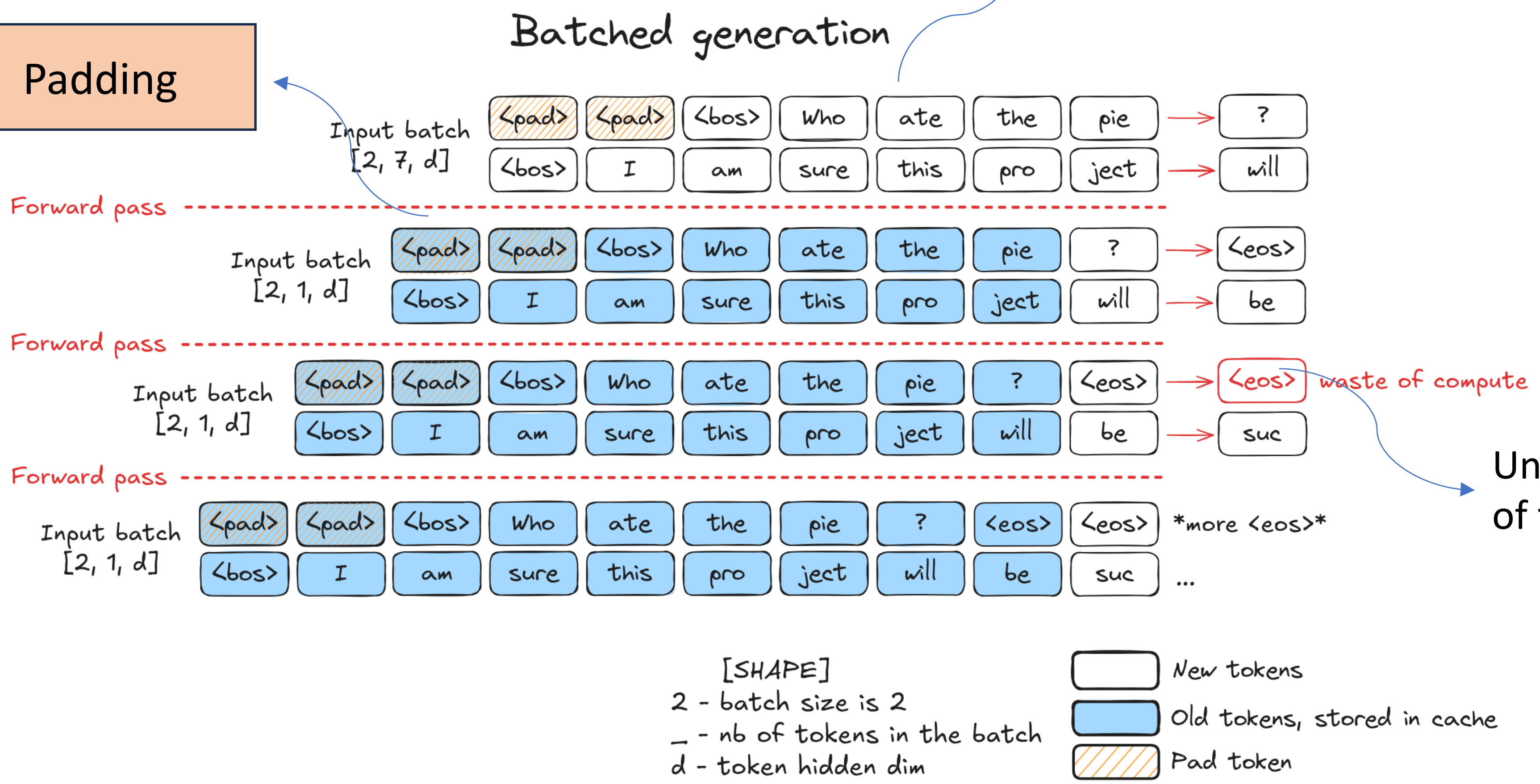
- Naively serving multiple requests simultaneously
  - Unequal input sequence lengths, unknown output sequence lengths



Static Batching

One prefill

Padding

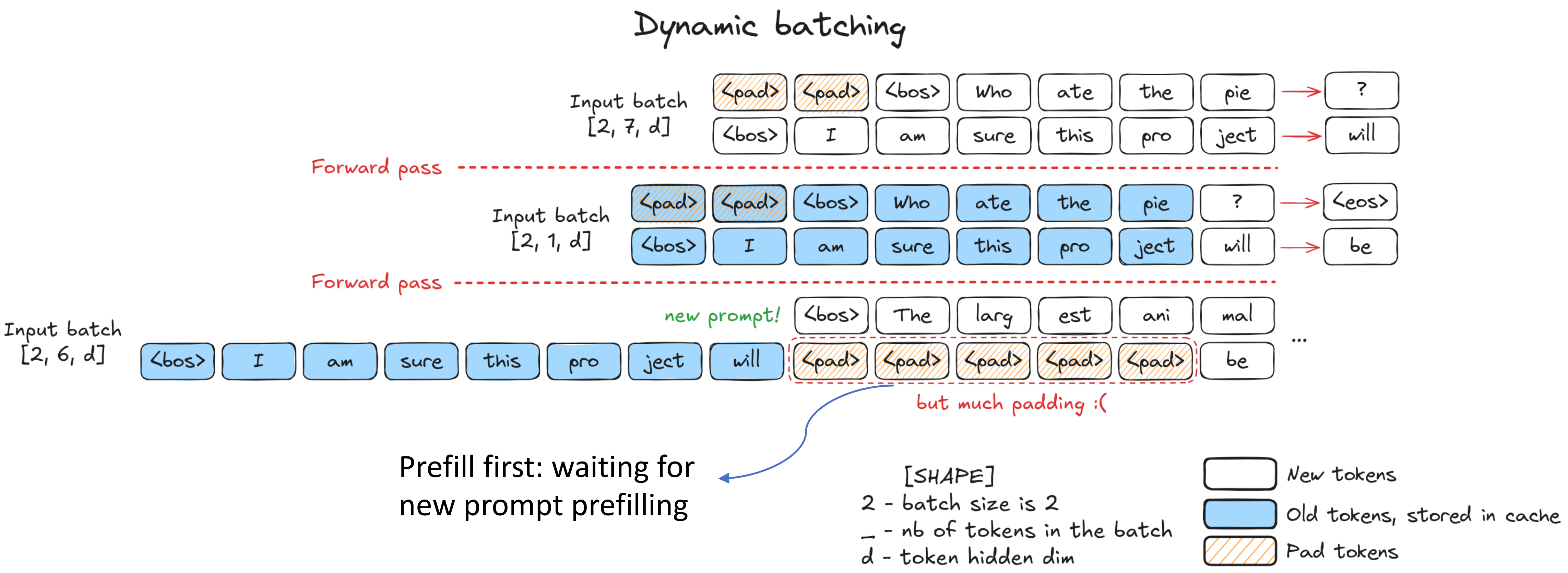


Until the longest request of the batch finishes

Source: https://huggingface.co/blog/continuous_batching

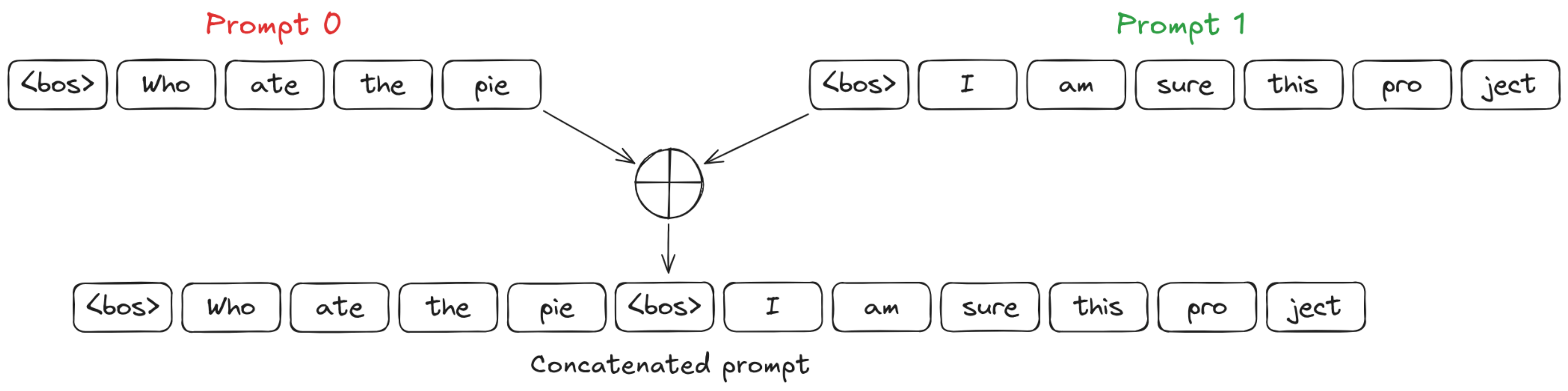
Dynamic Batching

- Insert an inference request in the queue



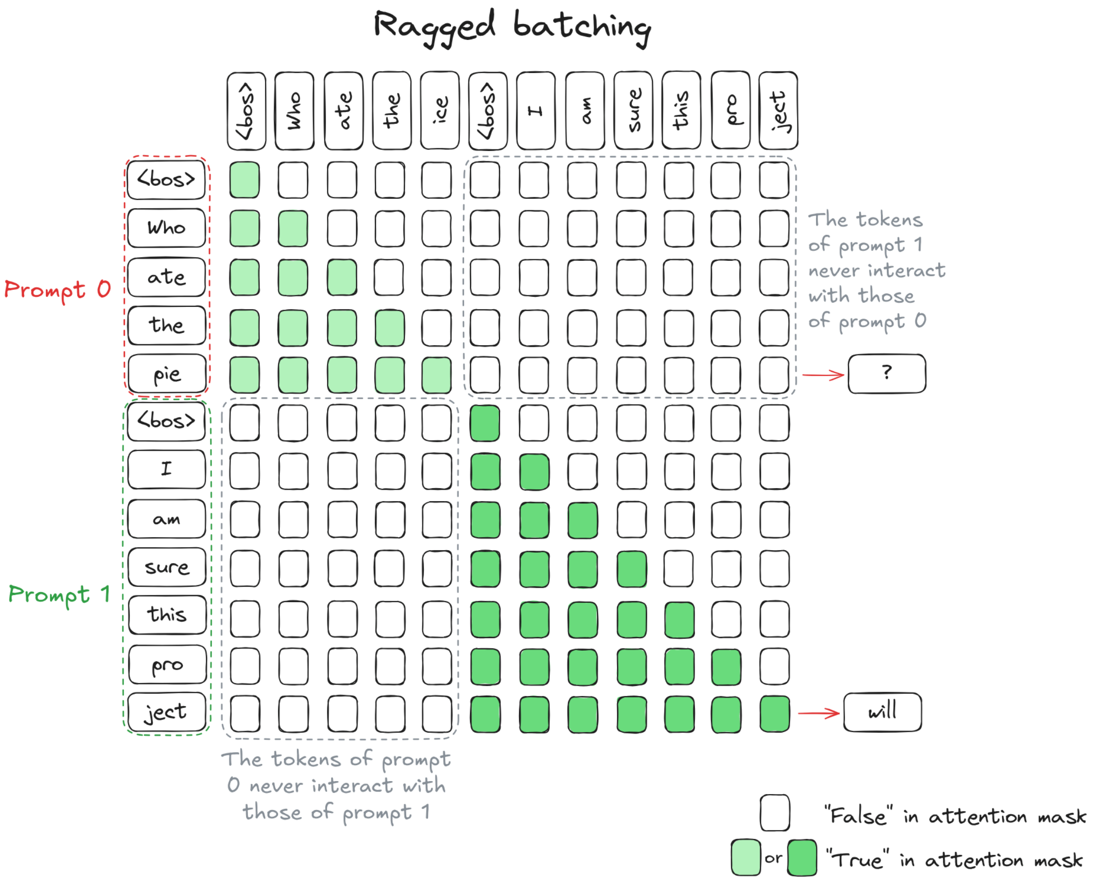
Idea Batching

- Concatenating prompts
  - ***Decode*** is equivalent to ***prefill*** with sequence length of 1
  - ***Decode*** and ***Prefill*** of different prompts operate simultaneously without bubbles



Continuous Batching

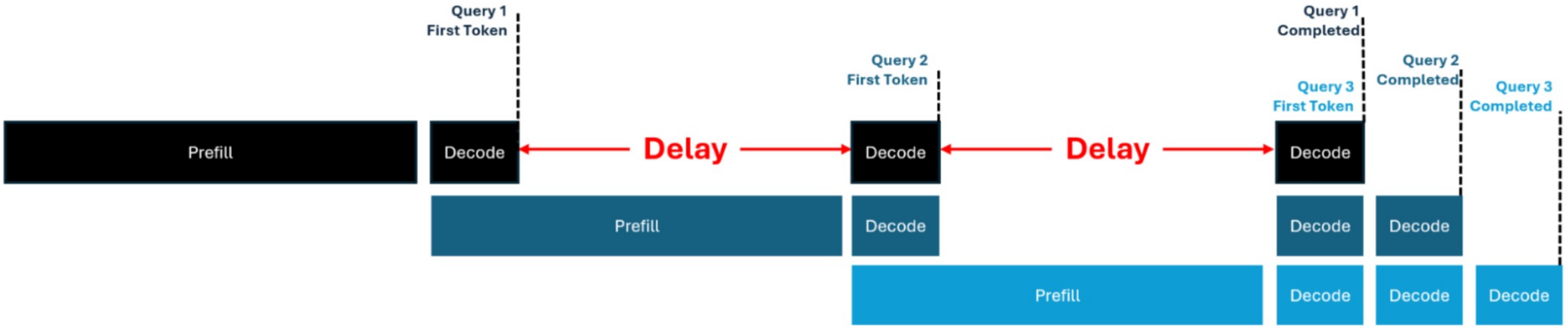
- Continuous batching = ragged batching + ideal dynamic batching
  - Attention scores of tokens belonging to *prompt 0* and *prompt 1* should not be computed together
  - Ragged batching because sequence lengths are 'ragged' or uneven, no need for padding tokens



Source: https://huggingface.co/blog/continuous_batching

Continuous Batching

- Question remains unsolved: how to make ***prefill*** of new requests and ***decode*** of existing requests operate ***in parallel***?

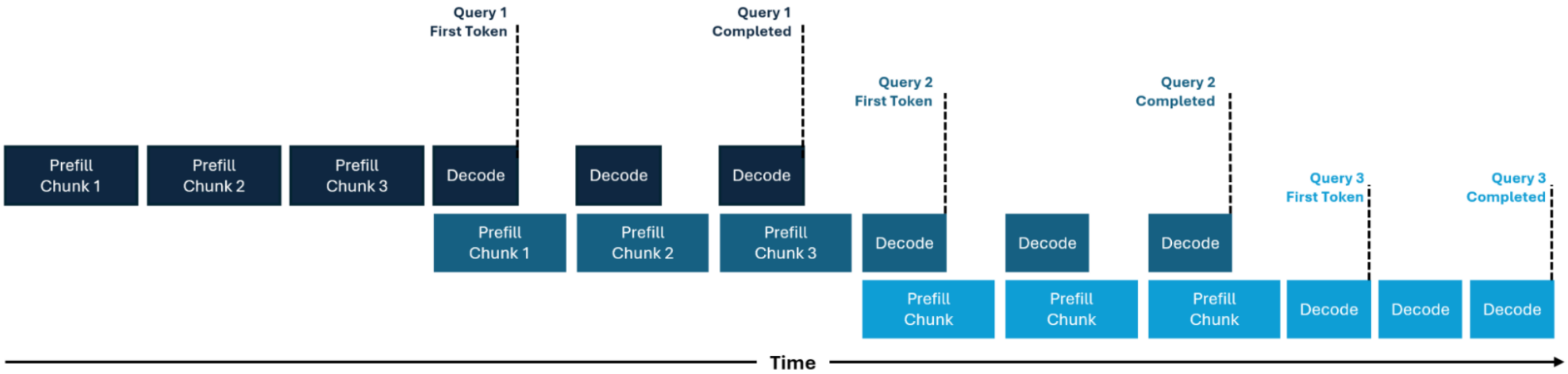


Continuous batching without chunked prefill

Source: Streamlining AI Inference Performance and Deployment with NVIDIA TensorRT-LLM Chunked Prefill

Chunked Prefill

- Chunked Prefill
  - **Elastic Scheduling**: Instead of processing the entire prompt at once, one can split it into smaller chunks: more scheduling flexibility and enables mixing prefill and decode.
  - **Decode-Maximal Batching**: one can combine a single prefill chunk with multiple decode requests in the same batch, allowing decode operations to "piggyback" on the more compute-intensive prefill operations



Source: Streamlining AI Inference Performance and Deployment with NVIDIA TensorRT-LLM Chunked Prefill

Chunked Prefill

- Combined together to acquire a global view of complete prefill

	k0	k1	k2	k3
q0	1	-	-	-
q1	1	1	-	-
q2	1	1	1	-
q3	1	1	1	1

attention mask during first chunk prefill

Mask is a lower triangular matrix

K/V being loaded 2 times by late

	k0	k1	k2	k3	k4	k5	k6	k7
q4	1	1	1	1	1	-	-	-
q5	1	1	1	1	1	1	-	-
q6	1	1	1	1	1	1	1	-
q7	1	1	1	1	1	1	1	1

attention mask during second chunk prefill

Mask is a trapezoid matrix

K/V being loaded 1 time

	k0	k1	k2	k3	k4	k5	k6	k7	k8	k9	k10	k11
q8	1	1	1	1	1	1	1	1	1	-	-	-
q9	1	1	1	1	1	1	1	1	1	1	-	-
q10	1	1	1	1	1	1	1	1	1	1	1	-
q11	1	1	1	1	1	1	1	1	1	1	1	1

attention mask during third chunk prefill

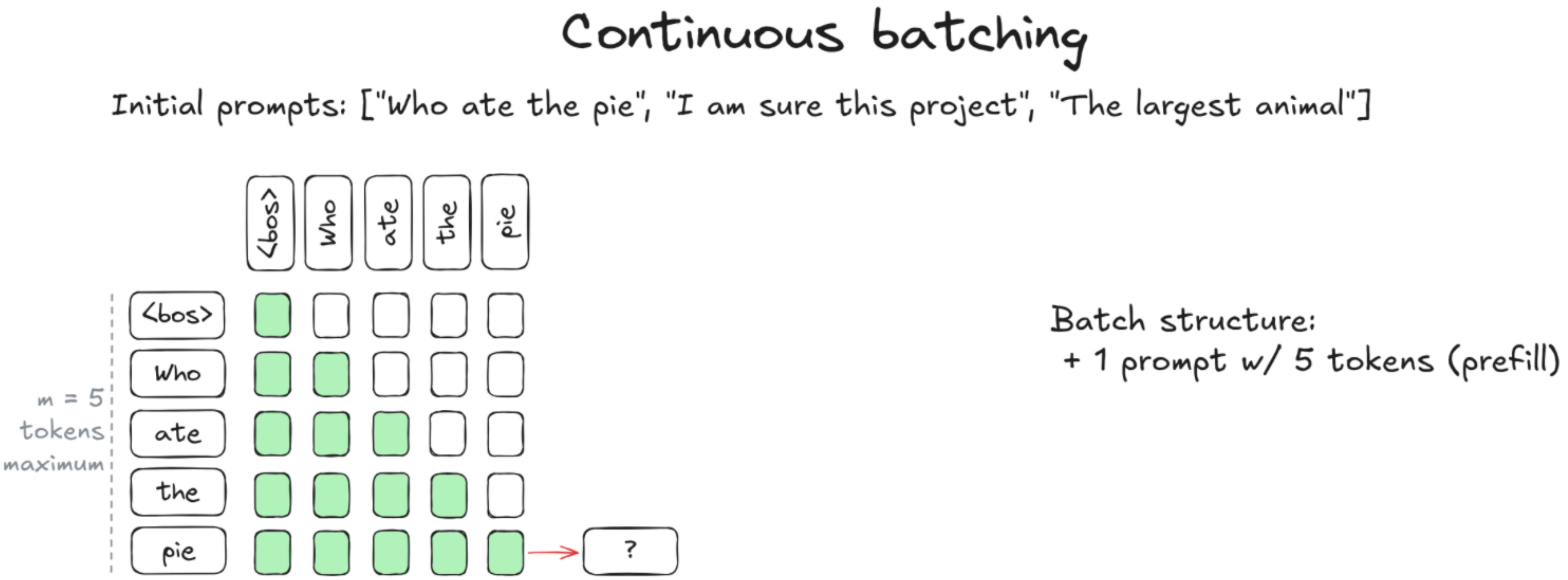
Mask is a trapezoid matrix

Attention mask matrices for successive prefill chunks

Source: SARATHI: Efficient LLM Inference by Piggybacking Decodes with Chunked Prefills

Continuous Batching

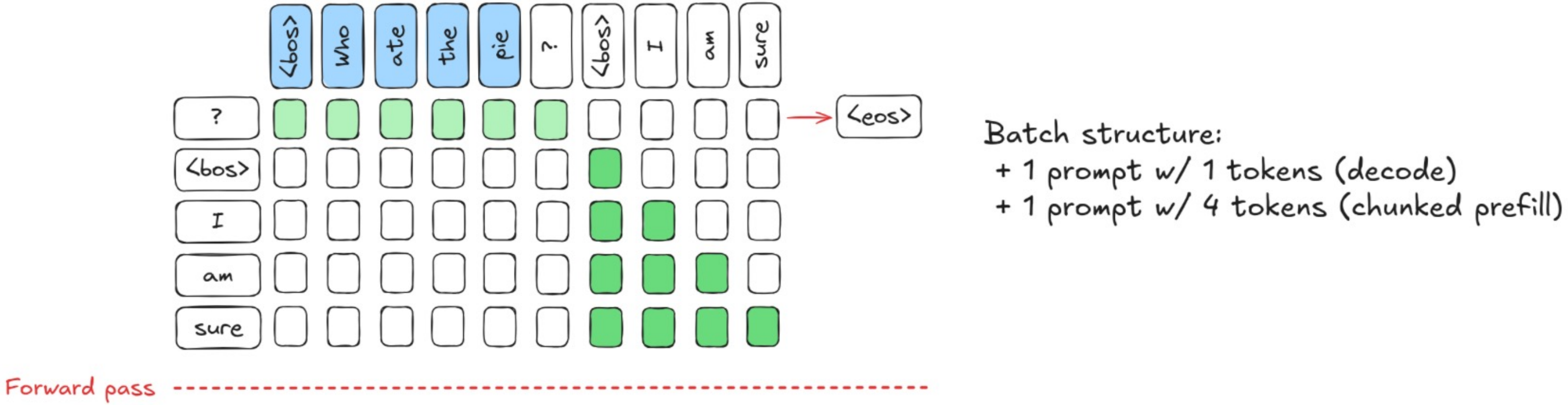
- Continuous batching = ragged batching + dynamic batching



Source: https://huggingface.co/blog/continuous_batching

Continuous Batching

- Continuous batching = ragged batching + dynamic batching



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Continuous Batching

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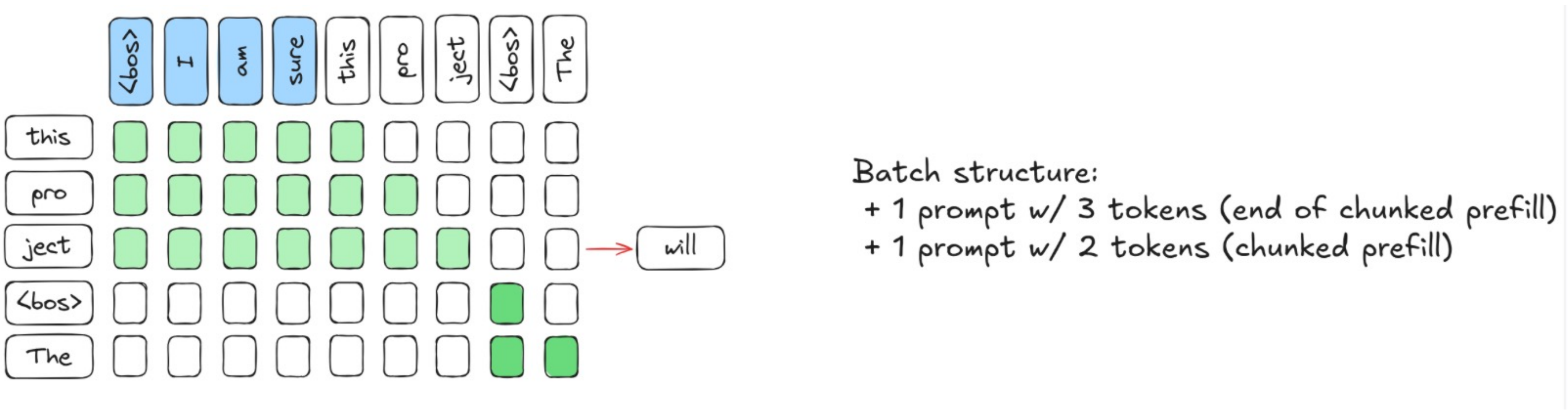

Source: https://huggingface.co/blog/continuous_batching

Thanks!